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A comprehensive review on key technologies toward smart healthcare systems based IoT: technical aspects, challenges and future directions

Muntadher Alsabah[1] · Marwah Abdulrazzaq Naser[2] · A. S. Albahri[3,4] · O. S. Albahri[5,6] · A. H. Alamoodi[7,8] · Sadiq H. Abdulhussain[2] · Laith Alzubaidi[9,10]

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Abstract
The unexpected death of humans due to a lack of medical care is a serious problem. Additionally, the number of elderly people requiring continuous care is increasing. A global aging population poses a challenge to the sustainability of conventional healthcare systems for the future. Simultaneously, recent years have seen remarkable progress in the Internet of Things (IoT) and communication technologies, alongside the growing importance of artificial intelligence (AI) explainability and information fusion. Therefore, developing smart healthcare systems based on IoT and advanced technologies is crucial. This would open up new possibilities for efficient and intelligent medical systems. Hence, it is imperative to present a prospective vision of smart healthcare systems and explore the key technologies that enable the development of these intelligent medical systems. With smart healthcare systems, the future of healthcare can be significantly enhanced, providing higher-quality care, improved treatment, and more efficient patient care. This paper aims to provide a comprehensive review of the key enabling and innovative technologies for smart healthcare systems. To this end, it will cover the primary goals of each technology, the current state of research, potential applications envisioned, associated challenges, and future research directions. This paper is intended to be a valuable resource for researchers and healthcare providers. Ultimately, this paper provides valuable insights for both industry professionals and academic researchers, while also identifying potential new research avenues.

Keywords Healthcare · Artificial intelligence (AI) · Internet of Things (IoT) · Explainable AI (XAI) · Data fusion

Extended author information available on the last page of the article

1 Introduction

The rapid growth of the world population and the increasing number of elderly and chronically ill patients have overwhelmed traditional healthcare systems, making them unsustainable and ineffective (Chen et al. 2023). Besides, the demand for hospitalization and patient care is rising worldwide. To this end, the need to provide quality care to patients while reducing healthcare costs becomes essential. To date, remote patient monitoring (RPM) and access to a reliable healthcare system are challenges in remote areas. Therefore, traditional healthcare systems are struggling to meet the growing demand for efficient and timely medical services, leading to significant challenges such as delayed treatments, incorrect medications, and errors in the management of many diseases, especially infectious diseases (Chen et al. 2021). Specifically, current healthcare systems are unlikely to sustain these ongoing healthcare requirements and demands. With such requirements and demands, healthcare systems need to be intelligent, efficiently managed, and sustainable, which has prompted a serious need to revolutionize healthcare systems in which smart healthcare becomes crucial.

Recent years have witnessed significant advances and developments in the utilization of devices and communication technologies from the Internet of Things (IoT) that can change the way we interact with our surroundings (Mansour et al. 2023). IoT can be defined as a network of physical devices that can be interconnected via the Internet, enabling them to communicate and share information (Ryalat et al. 2023). In particular, IoT enables physical objects to perceive, interact, and perform various tasks by allowing them to communicate and share information. IoT is a combination of ubiquitous communication, connectivity, and computing, along with ambient intelligence. The IoT can offer a range of benefits and capabilities, including identification, location tracking, sensing, and a cyber-physical paradigm that enables all real-world components to remain connected. IoT can include sensors, wearable devices, and implantable devices that perform various tasks. IoT enables these devices to gather and exchange information. The IoT has been shown to be a powerful tool for improving data management, decision-making processes, overall productivity, and operational efficiency (Allioui And Mourdi 2023).

The IoT has the potential to develop intelligent services in various sectors such as smart homes and smart healthcare systems. For instance, numerous IoT applications have already been successfully implemented in different areas, including smart traffic systems, fleet tracking solutions, logistics chain control, smart cities, smart metering, environmental monitoring, industrial automation, collision avoidance systems in cars, smart buildings, smart homes, and smart offices. Noting that there are somehow related terms to IoT that have been used in the literature, such as Internet of Everything (IoE), Machine to Machine (M2M), Cloud of Things (CoT), Web of Things (WoT) (Čolaković and Hadžialić 2018).

The number of IoT devices continues to grow steadily. The economic impact of IoT by 2025 is anticipated to be between \$3-6 trillion, with smart healthcare applications contributing the largest share, ranging from $\$1-2.5$ trillion (Shelke And Sharma 2016).

The global market size for IoT in healthcare is estimated at USD 53.64 billion in 2024 and is anticipated to reach around USD 368.06 billion by 2034, expanding at a compound annual growth rate of 21.24% from 2024 to 2034, as illustrated in Fig. 1 (precedenceresearch 2024). The integration of connected devices and the adoption of IoT systems are key drivers for this market expansion Fig. 2.

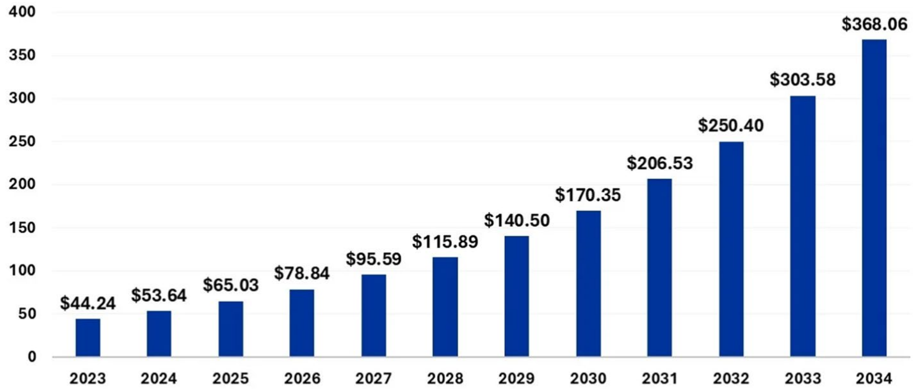
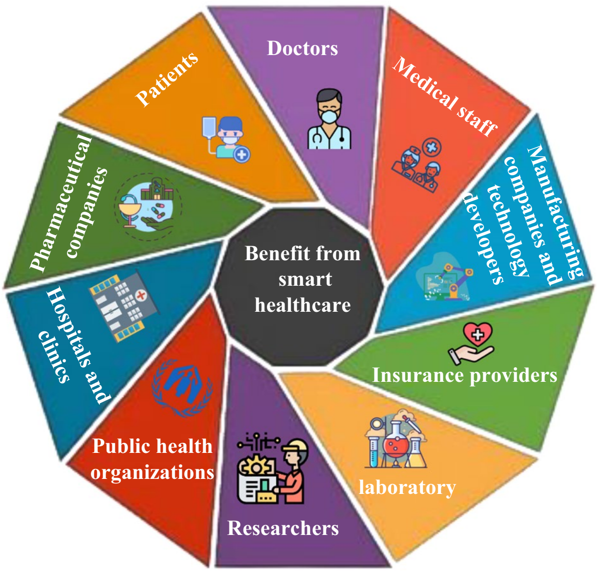


Fig. 1 The estimated IoT in healthcare market sizes, 2024 to 2034 (USD Billion) (precedenceresearch 2024)

Fig. 2 A list of those who can benefit from the smart healthcare system



1.1 IoT-driven smart healthcare systems

Currently, the healthcare system has transformed into a sophisticated ecosystem encompassing a diverse range of devices and communication technologies (Habibzadeh And Soyata 2020). These devices and technologies require interaction and collaboration to deliver various health services. In this regard, IoT has the potential to facilitate such interactions and enable an efficient connection for a global network of billions of devices, thereby improving healthcare systems (Zeadally And Bello 2021). Therefore, one of the most useful applications of IoT is in healthcare systems, which leads to what is commonly known as the Internet of Medical Things (IoMT), also known as the Internet of Healthcare Things (IoHT). In particular, IoMT integrates medical devices and IoT (Razdan And Sharma 2022; Muhammad et al. 2021). The IoT in healthcare systems visualizes a scenario in which all medical devices are connected to the Internet and can be remotely controlled by healthcare professionals (Razdan And Sharma 2022).

In the medical sector, IoT has several advantages, including improving drug management, reducing healthcare costs, enhancing patient experience, enhancing diagnosis and treatment, improving disease management, and achieving efficient RPM for chronic diseases most effectively (Sujith et al. 2022; Pradyumna et al. 2024). In healthcare systems, IoT serves as a bridge between doctors and patients, providing remote access that enables doctors to monitor patients and offer remote consultations continuously. We can enhance patient care by effectively integrating Internet of Things (IoT) devices into healthcare systems. Through interconnected sensors, wearable devices, and medical equipment, healthcare providers can remotely monitor patients' vital signs, track medication adherence, and predict health issues before they occur. This interconnected procedure enables healthcare professionals to make timely, informed decisions. Precisely, IoT-driven healthcare minimizes human error by networking all devices to a decision support system, hence empowering physicians to deliver more precise diagnoses.

In general, the development of IoT in medicine, with its diverse set of facilities, enables the delivery of smart healthcare, which can revolutionize healthcare delivery by integrating advanced technologies into medical systems. This integration will open up new possibilities for intelligent medical systems, leading to improved patient care, cost-effective medical solutions, and faster hospital treatments (Razdan And Sharma 2022). The following is a list of those who can benefit from a smart healthcare system.

1.2 Applications of smart healthcare systems

Smart healthcare can be defined as intelligent health management that utilizes advanced technologies to deliver efficient healthcare services anytime, anywhere. Smart healthcare systems can significantly address the evolving challenges of traditional healthcare systems by allowing real-time health monitoring, enabling secure access to healthcare, and facilitating accurate diagnoses.

In particular, smart healthcare systems efficiently utilize wearable sensors and medical devices, which can continuously monitor patients' health parameters and vital signs, such as temperature, heart rate, blood oxygen saturation, ECG, blood pressure, glucose levels, and activity levels (Mohanty et al. 2016). The information collected from patients can be transferred over wireless networks to healthcare providers. More importantly, smart healthcare systems involve the integration of several advanced technologies, including electronic health records (EHR), big data analytics, computing techniques, artificial intelligence (AI), information and communication technologies (ICT), data analytics, efficient security and privacy techniques, digital communication tools, and innovation techniques into healthcare systems to enhance the delivery of medical services and optimize healthcare processes. Based on these technologies, smart healthcare can reduce the burden on healthcare providers and improve the accuracy of diagnoses and treatments. In addition, it would have the potential to transform medical treatment, facilitate streamlined care delivery, and significantly improve patient' care and satisfaction. The following are the most effective applications of smart healthcare systems. Figure 3 demonstrates the most effective applications of smart healthcare systems.

- Cost reduction: Smart healthcare system-based systems could help in reducing patient treatment cost. Specifically, it can optimize healthcare operations and processes by automating routine tasks and reducing the requirement for expensive interventions. This would save time and reduce the need for an extensive administrative workforce, resulting in cost savings. Most significantly, the early detection and continuous monitoring capabilities offered by smart healthcare systems can help avoid expensive medical complications by reducing hospitalizations, thereby lowering healthcare costs significantly (Li et al. 2023).
- Health Data Analytics: By analyzing large amounts of medical data, primarily collected from IoT devices in smart healthcare systems, healthcare providers can gain valuable insights into patients' health issues, visualize patterns, and identify potential health risks. Therefore, analyzing patients' data in a smart healthcare system enables the development of personalized patient treatment plans, effective chronic disease management, informed decision-making, preventive healthcare interventions, and monitoring of daily physical activities (Talpur et al. 2024). Smart healthcare systems can tailor treatments and interventions to each patient's unique characteristics, genetic profile, and real-time health data.
- Sharing medical information: With a smart healthcare system, medical technicians, nurses, and doctors can access test data, patient information, and health recorders without losing time in transferring the same information physically from one office to another. This is because a smart healthcare system-based IoT facilitates seamless connectivity and interoperability between medical devices and healthcare providers, ensuring efficient data exchange and informed decision-making. Furthermore, different doctors can access patients' information to make informed judgments about a patient's condition, resulting in possible real-time decisions regarding patient health.

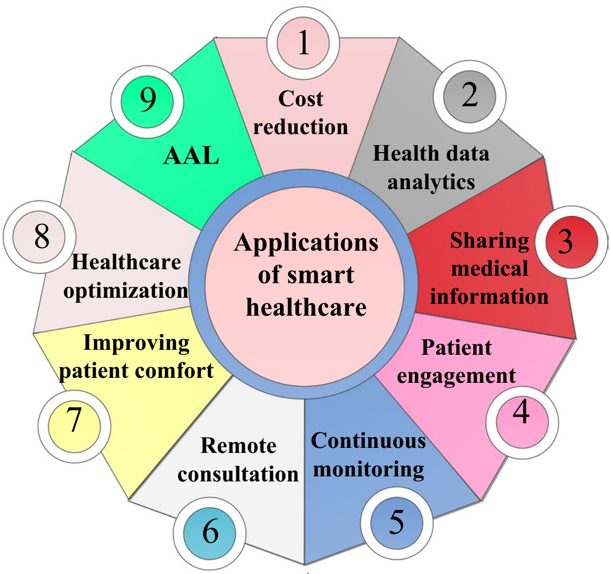


Fig. 3 A list of the most effective applications of smart healthcare systems

- Patient engagement: Smart health represents a transformative shift in healthcare delivery, empowering individuals to take a more active role in managing their health and enabling healthcare providers to deliver more personalized, accessible, and efficient care to all patients. Smart healthcare systems can also enhance patient engagement by providing patients with effective control over their healthcare, enabling them to record and communicate with healthcare providers easily. Smart healthcare empowers users to self-manage some emergencies (Mohanty et al. 2016). This type of patient engagement is referred to as patient- or human-centric IoT-based healthcare systems (Huang et al. 2023).
- Continuous monitoring: Smart healthcare system allows medical practitioners to extend their services without any geographical barriers (Krishnamoorthy et al. 2023). It enables telemedicine or telehealth by leveraging digital communication tools and telemonitoring platforms. Through the integration of IoT devices, wearable health monitors, and smart biosensors, patients can enjoy continuous, real-time health monitoring from the comfort of their own homes. Smart healthcare ensures continuously monitors health parameters, such as vital signs, activity levels, and medication adherence. This, in turn, would decrease the frequency of hospital visits and enable an early detection of health issues, hence facilitating quicker medical interventions.
- Remote consultation: Smart healthcare systems offer remote consultation and diagnosis between patients and healthcare providers via video conferencing, chat platforms, or mobile apps, facilitating access to healthcare services regardless of geographical location. Specifically, smart healthcare can enhance the sustainability and efficiency of healthcare delivery, ensuring that health services are accessible and affordable for everyone, particularly for those residing far from hospitals.
- Healthcare system optimization: Smart healthcare system aims to empower healthcare providers and organizations in managing and delivering healthcare services more effectively. For example, digital tools such as Electronic Health Records (EHRs) facilitate the seamless sharing of patient information, thereby reducing paperwork and administrative burdens. To this end, a smart healthcare system would enable streamlining of administrative processes, efficient resource allocation, and the use of digital tools and automation. This would significantly enhance operational efficiency, reduce costs, and improve patient satisfaction.
- Improving patient comfort and quick hospital treatments: Smart healthcare system can significantly enhance patient comfort and accelerate hospital treatments (Tian et al. 2019). Additionally, a smart healthcare system streamlines various processes, from patient admissions to post-treatment care. Hence, it would allow efficient workflows, shorter waiting times, and hence faster treatment.
- Ambient Assisted Living (AAL): Smart healthcare aims to achieve AAL. This is achieved by utilizing information and communication technologies (ICT) in people's living environments to support their quality of life, particularly for older adults and individuals with disabilities or chronic conditions. AAL aims to provide solutions that enable individuals to live independently and safely in their own homes, while also providing caregivers with tools to remotely monitor their patients. With a smart healthcare system, a significant improvement in the quality of life for elderly people based on AAL can be achieved (Alshammari 2023).

1.3 Classification of related review papers

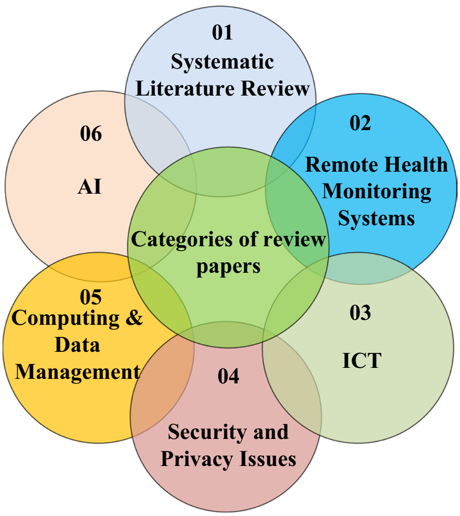
In recent years, there has been growing interest in integrating the Internet of Things (IoT) into healthcare systems. Hence, it's crucial to acknowledge current trends and highlight both technical developments and the challenges in IoT-based healthcare systems. In this regard, researchers have started to explore various technological solutions to enhance healthcare provision. An extensive search is conducted in this paper to identify the research works that considered IoT and advanced technologies in healthcare systems. In the following, we will classify the state-of-the-art review papers based on their primary focus area in healthcare systems into the following categories. Figure 4 shows a classification of related review papers in healthcare systems.

1.3.1 Systematic literature review papers

Several works have presented a systematic literature review (SLR) about healthcare system-based IoT. For example, the work in Qi et al. (2017) presented SLR about IoT architecture for personalized healthcare systems. To this end, a multi-layer architecture is presented. The work in Mieronkoski et al. (2017) delivered SLR aiming at investigating IoT's importance in enhancing basic nursing care in hospital environments by improving the quality of care and patient safety. The work in Jiang et al. (2017) provided SLR of AI applications in healthcare.

The work in Mutlag et al. (2019) presented SLR for fog computing in healthcare systems. The authors have identified three key aspects for controlling resources in healthcare systems, which are defined as computation offloading, load balancing, and interoperability. The work in Nazir et al. (2019) presented SLR to explore the role of mobile computing in enhancing IoT applications within healthcare systems. The work in Ahmadi et al. (2019) conducted SLR to identify the application areas of IoT in healthcare, the key components of IoT architecture within the healthcare systems, and the most critical technologies involved. The work in Hussien et al. (2019) conducted a comprehensive survey and classification of research articles on blockchain and their use in healthcare applications.

Fig. 4 A list of the review papers considering the healthcare system



The work in Al-Turjman et al. (2020) provided an SLR that investigated the IoT in healthcare systems and its potential to revolutionize healthcare by improving quality of life, care services, and cost-effectiveness. The paper proposed a generic IoT framework comprising data acquisition, communication gateways, and servers/cloud services. The work in Alshehri and Muhammad (2020) provided a systematic literature review (SLR) of IoT-based smart healthcare and discussed areas such as AI, security issues, and cloud/edge computing within smart healthcare systems. The work in Alamoodi et al. (2020) conducted a systematic literature review (SLR) on the assessment of medical mobile apps for healthcare systems. The study in Sadoughi et al. (2020) provided an SLR to identify developments in IoT-based healthcare systems, offering graphical and tabular classifications of experimental and practical IoT applications, aided by bibliometric analysis. The paper in Tortorella et al. (2020) provided a scoping literature review that explained the trends, challenges, and theoretical gaps in the implementation of healthcare 4.0 (H4.0) systems. The work in Aceto et al. (2020) presented SLR discussing the impact of IoT in healthcare systems, with a special focus on big data as well as cloud computing and fog computing.

The work in Muhammad et al. (2021) presented the systematic literature review (SLR) of multimodal medical signal fusion schemes in smart healthcare systems. To this end, the paper investigated how wearable devices and sophisticated machine learning (ML) algorithms can be integrated to enhance healthcare by accessing health records and intelligently managing health demands. The work in Secinaro et al. (2021) provided an SLR about AI strategies in healthcare, discussing their role in assisting physicians in diagnosis, disease prediction, predictive medicine, personalized treatment, and clinical decision-making. The work in Salman et al. (2021) conducted an SLR on the role of ML in electronic emergency triage (E-triage) and patient prioritization for expedited healthcare services in telemedicine applications. The work in Mendo et al. (2021) provided SLR with the application of ML in medical emergencies. The work in Aghdam et al. (2021) provided an SLR of IoT applications in healthcare systems. The paper highlighted the significant advancements in IoT technologies that can enhance the efficiency and intelligence of healthcare services, enabling physicians and hospital staff to make informed decisions and monitor patients more effectively. The work in Kashani et al. (2021) provided a systematic literature review (SLR) of the IoT in healthcare and classified the research into five categories: sensor-based, resource-based, communication-based, application-based, and security-based approaches.

The study in Sujith et al. (2022) provided the SLR of a smart healthcare monitoring system and focused on some related techniques for efficient healthcare monitoring systems such as the integration of cloud computing and cloud storage, the integration of blockchain to improve data security and data privacy, the involvement of deep learning (DL) and ML to analyze health data. The work in Ciecierski-Holmes et al. (2022) conducted an SLR about the role of AI in enabling efficient healthcare systems. The study in Antunes et al. (2022) presented SLR about federated learning in the context of EHR data for healthcare applications. The work in Ahsan et al. (2022) presented SLR to uncover the challenges associated with imbalanced data in heart disease predictions. To this end, a meta-analysis approach was employed, considering various factors such as heart disease type, algorithms, applications, and solutions.

The work in Mohsen et al. (2023) conducted SLR to apply AI-based models for predicting type 2 diabetes mellitus (T2DM). The work in Huang et al. (2023) investigated recent advancements in technologies that support healthcare systems, including sensors, communication, parallel computing, and AI. The paper in Rejeb et al. (2023) presented a bibliometric SLR of IoT applications in healthcare and discussed some key topics such as blockchain, AI techniques, 5 G telecommunications, and data analytics. The work in Malgaroli et al. (2023) conducted SLR of studies utilizing Natural Language Processing (NLP) to analyze mental health interventions (MHI), highlighting the potential of NLP to revolutionize the assessment and treatment of neuropsychiatric disorders.

1.3.2 Review papers focusing on remote health monitoring systems

Several review papers have focused on investigating remote health monitoring (RHM), also known as remote patient monitoring (RPM), within the healthcare system. For example, the work in Shah et al. (2016) reviewed the enabling Quality of Service (QoS) in terms of cyber-physical systems for RHM applications. The work in Mshali et al. (2018) provided an overview of smart RHM systems by offering context-aware electronic health (eHealth) services. The work in Rodrigues et al. (2018) reviewed the techniques used in IoT-based healthcare systems. The paper divided the healthcare systems based on IoT into four categories: (1) RHM, (2) healthcare solutions based on smartphones, (3) AAL, and (4) wearable devices, and discussed each one of these categories.

The work in Naresh et al. (2020) provided a review on the application of IoT in RHM, focusing on the enabling technologies such as sensors, cloud computing, fog computing, and Wireless Body Area Networks (WBANs), which can be used to allow real-time monitoring, telemedicine, chronic disease detection, and elderly care. The work in Singh et al. (2020) discussed the implementation of the IoT in healthcare systems to enhance RHM delivery, particularly for orthopedic patients. The work in Kelly et al. (2020) investigated how IoT can enhance health service delivery by enabling proactive prediction, diagnosis, treatment, and RHM of patients, especially in crises such as the COVID-19 pandemic.

The work in Pradhan et al. (2021) discussed how IoT technologies have connected medical devices, sensors, and healthcare professionals to enable RHM, enhance patient safety, reduce costs, and improve the accessibility and efficiency of healthcare services. The work in Tun et al. (2021) provided a detailed overview of the applications of IoT and wearable technologies for assisting RHM in elderly care, highlighting the types of data collected and devices used. The work in Kim et al. (2021) provided a review of mobile health (mHealth) applications designed to support caregivers of Alzheimer's disease patients.

The work in Alshamrani (2022) provided a review of RHM systems. The study clarified the integration of IoT and AI for RHM systems, including underlying technology, devices, systems, and use cases. The work in Li et al. (2023) highlighted several healthcare applications, including RHM, personalized treatment strategies, various sensor types, communication methods, and improved healthcare delivery processes. The work in Talpur et al. (2024) provided a review of IoT-based RHM for chronic diseases.

1.3.3 Review papers focusing on information and communication technologies (ICT)

Numerous studies have examined the role of ICT in healthcare systems. For example, the work in Islam et al. (2015) conducted a review of IoT-based technologies in healthcare systems, discussing the state-of-the-art communication network architectures, services, and applications. The work in Yuehong et al. (2016) provided a comprehensive literature review that summarized the applications of IoT in the healthcare system with a special focus on communication technology. The work in Alam et al. (2018) reviewed different communication protocols that are relevant to healthcare. The paper in Aceto et al. (2018) clarified the relationship between ICT and healthcare systems by identifying the prevalent ICT-based healthcare paradigms and the technologies that support them.

The work in Ahad et al. (2019) provided a review of the evolving technology of 5 G and IoT-enabled smart healthcare, outlining the shift from traditional hospital care to a patient-centric approach driven by advances in ICT. The work in Dhanvijay and Patil (2019) investigated the integration of the IoT with WBAN for healthcare applications. Additionally, several aspects related to IoT-based healthcare systems, including security and privacy, authentication, energy management, resource management, QoS, and real-time remote monitoring, were also discussed. The work in Ahad et al. (2020) presented a review of technological advancements of communication technologies with 5 G networks to assist IoT-based smart healthcare systems.

The paper in Poongodi et al. (2021) discussed the implementation of a smart navigation system for ambulances, integrating 5 G networks and IoT technologies to enhance emergency medical response in smart cities. The work in Nayak and Patgiri (2021) investigated the healthcare system based on future 6 G communication technologies and their role in enhancing telesurgery, as well as in preparing for epidemics and pandemics. The work in Srivastava et al. (2021) presented the challenges associated with adopting blockchain technology in eHealth, particularly in future 6 G networks. The work in Padhi and Charrua-Santos (2021) explained a Cognitive Internet of Healthcare Everything (CIoHE) framework, which was proposed to improve healthcare services and reduce costs. The authors in Tang et al. (2022) investigated the progression of communication networking paradigms and AI, establishing the foundation Internet of Intelligence concept. The paper in Krishnamoorthy et al. (2023) reviewed the advancements in WBAN and their role in developing next-generation healthcare applications. The work in Ahad et al. (2024) provided an overview of how 6 G technology can revolutionize wireless healthcare systems.

1.3.4 Review papers concentrate on security and privacy issues

Several review papers have investigated the security and privacy challenges in healthcare systems. For example, the work in Martínez-Pérez et al. (2015) investigated the security and privacy challenges in mHealth applications. The work in Gatouillat et al. (2018) conducted a review on the IoT in medical systems, focusing on enhancing its reliability, safety, and security through cyber-physical systems methodology. The work in Sun et al. (2018) focused on identifying and addressing the security and privacy requirements in healthcare systems.

The work in Sun et al. (2019) investigated the security and privacy challenges and requirements in IoT-based healthcare systems, starting from the data level and extending to the medical server level. The work in Khezr et al. (2019) presented a comprehensive survey of the emerging integration of blockchain technology in healthcare, offering a technical analysis of their strengths and weaknesses. The paper in Habibzadeh and Soyata (2020) presented a review of security and privacy issues in smart healthcare systems and highlighted how smart healthcare can be integrated with smart homes, environments, and transportation to tackle issues like population aging and rising healthcare costs. The work in Bhuiyan et al.

(2021) provided a review of IoT-based healthcare systems with healthcare applications and services with a special focus on security, privacy, and data protection.

The work in Yaqoob et al. (2022) presented a review that discussed the challenges faced by healthcare data management systems, such as issues with data transparency, traceability, immutability, auditability, provenance, flexible access, trust, privacy, and security. The work in Razdan and Sharma (2022) focused on reviewing security, privacy, and accuracy issues in healthcare systems. To this end, the paper investigated technologies such as Physically Unclonable Functions (PUF), blockchain, AI, and software-defined networking (SDN), which can be used to address the security, privacy, and accuracy challenges in healthcare systems. The work in Alhaj et al. (2022) provided a review discussing the challenges of security and privacy in IoT-based healthcare systems and introduced a new taxonomy for IoT-based security and privacy in healthcare systems, focusing on cybersecurity principles. The work in Mishra and Singh (2023) discussed the role of smart healthcare systems based on IoT, aiming to enhance healthcare efficiency, security, personalization, and accessibility. In this research (Mahdi et al. 2025), the authors presented a hybrid deep learning and machine learning methodology for intrusion detection in IoT networks, addressing critical limitations such as high false alarm rates, delayed detection, and outdated training data. Their approach combines correlation-based and sequential feature selection techniques with a cascaded LSTM and Naive Bayes classifier to enhance threat detection accuracy. Additionally, they introduced automated defense mechanisms to block malicious activities upon detection. Security and privacy are critical considerations in the design and deployment of IoT-based healthcare applications (Alzubaidi et al. 2025; Shyaa et al. 2024).

The work in Adil et al. (2024) provided an overview of the current authentication and data preservation techniques used in healthcare-based IoT applications. The work in Pradyumna et al. (2024) reviewed recent advances in IoT-based healthcare systems with a special focus on the role of ML and data security and privacy challenges in healthcare systems. The work in Ahmed et al. (2024) investigated the application of IoT in healthcare, highlighting opportunities in data storage, protection, and collection methods, advocating for standardized architecture and advanced security measures, including cryptography and blockchain technology.

1.3.5 Review papers focusing on computing and data management

Several works have investigated big data analytics and data management approaches in healthcare. For example, the work in Dimitrov (2016) reviewed the IoT and big data in healthcare systems and discussed how mobile apps can enhance communications between patients and doctors over a secured connection. The work in Kraemer et al. (2017) provided a comprehensive review of fog computing for data management in healthcare systems, categorizing and discussing various application use cases. The work in Baker et al. (2017) reviewed cloud computing for data storage in the healthcare sector and presented several works that focused on improving security in the cloud server.

The work in Rahmani et al. (2018) investigated the integration of IoT and cyber-physical systems in healthcare to develop intelligent and secure healthcare systems. Besides, the paper presented a fog computing-based solution to enhance energy efficiency, reliability, scalability, and seamless connectivity for mobile sensors in healthcare IoT systems. The work in Jagadeeswari et al. (2018) provided a detailed study of recent technologies in healthcare systems, with a focus on big data analytics, cloud and fog computing, and mobile-based applications. The work in Kumari et al. (2018) investigated how fog computing, cloud computing, and IoT can provide continuous context-aware services and deliver seamless services to healthcare systems.

The work in Saheb and Izadi (2019) provided a review of the convergence of IoT and big data analytics in the healthcare system. To address this, the authors posed two main questions: how IoT and big data analytics converge in healthcare, and what applications have emerged from this convergence. The work in Dang et al. (2019) provided a review aimed at investigating the latest IoT components, applications, and market trends in healthcare systems. The work in Ray et al. (2019) investigated the role of industrial standards and elements of edge computing in enhancing user experiences through its integration with IoT in healthcare systems. The work in Zhu et al. (2019) examined various aspects of smart healthcare, including health data management and patient-centered health management. The work in Elazhary (2019) presented a review that focused on various computing paradigms such as cloud computing, mobile edge computing, and fog computing. Additionally, the paper discussed the application of the Internet of Nano Things (IoNT) as an emerging technology.

The work in Qadri et al. (2020) provided an overview of IoT-based healthcare and presented various architectures tailored to specific use cases. In addition, the paper analyzed several technologies, including ML, edge computing, blockchain, big data, SDN, and IoNT. The work (Karatas et al. 2022) discussed the integration between industry 4.0, big data, and healthcare operations, focusing on big data for healthcare systems. The work in Ullah et al. (2021) provided a survey on IoT in healthcare, focusing on the secure aggregation and transmission of sensitive data using fog computing architectures. The work in Haleem et al. (2022) investigated the impact of medical 4.0, focusing on its role in revolutionizing patient care. The work in Adeghe et al. (2024) investigated the utilization of big data analytics in healthcare.

1.3.6 Review papers concentrate on AI in healthcare systems

Several review papers have investigated the application of AI tools in healthcare systems. For example, the work in Yu et al. (2018) described the developments in AI technology and its applications in healthcare and pointed out the obstacles to further advancements in smart medical systems. The work in Davenport and Kalakota (2019) provided a review that discussed the types of AI algorithms that are relevant to healthcare systems. The work in Schönberger (2019) investigated the decision-making capacities of AI technologies in healthcare. The work in Esteva et al. (2019) provided a review of the application of DL in healthcare systems. To this end, the paper explored the application of computer vision in medical imaging, the use of NLP in EHR data, the role of reinforcement learning in robotic-assisted surgery, and the application of generalized deep learning methods in genomics. The work in Zhao et al. (2019) conducted a review to summarize the emerging research on deep learning (DL) in health monitoring. The work in Piccialli et al. (2021) carried out an in-depth study of DL methodologies and applications in healthcare systems. The work in Bharadwaj et al. (2021) provided a review of the integration of ML algorithms into IoT-based healthcare systems, emphasizing their role in data gathering and processing for enhanced healthcare automation. The work in Bajwa et al. (2021) provided a multi-step approach to building reliable AI-augmented systems in healthcare systems.

The paper in Shehab et al. (2022) provided a deep analysis of ML applications in the healthcare systems, emphasizing standard technology and their impact on the diagnosis of diseases. The work in Nguyen et al. (2022) outlined the developments in federated learning, together with the reasons and prerequisites for applying federated learning in smart healthcare systems. The work in Ahsan et al. (2022) presented a systematic bibliometric study about the recent approaches in ML-based disease diagnosis, considering the following factors: algorithm, disease types, data type, application, and evaluation metrics. The work in Shmatko et al. (2022) discussed AI in histopathology and stated that with digital histopathology slides, AI can predict genetic changes, gene expression, therapy response, and cancer prognosis. The work in Rajpurkar et al. (2022) presented a review of the effectiveness of using AI techniques in medical systems. The paper in Manickam et al. (2022) reviewed the role of AI in enhancing the capabilities of the IoT in healthcare systems, including cardiac measurement, cancer diagnosis, and diabetes management.

The work in Neto et al. (2023) conducted a review discussing the role of ML in enhancing IoT security in healthcare systems. The paper in Kumar et al. (2023) reviewed several studies on the applications of AI to develop smart predictive maintenance models. The work in Ramgopal et al. (2023) provided a review discussing the role of AI-based clinical decision support in paediatric care. The work in Hamdoun et al. (2023) investigated the role of AI in digital mental health applications. The work in Ardahan Sevgili and Şenol (2023) discussed the role of AI and ML in the diagnosis, treatment, and management of paediatric oncological diseases. The work in Alowais et al. (2023) presented a review of the current state of AI in clinical practice, including its potential applications in disease diagnosis, treatment recommendations, and patient engagement. The work in Al Kuwaiti et al. (2023) focused on reviewing the role of AI in healthcare by investigating its impact across several key areas, including medical imaging and diagnostics, virtual patient care, medical research and drug discovery, patient engagement and compliance, rehabilitation, and administrative applications.

The work in Olawade et al. (2024) presented a review discussing the role of AI in mental healthcare. The work in Rahman et al. (2024) provided a survey of ML and DL applications in intelligent healthcare systems, highlighting their roles in addressing challenges like disease prediction, drug discovery, and medical image analysis. The work in Garcia et al. (2024) provided a review that discussed the integration of AI technologies into medical education. Recently, the authors in Naser et al. (2024) demonstrated the role of machine learning (ML) in predicting heart disease in healthcare systems. The work in Messinis et al. (2024) provided a review discussing how AI can enhance security in IoT-based healthcare systems.

1.4 Paper contributions

The reviewed papers presented above primarily concentrated on specific technologies and aspects of healthcare systems. However, despite extensive research, there remains a need to identify emerging trends and underscore the technological advancements and challenges in healthcare systems. Therefore, there is a lack of identifying what are the emerging trends and relevant technologies, which are essentially required to achieve smart healthcare systems. Furthermore, most review papers fail to provide a comprehensive understanding of certain relevant topics related to healthcare systems.

Therefore, unlike previous survey papers, this paper defines various components of smart healthcare systems and comprehensively investigates all relevant technologies, that are essentially required to obtain smart healthcare systems. This survey aims to offer a comprehensive review of recent advancements in healthcare systems. Following the identification of key relevant and innovative technologies, this survey paper explores the core definitions of each technology, emphasizes their essential benefits, reviews the current state-of-the-art research, and discusses the associated research challenges. This paper offers extensive insights and detailed research on several technologies that can transform healthcare systems.

We have reviewed the most significant research papers and contributions in the field of healthcare systems. Specifically, smart healthcare systems utilize several advanced technologies such as sensors and smart devices, RPM systems to enable telemedicine and AAL, EHRs, ICT for connection and networking, big data analytics and management, computing techniques, AI, and techniques to maintain security and privacy such as blockchain and watermarking. In addition, one of the key contributions of this survey paper is the presentation of innovative techniques for smart healthcare systems. In particular, this survey paper discusses the development and application of the following innovation techniques: IoNT, energy harvesting for energy efficiency, metaverse and digital twin, robotic and Tactile Internet, XAI, federated learning, data fusion, quantum communication, and edge learning. Furthermore, this paper identifies challenges, open issues, and future directions in the healthcare field, offering insights for future research to enhance the robustness of IoT technologies-based healthcare systems. This survey can be attracted to beginner, experienced researchers, professors, specialists, as well as health professionals who wish to know the latest and recent techniques studied in the healthcare systems. In summary, this paper serves as a valuable guide for academic researchers, engineers, health professionals, and policymakers working in the field of IoT and healthcare systems.

1.5 Key enabling technologies toward smart healthcare systems

The main objective of this survey paper is to provide an overview of the key enabling technologies toward smart healthcare systems. For each key enabling technology, we provide a basic definition, its benefits, the current state of research, and the associated challenges. Therefore, we identify research papers that align with this specific scope and the investigation's criteria. To this end, we aim to provide a comprehensive review of papers that address this research topic. Figure 5 provides the key enabling technologies for smart healthcare systems.

1.6 Methodology and search strategy

The study followed a structured methodology where the task of data collection precedes the task of data analysis (Nazir et al. 2019; Albahri et al. 2021). To enhance the probability of retrieving high-quality search results, a diverse range of well-known bibliographic citation digital databases, including scientific and social science journals from various disciplines, were utilized to conduct the research. The search for relevant papers involved four well-known and reliable digital databases: Science Direct (SD), IEEE Xplore (IEEE), Web of Science (WoS), Springer, Nature, and MDPI. These databases are essential for researchers because they offer comprehensive coverage of research conducted in scientific and tech-

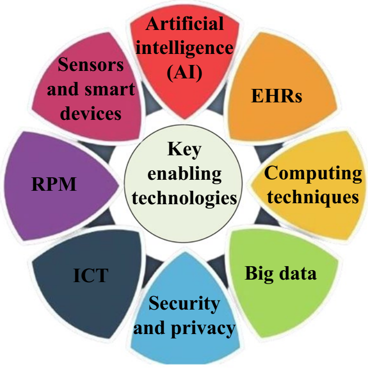


Fig. 5 A list of the key enabling techniques for the smart healthcare system

nological fields, providing valuable insights for further analysis and investigation (Albahri et al. 2021). To ensure a high-quality and focused literature base, the selected publications must meet the following criteria:

- The articles should be written in English and published in a journal or in the proceedings of a renowned conference.
- The articles should explore the integration of advanced techniques or methodologies aimed at developing or enhancing healthcare systems, with a specific focus on IoT-based applications.
- The articles should have a measurable citation record, indicating their recognition and impact within the research field.

We exclude publications such as book chapters, books, editorial materials, retracted papers, data papers, meeting abstracts, letters, corrections, and proceedings. Only peer-reviewed original research articles were considered. We obtained data from the above-mentioned online search databases and designed a structural format of our paper accordingly. The scope of this survey encompasses all scientific papers published over the last 10 years.

To conduct the search, a Boolean query was implemented to collect studies from different viewpoints; the query is general and utilizes the"OR"operator to connect the keywords,"healthcare", and"health care", and"smart healthcare", and the"AND"operator to connect these keywords with"Internet of Thing", and"IoT", and"Internet of Medical Things", and"IoMT", and"Artificial Intelligence", and"AI".

Initially, a comprehensive search was conducted, resulting in 1034 entries from various sources, including 246 from SD, 201 from IEEE, 217 from Springer, 187 from MDPI, 40 from Nature and 143 from WoS. We applied three filtering steps. The first step involved quick procedures, such as removing duplicates, books, and chapters, leading to the removal of 164 redundant entries and reducing the total number of reviewed papers to 870. In the second filter, we carefully assessed titles and abstracts, excluding 258 articles that did not meet our eligibility criteria. After this initial screening, 612 contributions remained for a detailed examination. In the final filter, 206 studies did not meet the specified criteria discussed above and were excluded. In comparison, 406 studies were deemed relevant and met the eligibility requirements, thereby securing their inclusion in the final collection of articles. The process and its outcomes are visualized in Fig. 6, which provides an overview of the steps in filtering and selecting the final set of articles for analysis. The final selected publications represented a diverse range of high-impact publishers, with the following distribution:

- IEEE (136 publications, 33%).
- Elsevier (113 publications, 28%).
- Springer (55 publications, 14%).
- MDPI (36 publications, 9%).
- Nature and Nature Publishing Group (25 combined publications, 6%).
- Other specialized publishers (41 publications, 10%).

1.7 Paper organization

The rest of the survey paper is organized as follows. Section 2 discusses the sensors and wearable devices that can be used for RPM in smart healthcare systems. Section 3 presents a review of RPM in smart healthcare systems. This includes a discussion related to telemedicine, eHealth, mHealth, connected health, and AAL. Section 4 explains the use of EHRs in hospitals and clinics and explains their role in enhancing patient care by reducing errors and improving care coordination. Section 5 describes the technologies used for efficient communication and networking in smart healthcare systems. Section 6 reviewed the role of big data analytics in smart healthcare systems. Computing techniques that are used in smart healthcare systems to address the long propagation distance are discussed in Sect. 7. These techniques include cloud computing, fog/edge computing, mobile edge computing (MEC), network functions virtualization (NFV), (software-defined networking (SDN), cloudlet, and network slicing. Section 8 highlights the role of AI tools and the applications related to ML and DL techniques. Section 9 provides an in-depth overview of the technologies necessary to ensure security and privacy in smart healthcare systems. To this end, this section discusses various security techniques in healthcare systems, including cryptography, zero-trust security, biometrics, watermarking, and blockchain. Section 10 provides a discussion related to the key innovation techniques for smart healthcare systems, which include IoNT, energy harvesting for energy efficiency, metaverse and digital twin, robotic and Tactile Internet, XAI, federated learning, data fusion, quantum communication, and edge learning. Finally, Sect. 11 provides a conclusion of this review paper. Fig. 7 outlines the structure of the paper, detailing each section and subsection.

2 Sensors and smart devices for RPM

The aging population and rising healthcare costs have garnered significant attention to wearable medical sensors. Recent advancements in communication, signal processing, AI, and biomedical sensing have brought continuous health monitoring closer to reality. A key development is Internet-connected wearable medical sensors that can collect and process various data (Mosenia et al. 2017). Sensors and smart wearable devices are enabling tools that can be effectively used to monitor the physiological parameters and activity levels of patients. Sensors are defined as computing devices that can be installed on people and objects. In particular, sensors are used for data acquisition and gather information related

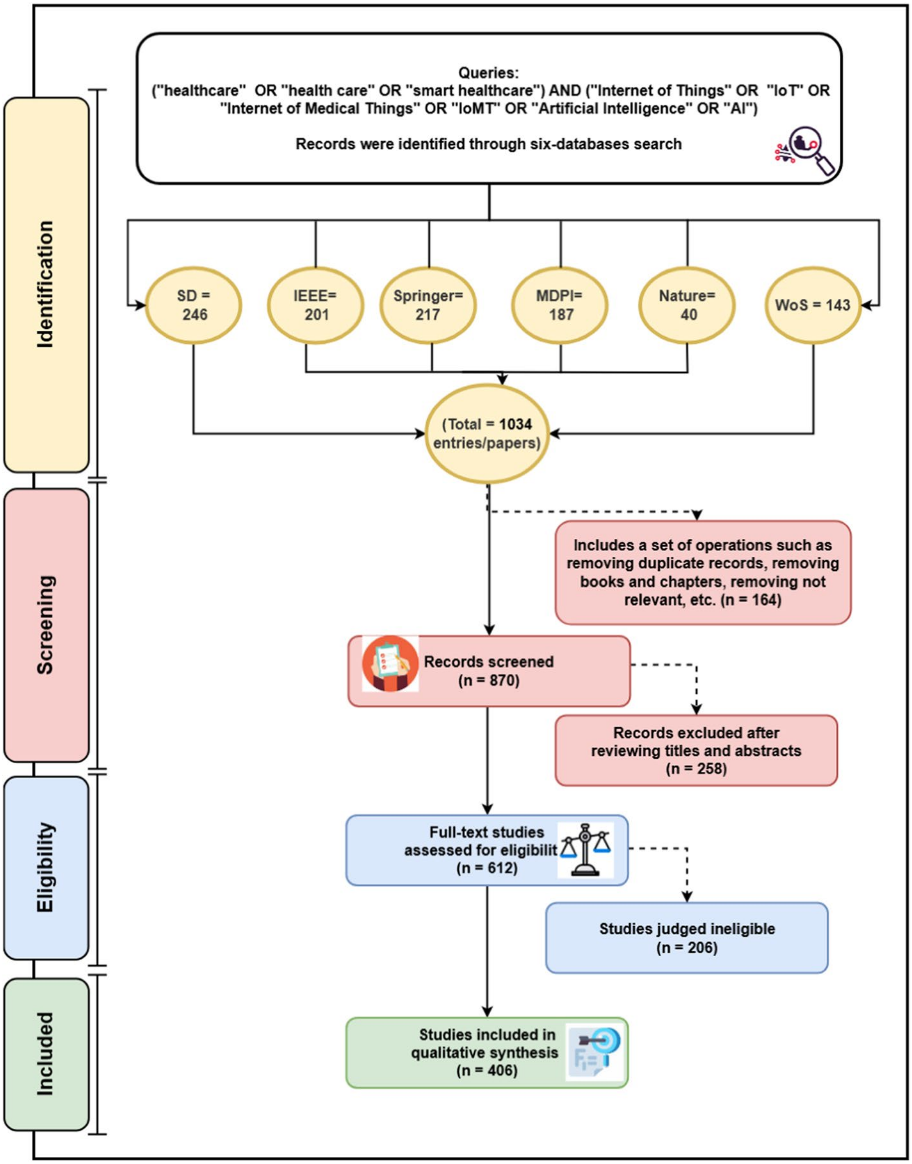


Fig. 6 Selection criteria considered in this paper

to the patient, environment, activities, and behaviors. Sensors are smaller in size, faster in process, and cost-efficient. Sensors can sense, transmit, process information, and perform various tasks, such as motion detection, image sensing, voice control, environmental perception, physiological signal monitoring, and gesture recognition.

Sensors receive their input signals from the physical surroundings and produce a response action. Typically, the input data can be collected through sensors or specific embedded or wearable devices and then processed. After the vital signs or signals are received by sensors,

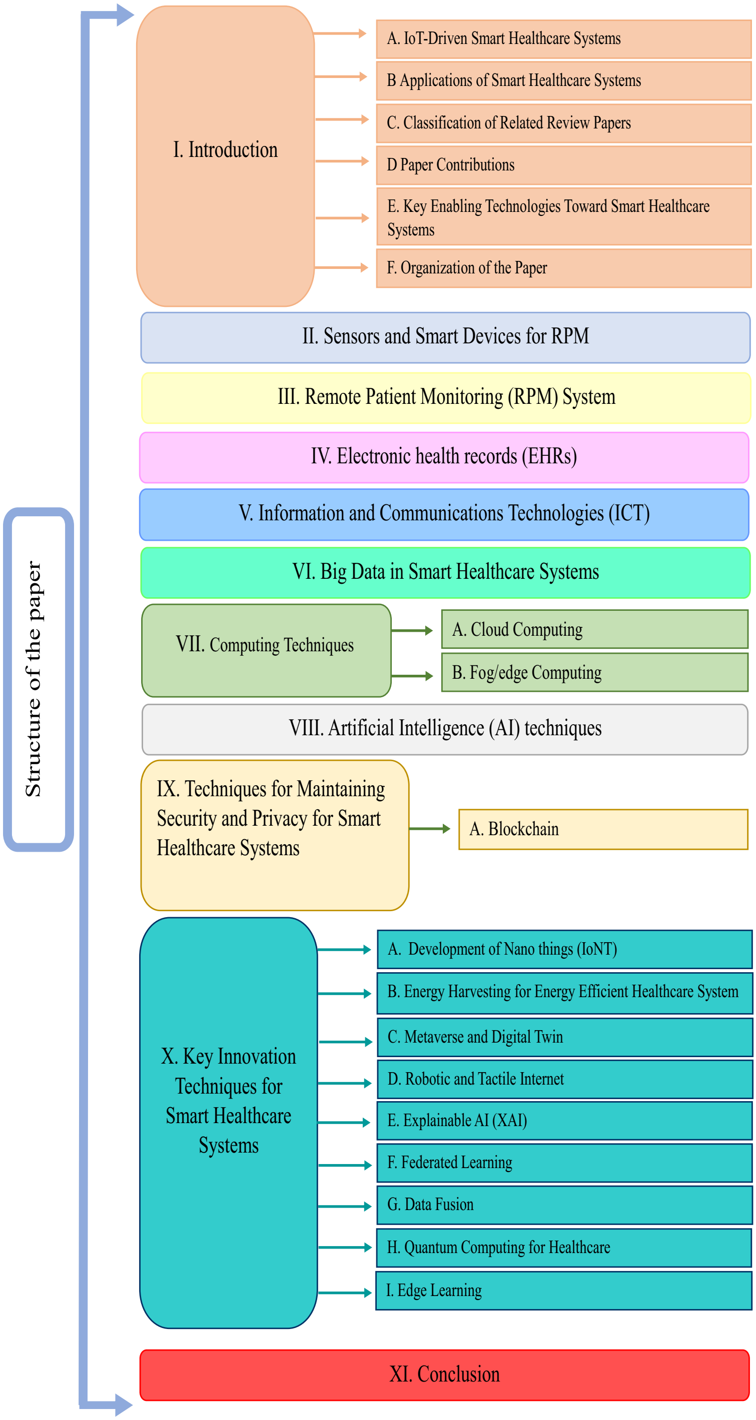


Fig. 7 An illustration of the paper structure

they are transmitted wirelessly to the control unit. Sensors send data to the microcontrollers, which then analyze the data and determine the process based on the algorithmic design implemented in the control unit. For example, if a temperature body sensor detects a change in the body temperature, the signal will be sent wirelessly to the specialist to sound an alarm and apply a quick treatment to the patient. The sensor can be implanted inside the human body, beneath the skin (in-body), placed close to the body (off-body), sewn into clothing (wearable), or applied to the human body as tiny patches (on-body) (Ghamari et al. 2016). Sensors can be divided into passive and active. The passive-type sensor does not require an additional energy source. It can generate an electric signal directly in response to an external stimulus. Examples of passive sensors are thermal, infrared, electric, and chemical. On the other hand, an active sensor type needs an external energy source for its response (known as an excitation signal). Examples of active sensors are thermistor sensors and resistive strain gauge sensors.

Wearable devices are typically attached to the human body or are integrated into elastic bands, textile fibers, or patient clothes. Wearable devices are used to measure the physiological signals of the patients and their activity. AI tools can be used to support wearable devices, especially in electrocardiogram (ECG).

The sensors need to be communicated through a personal-area network (PAN) or body-area network (BAN). This network is called Body Sensor Networks (BSNs), which is envisioned as a paradigm shift from the traditional healthcare system to the modern eHealth system (Narwal And Mohapatra 2021). The BSNs or what is commonly known as WBAN are systems of wearable or implantable sensors that wirelessly transmit health-related data in real-time to a central system for analysis, extract meaningful information, hence enabling continuous health monitoring, make efficient decision-making, capture physiological signals, context information, and medical diagnostics. WBANs utilize IEEE standards, such as IEEE 802.15.6 and IEEE 802.15.4j, which are specifically designed for medical WBANs. WBANs enable data to flow from sensors and drive communication to the closest processing node in an IoT architecture (Kim et al. 2017). WBANs can lower healthcare expenses and improve treatment quality by using various sensors to monitor patients' vital signs (Naresh et al. 2020). WBAN can be used to transfer patients' information into a wide communication range. Further explanation about the WBANs can be found in Narwal and Mohapatra (2021). Additionally, a review of low-power communication technologies that can be effectively utilized to support the deployment of WBAN systems is presented in Ghamari et al. (2016).

Several studies have investigated various sensors and wearable devices that can be beneficial for healthcare systems. To this end, the work in Jin et al. (2020) summarized the most advanced progress made in the key phases for future wearable and implantable technology from biosensing, and wearable biosensing to AI-biosensing. The work in Javaid et al. (2021) provided an extensive review of sensor technologies and their transformative impact on our lives. The paper (Tun et al. 2021) presented a hierarchical structure for application areas of wearable to monitor and support elderly people in the healthcare system. The work in Habibzadeh et al. (2019) provided a survey on IoT in healthcare systems, particularly focusing on three core components: sensing & data acquisition, communication and networking, and data analytics. Furthermore, the paper explained the importance of sensing technologies, which aim to improve power efficiency. The work in Castaneda et al. (2018) focused on reviewing the developments and challenges of wearable Photoplethysmography (PPG)-based monitoring technologies and their applications in healthcare. The work in Wang et al. (2023) reviewed the integration of AI with health monitoring sensors to improve healthcare capabilities, addressing some technical challenges such as noise, data processing, and feedback control. Additionally, this paper explored advances in wearable and implantable sensors for monitoring vital signs, soft electronics for therapeutic applications, and the detection of volatile organic compounds. The work in Wang et al. (2023) focused on the development of healthcare intelligent sensors enhanced by AI tools from the aspects of materials, device structure, system integration, and application scenarios. The work in Seneviratne et al. (2017) provided a comprehensive review of commercially wearable devices. The paper also discussed the security issues related to communication faced by popular wearable devices. Additionally, the paper categorized and explained various techniques designed to enhance the power efficiency of wearable devices. The work in Tettey et al. (2024) investigated the future-oriented patterns in biomedical device development and their implications on patient care.

The work in Baker et al. (2017) discussed different wearable and non-intrusive sensors that can be useful for healthcare systems. The paper in Naresh et al. (2020) highlighted the importance of wearable sensors due to their non-intrusive nature and popularity. The work in Baali et al. (2017) presented a review of healthcare-based IoT sensors, explaining their working mechanism in detail. The paper investigated the application-specific sensors, exploring their power systems, on-sensor pre-processing capabilities, and the communication systems they utilize. The paper in Channa et al. (2021) provided SLR that discusses wearables, or on-body sensors, their types and locations, and how AI can be used with eHealth devices.

Companies are trying to incorporate as many sensors as possible to offer ubiquitous monitoring. There are many types of sensors used to detect a patient's vital signs and physiological conditions. Generally, the common types of sensors that are used in healthcare systems for activity monitoring are listed below Mosenia et al. (2017); Muhammad et al. (2021). Figure 8 shows the most common types of sensors, which are placed in the human body to gather physical information and perform the required processing as illustrated in Salayma et al. (2017).

- Electrocardiogram sensor (ECG) sensor a heart rate monitor: Measures the electrical activity of the heart. This can be used to detect abnormalities such as arrhythmias or heart attacks. Additionally, wearable devices such as smartwatches or chest straps can also measure heart rate and detect irregular heartbeats.
- Electromyography (EMG) sensor: Used to track muscle contraction.
- Electroencephalography (EEG) or magnetoencephalogram (MEG) sensor: measures brain electrical activity.
- Electroglottography (EGG) sensor: Measures changes of relative vocal fold contact area during laryngeal voice.
- Electromyogram (EMG) sensor: Records electrical activity produced by skeletal muscles
- Photoplethysmogram (PPG) sensor: Detects blood volume changes in the microvascular bed of tissue.
- Glucometer sensor: Measures and monitors blood glucose concentration in the body, commonly used in diabetes management to provide real-time data for insulin dosing

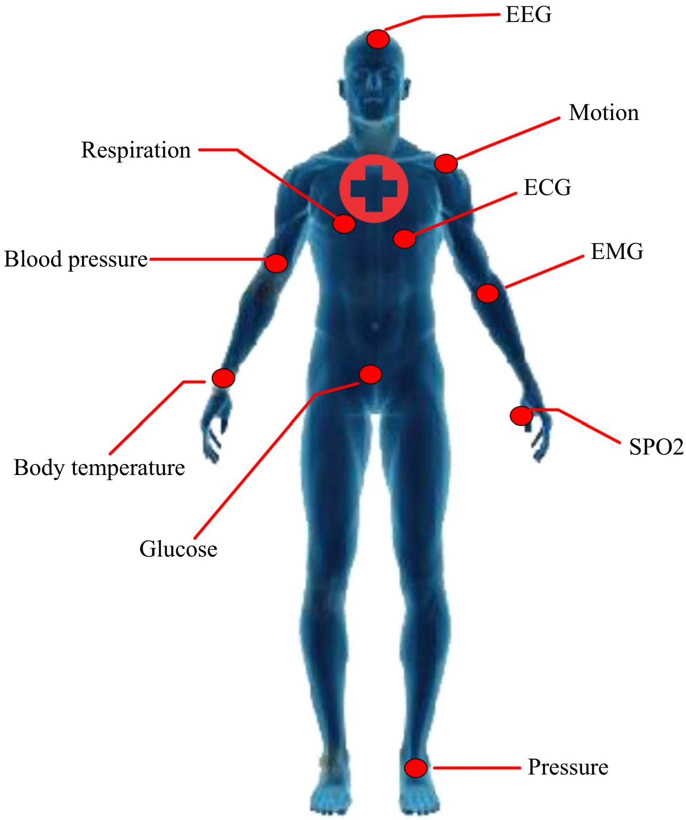


Fig. 8 The most common types of sensors that are utilized in healthcare systems as given in Salayma et al. (2017)

and health monitoring.

- Blood pressure sensor: Monitors systolic and diastolic blood pressure of the body.
- Oximeter or pulse oximeter (SPO2): Measures the fraction of oxygen-saturated hemoglobin relative to the total hemoglobin count in the blood.
- Mechanomyogram (MMG) sensor: Measures skeletal muscles' activity.
- Thermometer sensor: Measures an individual's body temperature.
- Respiration rate (RESP) sensor: Used for detecting and measuring respiratory parameters such as breathing rate, volume, or patterns to monitor respiratory function and count how many times the chest rises in a minute. This type of sensor can be used in sleep monitoring systems.
- Near-infrared spectroscope: Provides neuroimaging technology to examine an aspect of brain function
- Accelerometer motion sensor: Measures changes in the acceleration and uses it to estimate the user's activity.
- Electrooculography (EOG) sensor: Measures the electrical potential between the cornea and the retina of human eyes.
- Pedometer: Counts each step a person takes by detecting the motion of the person's

hands or hips.

- Wearable biometric sensor: Measures biometric data such as electrodermal activity (EDA), skin temperature, galvanic skin, and heart rate variability (HRV). Used in stress management and wellness monitoring applications.
- Environmental sensor: Measures environmental factors such as air quality, humidity, and temperature. Used in healthcare facilities to ensure optimal conditions for patient comfort and safety.
- Fall detection sensor: Detects falls and alerts caregivers or emergency services. Commonly used in wearable emergency alert systems for seniors and individuals at risk of falls.
- Sleep monitor sensors: Tracks sleep duration, quality, and patterns. Often integrated into wearable devices and sleep-tracking apps.
- Body composition analyzer: Measures body composition metrics, including body fat percentage, muscle mass, and hydration levels. Available in smart scales and handheld devices.
- Medication adherence sensor: Monitors medication adherence by tracking pill bottle openings or doses taken, which can help to improve medication compliance and management.
- Activity Tracker: Measures physical activity levels, including steps taken, distance traveled, and calories burned. Found in wearable fitness trackers and smartwatches.

2.1 Benefits of sensors and smart devices and key related point

Sensors are used to transmit vital signs to a remote server for diagnostic purposes. This process facilitates continuous health monitoring and tracking, and personalized care interventions. Sensors play a crucial role in capturing various physiological and environmental data that are essential for monitoring health parameters and activity levels. By adopting low-cost ambient sensors that can be installed in patients' homes, patient behavioral and activity patterns can be captured to detect anomalies. Various types of sensing devices, including vision sensors and embedded sensors can be effectively used for human activity recognition (HAR), which allows machines to analyze and comprehend several human activities from input data sources collected by such devices (Dang et al. 2020). Further explanation of HAR can be found in Dang et al. (2020). Drug-delivering sensors can aid medication adherence by reminding individuals to take their medicine and providing doses at specific times, benefiting older adults, athletes, and patients at risk. One of the most common causes of elderly people's injuries, deaths, and loss of autonomy among elderly people is an accidental fall. Fall detection sensors can help lower the death toll, the number of injuries, and the expense of healthcare systems by offering instant emergency assistance to those who have fallen accidentally. By monitoring vital signs, wearable sensors provide enormous potential for the early detection of diseases. Patients can have a variety of sensors attached to them on the client side. These sensors enable data about the patients to be sent to a central management console and notify medical staff when conditions are identified remotely from the server side.

2.2 Current challenges in sensors and smart devices and future research directions

For future research, it is crucial to develop advanced algorithms that can efficiently handle the challenges of processing heterogeneous, noisy, and inconsistently represented data. This will ensure the effective use of wearable medical sensors in continuous health monitoring, enhancing the reliability and accuracy of healthcare systems.

Data privacy and security are the most critical issues in healthcare systems. Hence, ensuring the safety and non-exposure of data sensed by WBAN sensors to unauthorized entities and security threats is crucial. Besides, robust authentication methods and security solutions are essential, which need to be investigated (Narwal And Mohapatra 2021). Furthermore, secure communication protocols and data encryption are critical.

Wearable devices must be energy-efficient to ensure long-term use without frequent recharging. This includes optimizing hardware and developing low-power algorithms. Wireless power transmission could be a promising solution for energy harvesting, but it needs further investigation. Future research should focus on developing energy efficiency algorithms, resource management techniques, and hardware-software architectures that extend the lifespan of sensors and devices.

The performance of wearable and sensor devices remains suboptimal due to high false alarm rates. Therefore, further investigation into improving the false alarm rate could be possible in the future.

Calibration and noise cancellation for sensors and wearable devices are crucial and warrant further investigation in the future. The sensors collect substantial amounts of health-related data every second, which is then processed and analyzed in real-time to gain insights into the user's health status. Processing the continuous and large amount of data generated in real-time is a challenge. Therefore, there is a critical need to find efficient processing algorithms to extract meaningful information from this data. A significant challenge lies in extracting all helpful information, necessitating the development of algorithms that can identify various features and consolidate redundant ones during runtime. Furthermore, to date, feature extraction and classification processes are crucial for signals generated by sensors and smart wearable devices, which are challenging to process due to their non-stationary and high-dimensional characteristics. Therefore, finding accurate and fast methods for feature extraction and classification of sensor signals is crucial.

Furthermore, although wearable sensors can detect and recognize human physical activities, they can be infeasible in a pandemic crisis. Hence, a smart non-contact sensing approach with medical applications can be considered as a promising solution to address the challenge of monitoring physical activities and assisting patients suffering from serious health issues. To this end, leveraging AI techniques with smart, non-contact IoT sensors could be a feasible solution.

Wearable sensors often face limitations due to the need for frequent battery recharging and their limited energy capacity. Emerging energy harvesting technology, which will be discussed later in this paper, could be a promising solution to offer continuously supplying power, thereby extending the sensors' operational lifetime and reducing reliance on manual charging.

For future research, addressing environmental concerns caused by non-degradable medical and electronic waste is essential. Investigating eco-friendly sensors made from biocompatible and biodegradable materials offers a promising solution. These sensors degrade naturally without harming the environment and can replace conventional sensors in both non-invasive and invasive health monitoring.

3 Remote patient monitoring (RPM) system

In recent years, significant advancements have been made in the development of wearable sensors that offer flexibility, excellent mechanical stability, high sensitivity, and accuracy. These advancements introduce a new approach to remote and real-time health monitoring (Vaghasiya et al. 2023). In addition, RPM becomes crucial due to the increasing need to manage aging populations outside traditional healthcare settings (Shah et al. 2016).

Remote Patient Monitoring (RPM) refers to the use of wearable devices and sensors, as discussed in the previous subsection, to collect and transmit health data from patients to caregivers. There are various terms used in the literature to describe RPM, including RHM, remote healthcare systems (RHS), mobile patient monitoring (MPM), patient monitoring system (PMS), mobile health monitoring (MHM), and wearable health monitoring (WHM). Note that each one of these terms denoted a similar concept that serves similar purposes (Sujith et al. 2022). Advancement in RPM allows a smart healthcare system to be achieved. Specifically, real-time monitoring of a person's health status through RPM enables the early identification of possible health problems, timely medical interventions, and personalized healthcare. RPM is beneficial for managing chronic diseases and reducing hospital visits. The primary objective of RPM is not only to reduce costs but also to provide timely healthcare services to elderly individuals who wish to maintain their health independently. By doing so, elderly people can avoid, for as long as possible, any interaction with healthcare institutions (e.g., nursing homes and hospitals), which in turn reduces pressure on the healthcare system.

Several studies have investigated the advancement of RPM in healthcare systems. For example, the work in Pradhan et al. (2021) investigated the architecture and applications of IoT for efficient RPM in healthcare systems, thereby shifting the focus from hospital-centric (or clinic-centric) to patient-centric care. The work in Mamdiwar et al. (2021) presented several studies on IoT-assisted wearable sensors for RPM systems. The work in Shaik et al. (2023) provided a review of using AI tools in RPM. The works in Baker et al. (2017); Wu et al. (2018); Khan et al. (2023) presented detailed research on RPM systems based on IoT. An efficient IoT biomedical device based on the Raspberry Pi 3 has been proposed in Garbhapu and Gopalan (2017) for RPM, which can quickly monitor the vital signs of many people and send the collected information wirelessly to doctors. The work in Durán-Vega et al. (2019) created a real-time RPM system for the elderly. The application of dual-arm motion mapping for telerobotics in RPM was discussed in Zhou et al. (2019). The work in Yang et al. (2016) investigated the use of wearable ECG for RPM in smart healthcare systems. The work in Rani et al. (2023) proposed an IoT-based framework for RPM services, addressing challenges related to data security, response time, and system availability. The work in Taiwo and Ezugwu (2020) proposed a smart home healthcare system leveraging IoT technology for RPM, hence reducing the need for hospital visits.

Several studies have investigated the use of cloud computing for efficient RPM in healthcare systems. For example, the work in Shah et al. (2016) introduced RPM as an effective solution for healthcare and suggested the exploitation of semantic web, data analytics, and cloud computing to address the QoS challenges and limitations of big data processing. The work in Abawajy and Hassan (2017) presented a pervasive patient RPM system infrastructure based on integrated cloud computing and IoT. The work in Hu et al. (2017) presented a secure RPM system using IoT sensor based on cloud computing. The work in Yang et al. (2017) presented a wearable device with a bio-sensing facial mask to monitor pain intensity through facial surface electromyogram (sEMG), where the data is transmitted to a cloud server in real time. The authors found that a mobile web application can provide real-time sEMG data to caregivers, thereby enabling digital signal processing, interpretation, and visualization. The work in Shahzad et al. (2018) proposed the use of a cloud-based RPM system within designated boundaries for cardiology patients. The authors in Verma et al. (2018) proposed a cloud-based IoT framework for intelligent student healthcare monitoring. In addition, the authors suggested utilizing mobile computing technology in IoT-based healthcare systems to transform the reactive care system into proactive and preventive healthcare systems. The work in Verma and Sood (2018) used fog computing to speed up health event decisions in a real-time RPM system. The work in Mohapatra et al. (2019) presented a prototype model for RPM system utilizing cloud computing. The proposed system also included multiple bio-sensors attached to patients, which capture and record data before transmitting it to a cloud server. The work in Kaur et al. (2019) proposed to use of different ML techniques, which are stored in the cloud to build an efficient healthcare system. The work in Zhang et al. (2020) used mobile edge computing and 5 G communication technology for RPM systems. The work in Sahu et al. (2022) presented RPM approach that uses cloud computing for accurate and continuous RPM and tracking. To this end, the authors proposed a model called"k-Healthcare,"which incorporates four layers: sensor, network, Internet, and services, enabling real-time data collection and storage in the cloud. The work in Ali et al. (2023) discussed the use of private server software for the RPM system, in which the proposed approach uploads vital signs data to a cloud server to achieve enhanced accuracy of RPM.

Furthermore, several studies have investigated the exploitation of advanced techniques to achieve smart and secure RPM systems. For example, the work in Kaur et al. (2019) proposed an RPM system to predict diseases. The work in Ali et al. (2020) proposed DL and feature fusion for RPM based on an innovative healthcare system to predict heart disease. The work (Karatas et al. 2022) demonstrated how AI-enabled medical equipment and blockchain can facilitate RPM, improve data management, and enable early disease detection, hence optimizing patient care. The work in Pathak et al. (2023) discussed the role of digital technology in the surgical home hospital system. The work in Balasubramanian et al. (2023) proposed a convolutional neural network (CNN) framework to provide an RPM system for accurate human action recognition. The work in Alabdulatif et al. (2018) presented a secure surveillance system for the healthcare system. The work in Ahmed and Kannan (2022) proposes a secure and private RPM system that enables secure authentication and communication methods, utilizing a smartwatch, server, and smartphone app to monitor vital signs.

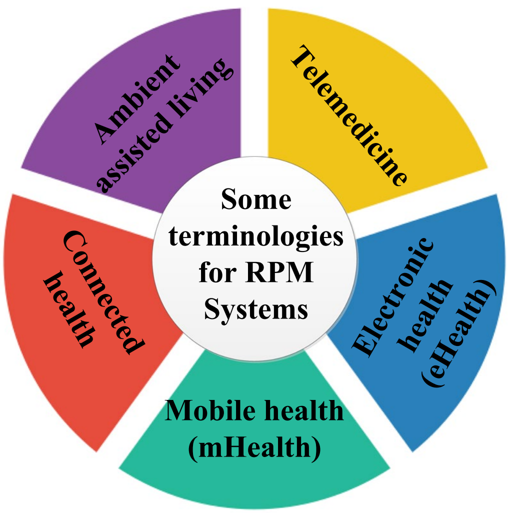
The work in Wang et al. (2020) developed a detection fall system using a millimeter-wave radar. The work in Parvathy et al. (2021) proposed the use of cognitive IoMT for RPM. The work in Mshali et al. (2018) reviews several studies investigating smart RPM, focusing on systems for the elderly and dependent individuals, and identifies the technological, design, and requirements. To this end, a case study on the real-time RPM of a patient with heart failure using ECG was presented. The work in Al-Khafajiy et al. (2019) focused on developing a smart healthcare RPM system for remotely observing elderly people. The proposed approach tracked the physiological data of elderly people to detect specific disorders, supporting early intervention. The work in Talal et al. (2019) provided a review of real-time RPM systems for triage and priority systems based on multi-layer architecture. The implementation of multi-layer architecture in the RPM system was also explained in Farahani et al. (2018). To this end, the paper discusses the application areas of this architecture for mHealth, AAL, e-health, implants, early warning systems, and population monitoring in smart cities. The work in Negra et al. (2016) demonstrated that the RPM system can be implemented in real-time with multi-tier pervasive systems based on the use of WBAN.

The following are some related terminologies that aim to achieve efficient RPM for smart healthcare system. Figure 9 shows a list of some associated terminologies for RPM in smart healthcare systems.

3.1 Telemedicine

Telemedicine, also known as Telehealth, is a spatial form of RPM. The World Health Organization (WHO) defined telemedicine as the delivery of healthcare services using telecommunications (information and communication) technologies (Lai And Lai 2022). Telemedicine utilizes various communication tools to facilitate interactions between patients and healthcare providers. For example, telemedicine facilitates virtual appointments (through video conferencing), allows real-time data sharing, and promotes direct communication with healthcare providers. Therefore, it will enable healthcare professionals to evaluate, receive medical advice, diagnose, and treat patients remotely, often overcoming geographical barriers and improving access to medical care (Talal et al. 2019). Telemedicine makes patient care safer and more accessible by eliminating the need for travel, which not only increases convenience but also significantly lowers the risk of infection. Telemedicine is envisioned to provide urgent care in emergencies and can save lives in such critical circumstances. Additionally, telemedicine plays a crucial role in monitoring patients at home, facilitating faster, more efficient, and cost-effective patient mobilization and rehabilitation.

Fig. 9 A list of some related terminologies for RPM in smart healthcare system



3.2 Electronic health (eHealth)

EHealth refers to the use of ICT to improve healthcare services. EHealth encompasses a diverse range of digital health solutions, including Electronic Health Records (EHRs), telemedicine, health informatics, and online health services. eHealth uses digital technology to enhance the quality, accessibility, and efficiency of health services. Health data are transmitted, stored, retrieved, shared, and applied for clinical and other purposes. IoT, big data, and AI technologies are used to enable powerful and effective eHealth systems. The eHealth can increase the possibilities of patient surveillance and it is expected to become an essential part of the future self-management programs (Lloret et al. 2017). EHealth will process information about patients at various levels, including their DNA, blood cells, and organ activity, using a combination of wearable and microfluidic biosensors. This system will allow more real-time experiments as labs and researchers share data and collaborate. This shared information will help guide follow-up practices and procedures in biochemistry. EHealth has the potential to enhance and refine the precision of medical decision-making. EHealth can combine physical devices with big data infrastructure, thereby allowing for the streamlining of connections between patient data and treatment. This combination would allow thousands of devices to work together efficiently.

3.3 Mobile health (mHealth)

mHealth is a subset of eHealth, focusing on mobile technologies that utilize mobile devices, personal digital assistants (PDAs), and other wireless and multimedia devices for the delivery of healthcare services and health information. mHealth encompasses a wide range of applications, including health monitoring, disease management, and remote diagnostics. mHealth seeks to enhance access to healthcare, particularly in rural areas. This can be achieved by providing real-time health information, enabling effective care, and improving communication between patients and healthcare professionals. With mHealth, patients can easily call or receive messages about their health treatments and communicate with medical staff. Medical staff can also utilize mHealth to access the latest clinical guidelines, receive diagnostic support, and communicate with patients. Medical devices and body sensor networks are key enablers of mHealth. Using these devices in a patient's environment can help in predicting health issues. In addition, advances in micro- and nanotechnologies, information processing, and wireless communication enable smart mHealth systems. A unique type of mHealth service, called MPM, is used to measure, gather, and transmit bio-signal data from patients, collected via wearable sensors, to hospitals or other healthcare facilities. A mHealth enhances RPM healthcare through advancements in mobile technology, communication networks, sensors, and embedded devices. This mobile technology enables direct access to patient information and the effective analysis of this information through connected sensors (Jagadeeswari et al. 2018). Furthermore, mHealth offers quick information delivery at reduced costs and provides unlimited internet-based services (Jagadeeswari et al. 2018).

3.4 Connected health

The term "connected health" refers to remote-operating digital healthcare technologies with capabilities that include emergency notifications and health monitoring. Through the integration of many health technologies, connected health emphasizes data exchange and real-time monitoring to deliver complete and ongoing remote care (Mishra And Singh 2023). By enabling remote care, connected health seeks to improve the efficiency of the healthcare system. Connected health provides individuals with information about their health and offers support when needed. Additionally, connected health emphasizes interoperability and data sharing between devices, incorporating clinician feedback.

3.5 Ambient assisted living (AAL)

AAL describes the services and tools that help older people and people with disabilities live independently and maintain a good standard of living (Cicirelli et al. 2021). This is achieved by utilizing advanced technologies to create supportive environments. AAL encompasses a range of systems, including wearable technology, smart home technologies, and telehealth services, that support everyday tasks such as medication management, fall detection alarms, health monitoring, and emergency response. The primary objective of AAL is to enhance the quality of life, ensure safety, and provide timely support when needed (Islam et al. 2015). With AAL, IoT enables indoor positioning as well as location-aware real-time monitoring of living parameters (e.g., heart rate) and environmental conditions (Farahani et al. 2018). AAL systems monitor health, send reminders, and assist with everyday tasks by integrating sensors, smart devices, and communication networks into the home or living space. Furthermore, one of the most critical tasks of AAL systems is HAR, which is responsible for recognizing human activity patterns from various types of low-level sensor data. HAR can identify abnormal circumstances and mitigate the impact of unpredictable events, such as sudden falls, by continuously monitoring the actions of older individuals. To keep senior citizens from straying, HAR may also locate them and offer navigational support. For the elderly, object locators might make it easier to discover their personal and home belongings (Tun et al. 2021). An innovative healthcare system for AAL is presented in Syed et al. (2019), which monitors the physical activities of elderly individuals using IoT and ML algorithms, thereby enabling faster analysis, enhanced decision-making, and improved treatment. Further discussion about AAL can be found in Cicirelli et al. (2021).

3.6 Benefits of RPM and key related points

RPM system can be very useful for all patients suffering from serious illnesses such as diabetes, cardiovascular disease, mental illness, cancer, and hypertension (Vaghasiya et al. 2023). RPM in smart healthcare applications presents various facilities and benefits by adopting IoT and communication technologies like Radio Frequency Identification (RFID) (Catarinucci et al. 2015). The RPM system provides efficient healthcare from a remote distance. In addition, RPM can play a crucial role in addressing the limitations of traditional healthcare systems during epidemics and disease crises, such as the COVID-19 pandemic, and for managing chronic health conditions such as diabetes mellitus and heart disease, as well as for elderly people and those with disabilities, who wish to maintain their health independently. RPM not only reduces mortality rates and hospital admissions but also effectively treats chronic diseases with the aid of smart body sensors, built upon modern connectivity standards (Khan et al. 2023). RPM can facilitate the efficient transmission of patient data to healthcare providers, enabling timely interventions and necessary actions based on real-time information. RPM has the ability to continually gather information on health conditions, including physiological parameters, metabolic status, and body movements. These real-time data recordings enable close monitoring of health conditions and prompt action recommendations as needed (Vaghasiya et al. 2023). In RPM, sensor fusion can also incorporate data from wearables and mobile apps to facilitate early treatment initiation and improved chronic condition management. By providing a more comprehensive picture of patient information and medical resources, data fusion can significantly enhance the RPM system (Ahmed et al. 2024). In addition, RPM reduces healthcare costs for patients by avoiding hospital admissions, thereby providing healthcare services to patients in remote areas (Tun et al. 2021). By enabling quicker detection and treatment of health issues, RPM helps in minimizing the need for hospital visits. RPM transits the routine medical checks and other services from hospitals to home environments, hence shifting the focus from hospital-centric to patient-centric care. This transition can significantly alleviate the burden on hospitals by transferring simpler tasks to home settings.

3.7 Current challenges in RPM and future research directions

There are various challenges associated with the development of RPM systems, including data aggregation, filtering mechanisms, and the need for intelligent processing to optimize monitoring detection accuracy (Mshali et al. 2018). Therefore, it is crucial to develop advanced tools to enable efficient RPM systems (Mohapatra et al. 2019). Furthermore, it is essential to address security and privacy challenges in RPM. Further investigations in this regard are required.

In addition, one of the significant challenges in RPM healthcare systems is the need for an efficient and reliable bridging point (i.e., gateway) between sensor networks and the Internet. Traditional gateways often perform only basic functions, such as protocol translation, which limits their potential in managing complex healthcare data and ensuring seamless integration. Therefore, to deliver higher-level services such as local storage, embedded data mining, and real-time data processing, more suitable health gateways are needed that capitalize on the strategic location of gateways at the network edge. This might be achieved by integrating fog computing into IoT systems, which can address issues related to energy efficiency, reliability, mobility, and scalability in ubiquitous RPM healthcare systems. Further investigations in this regard can also be considered in the future.

Power consumption is also considered a major issue in sensor networks used for RPM. In particular, RPM applications often require continuous and real-time data transmission (Rahmani et al. 2018). This process uses a significant amount of energy. Rapid sensor battery drain causes system disruption. In such system setups, battery changes or recharges are inefficient and time-consuming, particularly when there are numerous sensors. We also need to investigate novel wireless energy collection and charging methods. Developing power-efficient algorithms for sensor networks may also be beneficial in overcoming this energy issue. Further investigation is needed to develop more effective energy harvesting techniques and gain a better understanding of energy utilization in RPM.

The chosen sensor networking techniques must be resource-efficient and customized for RPM applications. Additionally, it is crucial to develop small, long-lasting circuits to monitor human health data. Future research should focus on developing new 2D material-based sensors and understanding how these sensors can track various health markers (Vaghasiya et al. 2023).

To utilize wearable sensors more widely in healthcare, we need to understand the composition of body fluids and their relationship to health and disease. Therefore, there is a need to develop systems that facilitate patient mobility while also improving their safety and increasing their autonomy.

Huge amounts of data are generated by many connected IoT devices in the RPM system, which need processing. The data are generated from various sources, leading to data heterogeneity. Further research is needed to investigate how AI and ML can effectively manage this data variability. In particular, one of the key challenges is efficiently managing such a vast amount of data, particularly under time constraints, limited computational capacity, processing power, and resource availability. Addressing this issue would require the development of standardized sensor data models alongside the integration of advanced AI algorithms. Therefore, further research is needed to further explore how AI and ML-based algorithms can be effectively used to address data variability and enhance the overall robustness of smart healthcare systems.

A common challenge in RPM development is transferring sensor data to servers for analysis, which might cause severe delays. Data transmissions are categorized into multicast, broadcast, unicast, and anycast. Existing RPM systems often use multicast or broadcast to send packets to multiple recipients, which can result in increased network traffic and delays. Unicast sends packets to a single recipient with minimal traffic, but requires a backup plan for failures. Anycast, a newer method, sends data to the nearest recipient, reducing traffic but increasing device and network complexity. Addressing these challenges involves analyzing transmission methods, data volume, and packet size.

4 Electronic health records (EHRs)

The EHR technique is a popular method for retaining patient encounter data, including demographics, diagnoses, medical history, medications, lab results, radiological images, prescriptions, and clinical notes (Shickel et al. 2017). Traditionally, patient data was collected manually and entered into spreadsheets. These spreadsheets were frequently deleted when the patient was released from the hospital, offering no long-term advantages. Implementing EHR systems is essential to prevent the loss of patient information, as they enable the storage and retrieval of valuable patient data. This would provide several benefits, including improved patient care, enhanced research opportunities, and data-driven decision-making. Digital medical imaging technologies, such as computed tomography (CT) scans, magnetic resonance imaging (MRI), X-rays, and ultrasounds, can all be stored in an electronic health record (EHR) system. These records can be analyzed to aid in diagnosis, treatment planning, and disease monitoring. Since hospitals originally introduced EHR for internal administrative purposes, a variety of classification systems and controlled vocabularies have been developed to record relevant medical events and information. For example, diagnosis codes were used, such as the International Statistical Classification of Diseases and Related Health Problems (ICD), procedure codes, including the Current Procedural Terminology (CPT), laboratory observations identified by Logical Observation Identifiers Names and Codes (LOINC), and medication codes, like RxNorm (Shickel et al. 2017).

EHR systems offer various potential benefits, including improved public healthcare management, online patient access, and the sharing of medical data. EHRs have attracted significant research interest, with numerous studies investigating their various aspects and implications. To this end, the authors in Bahga and Madisetti (2013) developed an EHR system called Cloud Health Information Systems Technology Architecture (CHISTAR), which ensures semantic interoperability by using a reference model for data structures and an archetype model for clinical data attributes. An effective and robust architecture for heart failure prediction using EHR was investigated in Jin et al. (2018). The work in Dagher et al. (2018) proposed a blockchain-based framework for secure, interoperable, and efficient EHR. The proposed approach enabled efficient access to medical records for patients, providers, and third parties while maintaining the privacy of patients' sensitive information. The work in Madine et al. (2020) proposes an Ethereum blockchain-based smart contract for the Personal Health Records (PHRs) system, which ensures reliable, immutable, decentralized, transparent, traceable, trustworthy, and secure control over medical data. Besides, the proposed approach utilized decentralized storage through Interplanetary File Systems (IPFS) and employed trusted reputation-based re-encryption oracles to securely fetch, store, and share medical data. Noting that, unlike EHRs, which are controlled by doctors, a PHR is controlled by patients (Madine et al. 2020). In PHRs, patients can own and manage their data collected from healthcare providers or medical institutions. To address the privacy-sensitive data concern in EHR systems, the work in Parah et al. (2020) proposed a secure and efficient EHR hiding technique in medical images within an IoT-driven healthcare system. The work in Shahnaz et al. (2019) developed a model considering the blockchain technology in healthcare systems to manage EHR. The proposed framework aimed to securely store electronic records and set detailed access rules for users. Additionally, the work addressed blockchain scalability issues by utilizing off-chain storage, which provides a scalable, secure, and integral solution for EHR systems. The work in Girardi et al. (2020) discusses the management and protection of EHRs and data exchanged through medical or diagnostic devices. To address EHR confidentiality concerns, a secure, fine-grained EHR access control scheme was proposed in Liu et al. (2018). To this end, the authors proposed a decisional parallel bilinear Diffie-Hellman exponent assumption for security, which enables EHR owners to conduct offline ciphertext operations without being aware of specific EHR access and data policies, thereby speeding up the encryption process. The work in Singh et al. (2025) introduced a robust framework to address critical security requirements in digital healthcare systems, ensuring the confidentiality and integrity of medical data.

4.1 Benefits of EHR and key related points

The implementation of EHR in hospitals and clinics can significantly enhance patient care by reducing errors and improving care coordination (Knake et al. 2016). Among the main advantages of EHR systems is the availability of large volumes of data, which can be used to facilitate data analysis to inform other medical research efforts, such as disease prediction. Wearable and other IoT devices can collect and upload relevant information to the EHR systems, which can facilitate healthcare monitoring and personalized health services. With the availability of comprehensive EHRs on cloud platforms, healthcare providers can better track patient health and provide appropriate medical care during the diagnosis and treatment processes. EHR provides remote access to patient data, hence improving efficiency and coordination of care. In addition, EHR provides a valuable data resource for researchers, enabling the examination of health data across demographics and helping to detect health patterns and implement public health initiatives. EHRs can be very useful for predicting heart disease. For example, mass unstructured data can be expected from EHR time series. By analyzing and mining these time-based EHRs, the links between diagnostic events can be identified.

EHR also allows coordination of care among different healthcare providers by sharing patient information easily and quickly, which can enhance disease diagnosis and treatment. Cloud-based EHR systems can also be beneficial, as they enable the exchange of medical documents between medical institutions, which is expected to contribute to improvements in various medical services. Besides, strict control over who has access to patient data is made possible by the EHR system, hence improving the security and privacy of the healthcare system. The EHR is crucial for remote patient diagnosis. The multispeciality EHR solution brings higher efficiency in managing patient health records and serves various clinical requirements (Dang et al. 2019).

4.2 Current challenges in EHR and future research directions

The most frequent challenge encountered while implementing an EHR is its heterogeneous representation of the data. Specifically, EHR data is highly heterogeneous, containing both structured and unstructured components. In particular, it is challenging to draw practical insights from such heterogeneous EHR data because it contains structured and unstructured information. Structured data includes predefined fields like demographics, diagnoses, lab results, medications, and sensor readings. On the other hand, unstructured data encompasses everything else, such as imaging and text. Extracting valuable features from unstructured EHR data is a challenging task and, hence, necessitates the use of supervised and unsupervised AI techniques. For example, one of the most effective solutions to address this challenge is to make use of the DL algorithm, see e.g., Shen et al. (2017). Further investigation is needed to find efficient deep learning methods that address the heterogeneous representation of EHR data.

In an EHR system, the patient data stored in electronic health records constitutes a significant portion of big data. This poses a considerable challenge due to the enormous amount and diversity of this data. Therefore, future research should investigate advanced data processing and storage techniques to manage and analyze such large data. Besides, an effective data management strategy for EHRs on the cloud is needed. In addition, improving data quality through a preprocessing methodology is necessary to leverage the potential advantages of the EHR system fully.

EHR often uses different specifications, making data sharing a challenging issue even between systems on the same platform. Hence, standardized coded data is needed for interoperability. A blockchain-based healthcare data management technique can be used to ensure all EHR data follows a standardized format, hence allowing easy access and use by any healthcare facility (Dagher et al. 2018).

Privacy and security issues are another concern that, to date, the EHR implementation (Hussien et al. 2019) faces. Handling EHR in a secure manner has become very challenging because the data is spread across various medical facilities (Shahnaz et al. 2019). Therefore, finding efficient techniques is necessary to address the privacy and security issues associated with patient databases stored in EHRs, where patient data can be accessed and retrieved efficiently and securely. Furthermore, storing EHRs on the cloud presents security challenges. Therefore, further investigation is needed to address security in cloud environments.

In EHR systems, missing data is common, which can have detrimental effects on performance, particularly if the data are missing not at random or if the missing values occur with a large percentage. Therefore, finding efficient techniques to address the missing data in EHR is required.

Imaging and sensor-based data are typically collected using various acquisition parameters and equipment, resulting in measurement variability across different EHRs over time. Additionally, data collection often employs diverse coding systems, which may be subject to periodic updates and changes. These factors can complicate external validation and contribute to data drift, potentially leading to a degradation in information extraction performance. Further investigation is necessary to address this inconsistency issue in the EHR system.

5 Information and communications technologies (ICT)

The human–machine interface has undergone significant progress in recent years. This is due to advancements in sensing technologies and the development of electronics equipped with wireless data transfer capabilities. Specifically, wireless data transfer involves the transmission of digital information between devices without the use of physical cables. Wireless data transfer utilizes ICT, such as radio waves, infrared, and microwave signals, to transmit data over varying distances. This enables easy and flexible connectivity between devices, allowing the exchange of information among computers, smartphones, tablets, IoT devices, and other electronic gadgets without the need for direct physical connections. Wireless networks connect patient devices and sensors to remote locations, managing data transmission. The patient's signals are transmitted using ICT communication protocols.

In health systems, data generated by IoT devices, such as wearables and sensors, is sent to the nearest node using communication techniques that enable the networking of these devices. The flowing data from IoT devices to the nearest processing node can be accomplished through WBAN (Kim et al. 2017). ICT can be classified as either short-range or long-range based on their coverage distance (Alam et al. 2018; Huang et al. 2023). Long-range distance typically involves communication techniques like cellular networks (3 G, 4 G, 5 G, and 6 G), satellite Communication, and Long Range Wide Area Network (LoRaWAN), while short-range distance includes Bluetooth, RFID, Wi-Fi, Ultra-wideband (UWB), and ZigBee, both communication techniques support data exchange. If a personalized monitoring device, such as a wristband, is used, Bluetooth, Internet Protocol version 6 (IPv6) over low-power wireless personal area networks (6LoWPAN), or radio-frequency identification (RFID) can be utilized to connect the device to the Internet. However, in hospitals where a healthcare network is managed, Wi-Fi can be used to maintain continuous Internet connectivity and support high data traffic. These communication technologies differ in terms of radio frequency and security standards, resulting in variations in transmission rates, working distances, the number of allowable nodes, power consumption levels, installation, and maintenance costs.

WBAN mainly collects a large amount of real-time data and sends it to the control unit. WBANs utilize IEEE standards, such as IEEE 802.15.6 and IEEE 802.15.4j, which are specifically designed for medical purposes. These standards ensure reliable communication, fast transmission, and energy efficiency between sensors and actuators, thereby facilitating seamless operation. Efficient WBANs could enhance healthcare quality and reduce costs by networking various sensors to monitor patients' vital signs (Naresh et al. 2020). For efficient and reliable ICT systems, WBANs can be integrated with advanced technologies such as AI, blockchain, cloud computing, edge/fog computing, and big data analytics (Krishnamoorthy et al. 2023).

Over the past decade, the field of wireless sensor networks (WSNs) has witnessed notable advancements, particularly due to their integral role in enabling various IoT applications. A critical challenge within this domain is accurately localizing nodes, which plays a fundamental role in enhancing the reliability and effectiveness of monitoring systems. Localization involves determining the exact geographical coordinates of sensor nodes, a task essential for interpreting contextual data and optimizing the network. In response to this, recent research in Mihoubi et al. (2020) focused on applying swarm intelligence methods to treat localization as an optimization problem. To this end, the authors presented the Enhanced Fruit Fly Optimization Algorithm (EFFOA), specifically designed to improve localization accuracy in Wireless Sensor Networks (WSNs). By leveraging iterative optimization strategies, EFFOA demonstrates a strong ability to estimate the positions of unknown nodes with high precision, effectively navigating the search space and converging toward optimal solutions. This advancement significantly contributes to the ICT framework within IoT systems, where dependable node positioning is crucial for enabling real-time decision-making and optimal system performance. In addition, the work in Miloud et al. (2019) developed an approach for node localization using the Moth Flame Optimization Algorithm (MFOA). The algorithm estimates node positions by utilizing Euclidean distance as the fitness criterion within the optimization framework. When applied to large-scale WSNs consisting of hundreds of nodes, the proposed method demonstrates high effectiveness in accurately determining node locations. Simulation results indicate that MFOA achieves fast convergence toward optimal positions. Besides, its performance significantly surpasses that of other popular swarm intelligence algorithms such as the Bat Algorithm (BAT), Particle Swarm Optimization (PSO), Differential Evolution (DE), and the Flower Pollination Algorithm (FPA), highlighting its robustness and efficiency in solving the localization problem. The work in Mihoubi et al. (2018) introduced an enhanced BAT algorithm, named Dopeffbat, for efficient node localization in WSNs based IoT. By incorporating the Doppler effect to adjust the bats' velocity, the algorithm enhanced accuracy and convergence speed. Using Euclidean distance as the fitness function, Dopeffbat effectively determined node positions across large-scale deployments. Simulation results confirmed the superior performance of the proposed method compared to the original BAT algorithm and PSO, demonstrating higher precision and faster convergence. The works in Mihoubi et al. (2020); Supreeth and Akhil (2025) presented a hybrid localization method named Enhanced Bat Algorithm with Doppler Effect (EBADE) in large WSN with hundreds of IoT sensors to achieve precise sensor node localization. By modifying the frequency equation to improve parameter tuning, the algorithm enhanced the Bat algorithm's exploration ability. EBADE efficiently estimated node positions by iteratively optimizing Euclidean distances, offering improved accuracy in multidimensional localization problems. The work in Mihoubi et al. (2021) introduced a multi-criteria localization approach using the Harris Hawks Optimization Algorithm (HHOA) to identify the position of the triggering sensor node while reducing energy consumption. By enabling cooperative information sharing among nodes, the algorithm enhanced localization accuracy and extends network lifetime in WSN-based IoT. Simulations on large-scale WSNs demonstrated the high efficiency and performance of the proposed method.

Furthermore, several studies have investigated the application of ICT techniques in the healthcare system. For example, the work in Alam et al. (2018) discussed the existing and emerging ICT and standards, with a special focus on short-range and long-range communications. The short-range and long-range communications technologies have been compared in terms of suitability for healthcare applications in Baker et al. (2017). The work in Ghamari et al. (2016) investigated the existing low-power ICT that can support the rapid development and deployment of WBAN systems in the healthcare system. The authors were particularly focused on the RPM of elderly or chronically ill patients in residential environments.

In the following, we discuss briefly the ICT protocols that can be used for smart healthcare systems. Figure 10 shows a list of some related ICT protocols that can be used for smart healthcare systems.

- Classic Bluetooth technology, based on the IEEE 802.15.1 standard, is a wireless personal area network (WPAN) used to exchange data over short distances. Bluetooth operates in a 2.4GHz unlicensed ISM band using frequency-hopping spread spectrum (FHSS) (Chen et al. 2023) and provides a maximum data rate of 3 Mbps. Bluetooth has a transmission range of up to 100 m and is commonly used to connect portable personal devices, such as medical devices, sensors, smartphones, and tablets. Bluetooth is widely used in telemedicine to enable devices to send vital signs directly to monitoring systems, allowing for real-time data collection and analysis.
- Bluetooth Low Energy (BLE) is an enhanced version of classic Bluetooth, designed to

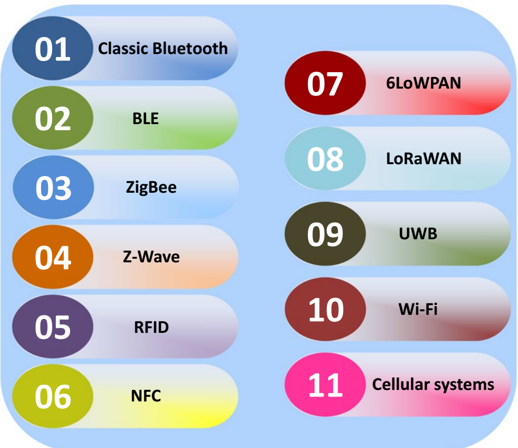


Fig. 10 A list of some related ICT protocols that can be used for smart healthcare systems

be suitable for WBAN devices due to its low energy consumption and minimal duty cycling. BLE was initially introduced to connect small devices with mobile devices. Currently, BLE is considered ideal for wearable and healthcare devices due to its low power consumption. BLE has a faster synchronization time than classic Bluetooth and uses FHSS with fewer channels for pairing. Similar to classic Bluetooth. BLE operates in the 2.4GHz unlicensed ISM frequency band. BLE offers a 1Mbps data rate and works with a range of less than 50 m, hence it is effective for high data rates and short delay scenarios with low power consumption, such as peer-to-peer communication between wearable sensors and access points, or between WBAN devices.

- ZigBee is a wireless personal area network technology based on IEEE 802.15.4 designed for low data rate networks, typically supporting up to 250kbps (Negra et al. 2016). ZigBee offers long battery life, potentially lasting several years, although its power consumption is higher than Bluetooth. The transmission range of ZigBee can reach up to 100 m, depending on power and environmental factors (Wu et al. 2018), and can extend up to 1,600 m with ZigBee-Pro. A ZigBee network comprises three types of devices: coordinators, routers, and end devices. ZigBee utilizes 128-bit symmetric encryption for security and operates on one channel in the 868 MHz band in Europe, ten channels in the 915 MHz band in Australia and the US, and sixteen channels in the 2.4 GHz ISM band. Additionally, ZigBee supports a mesh network architecture, which enhances coverage and communication reliability.
- Z-Wave is a low-power wireless technology designed for short-range communication. Z-Wave allows a data rate of up to 40 kbps and a coverage distance of approximately 30 m, supporting mesh networking, which is advantageous for RPM systems. In regions such as Australia and North America, Z-Wave operates in the 900 MHz ISM unlicensed frequency range, offering improved coverage and lower power consumption compared to the 2.4 GHz band. Z-Wave is resistant to interference and ensures reliable communication alongside other wireless devices, making it a stable choice for smart home automation and potentially for WBANs in healthcare systems (Khan And Pathan 2018). The ability of Z-Wave to penetrate walls and solid objects, as well as low power consumption, enhances its suitability for deploying sensors and monitoring devices in healthcare.
- Radio-frequency identification (RFID) technology consists of a battery-assisted passive (BAP) tag and a reader, which can be connected to personal or handheld computers. RFID uses electromagnetic waves to identify and track tags attached to items or people automatically. Each tag includes an antenna for communicating with the reader and a chip that stores a unique identifier. The reader creates a radio frequency field, allowing the tags to be detected through their reflected radio waves. Passive RFID tags, which have no internal power source, can last for decades and can be integrated with other technologies, such as Wi-Fi and ZigBee, making them suitable for healthcare applications. In healthcare, the advent of ultra-high frequency (UHF) tags with improved sensing and computing capabilities promises cost-effective, easy-to-implement, and battery-free monitoring of patients' vital signs (Catarinucci et al. 2015). In particular, RFID is externally useful for real-time patient tracking within medical facilities and monitoring the location and usage of medical equipment. RFID operates across a broad frequency range, from 120 kHz to 10 GHz, with a communication range of 10 cm to 200 m, and can deliver data rates of up to 4 Mbps.
- Near Field Communication (NFC) is well-suited for wearable devices that require mini-

mal power and short-range data transfer, typically within a range of 4 cm to 10 cm. NFC devices operate in two modes: passive, which consumes no energy, and active, which uses power comparable to BLE. NFC can be integrated with wearable items, such as smart suits and jackets, for contactless payments, unlocking doors, and transmitting health information. In addition, NFC is widely used in smart jewelry, wrist-worn wearables, E-Skin, and E-tattoos (Seneviratne et al. 2017). NFC enables seamless and secure communication between nearby devices, making it well-suited for healthcare scenarios where rapid data exchange and authentication are essential. NFC enables the simple pairing of medical devices with smartphones or tablets, providing medical professionals with effective access to real-time patient data.

- IPv6 over Low-Power Wireless Personal Area Networks (6LoWPAN) enables efficient transmission of IPv6 packets using short link-layer frames, as defined by the IEEE 802.15.4 standard. 6LoWPAN was originally designed for low-power wireless networks operating at 2.4 GHz, 6LoWPAN. It is now also implemented over sub-1 GHz low-power RF. There are several advantages to 6LoWPAN, including extensive mesh network support, reliable communication, and extremely low power consumption. It utilizes strong AES-128 link-layer security. These advantages make 6LoWPAN suitable for various IoT applications, including smart metering, residential lighting, streetlight monitoring, and home automation with the use of sensors and actuators. The 6LoWPAN supports data rates ranging from 20 to 250 kbps, with a transmission range of up to 100 m. 6LoWPAN is well-suited for short-distance communication, enabling the integration of multiple sensors and devices into IP networks in healthcare systems, thus facilitating comprehensive sensor connectivity. 6LoWPAN offers several advantages to the healthcare system platform, including enhanced capability, improved mobility support, and management by multiple stakeholders, thereby enabling better control of smart objects (Islam et al. 2015).
- Long Range Wide Area Network (LoRaWAN) is a wireless communication technology highly suitable for healthcare systems. LoRaWAN enables long-range, low-power communication between sensors and gateways, making it ideal for collecting data from remote IoT devices in monitoring systems (Sundaram et al. 2019). LoRaWAN ensures secure and reliable transmission of health data to healthcare providers. This communication technology is also known for its robustness, which facilitates the deployment of sensor nodes across various healthcare environments, including hospitals, nursing homes, and patients' homes, ensuring continuous and real-time monitoring without the need for frequent battery replacements.
- Ultra-Wideband (UWB) is a low-power, high-precision wireless communication technology based on IEEE 802.15.3a standard, which operates across a wide range of frequencies (greater than 500MHz) to transmit data over short distances (approximately 2-30 m). UWB is ideal for real-time applications in RF-sensitive environments, such as hospitals. It is used for accurate positioning, asset tracking, and short-range data transfer. UWB can support data rates ranging from 20 Mbps to 1.3 Gbps, making it ideal for transmitting multimedia data, such as images and videos. This high data rate capability supports various wearables and body sensors used for RPM in healthcare systems (Dang et al. 2019). UWB can also be implemented in two forms: multi-band Orthogonal Frequency Division Multiplexing (OFDM) UWB and Impulse Radio (IR) UWB. IR-UWB, which uses very short pulses to transmit data, is more suitable for

low-resource wearables and WBAN devices due to its lower energy consumption and efficiency in short-range data transmission.

- Wireless local area networks (Wi-Fi) are based on IEEE 802.11 standards, including 802.11b (11 Mbps), 802.11g (54 Mbps), 802.11a (54 Mbps), and 802.11n (300 Mbps), which are extensively used in various devices such as mobiles, smartphones, tablets, PCs, and laptops. This technology enables devices to access the internet and communicate within a local area. Wi-Fi operates in the 2.4 GHz, 3.5 GHz, and 5 GHz unlicensed Industrial, Scientific, and Medical (ISM) frequency bands, supporting short-range communications up to 100 m. Wi-Fi offers reliable, secure, and high-speed data communication. Wi-Fi is extremely suitable for large-scale adoption in wireless wearable and WBAN medical applications, facilitating substantial data transfers at high bit rates. For example, Wi-Fi technology has been successfully used in a network of 45 essential medical devices, such as infusion pumps, defibrillators, lung ventilators, and anesthesia machines, showing its safe and effective application for device communication. However, its energy consumption cannot be overlooked. To address this, a modified version known as low-power Wi-Fi has been developed, featuring reduced transmission power, low-duty cycling, and optimized energy use. Unlike NFC, Bluetooth, Zigbee, and LoRaWAN, Wi-Fi uniquely supports multimedia applications such as live audio and video streaming.
- Cellular wireless systems are a promising technology for connecting smart wearable WBAN devices without requiring additional bridging mechanisms. Cellular networks facilitate voice and data communication over long distances. Cellular networks include 2 G (Global System for Mobile Communications (GSM) and Enhanced Data rates for GSM Evolution (EDGE)), Universal Mobile Telecommunications System (UMTS), and Code Division Multiple Access (CDMA) for 3 G, 4 G (Long-Term Evolution Advanced (LTE-A)) mobile communication, and 5 G. Future 6 G networks are also expected to support a wide range of applications in smart healthcare systems. While conventional networks focus on human communication, they face significant challenges with wearable WBAN devices, especially as applications move from basic vital sign monitoring to advanced virtual or augmented reality experiences, which require data rates in the gigabit per second (Gbps) range. Additionally, due to the need for low energy consumption, cellular wireless technology has not yet been widely adopted in wearable WBAN devices. However, future cellular technologies will be better suited for wearable applications, enabling more advanced healthcare and entertainment experiences. 5 G networks improve 4 G networks in terms of throughput, stability, delay, and high bandwidth and can support diverse use cases by providing ultra-reliable low-latency communications (URLLC), mobile broadband (eMBB), and massive machine-to-machine communications (mMTC). The next generation of communication technology, such as 6 G, is expected to boost performance, coverage, and security for intelligent healthcare systems (Ahad et al. 2024). Furthermore, satellite communication can provide coverage in remote areas or extend the reach of cellular networks.

5.1 Benefits of ICT and key related point

ICT is essential to support IoT in healthcare systems. By utilizing efficient and reliable wireless communication networks, data can be transmitted safely and reliably to multiple locations, such as doctors and caregivers. ICT facilitates remote consultations, ultimately leading to faster service delivery and reduced healthcare costs. Additionally, ICT enables telemedicine, thereby improving access to healthcare services, particularly in remote areas.

Wireless technologies can play a crucial role in exchanging information among various physical elements that are configured to form a healthcare network. ICT enables better coordination between healthcare practitioners, ensuring that all patient data is readily available, thereby reducing error rates and enhancing treatment continuity. ICT helps improve patients' quality of life by enabling continuous health monitoring and individualized medical care. The efficient integration of small devices through wireless technologies can help achieve effective RPM systems.

5.2 Current challenges in ICT and future research directions

Transmission reliability and low latency are essential factors in healthcare applications. Specifically, ensuring high reliability and low latency in wireless data transfer is essential for real-time transmission and immediate accessibility of data to healthcare providers. Further investigation is needed to identify feasible techniques for ensuring reliability and low latency in the healthcare system. Further investigation is also required to enhance network performance and expand cellular coverage (Ahad et al. 2024). There is a need for an ultra-high-speed communication network to seamlessly connect homes, hospitals, healthcare professionals, and medical devices. Further investigation is required to develop advanced techniques that meet the ultra-high-speed communication network requirement.

Mobility poses a significant challenge in IoT-based healthcare systems, especially when providing services to mobile users. IoT devices can move, resulting in frequent changes to the network topology. The objective is to establish a resilient system that leverages ICT, which can effectively adapt to these dynamic changes. Therefore, efficient mobility management mechanisms are crucial for addressing these challenges.

From an ICT point of view, each IoT device must have a unique identifier, like an IP address or Uniform Resource Identifier (URI), which is essential for the success of IoT communication systems. Effective identity management, featuring unique identifiers and efficient key distribution schemes, is an essential issue that needs to be considered.

Furthermore, security and privacy pose significant challenges in the deployment of IoT for healthcare systems. Ensuring the security and privacy of health-related data transmitted from on-body sensors and monitoring devices in WBAN systems to hospitals is crucial. Numerous threats, vulnerabilities, and risks have been identified, prompting the development of security models and taxonomy frameworks specifically tailored for IoT systems. Privacy concerns often stem from inadequate authentication and authorization, lack of transport encryption, insecure web interfaces, and vulnerabilities in software and firmware. To address these issues, it's essential to encrypt the information to protect patient privacy. Additionally, security functionalities must be integrated across all layers of the IoT architecture, accompanied by robust trust management mechanisms. These measures should encompass authentication, access control, data integrity, privacy protection, encryption, and other capabilities, facilitating automated data processing based on user-defined policies and rules. Furthermore, implementing efficient mechanisms for detecting malicious activities is essential for mitigating these security risks.

Encryption plays a crucial role in securing information, particularly for large volumes of real-time data transfers, which necessitate energy-efficient encryption algorithms. Therefore, developing interoperable protocols with limited processing power and energy requirements, as well as encryption algorithms, represents a promising research area for enhancing security in IoT environments while effectively managing energy consumption. Addressing these challenges will be essential for advancing the security and reliability of IoT systems in the future.

Wireless communication consumes a significant amount of energy in the sensor nodes. Thus, a suitable selection of low-power communication technology and optimizing communication stacks for low-power usage is crucial. Optimizing routing techniques to minimize energy consumption is a crucial research area that warrants investigation. Techniques such as reducing data redundancy can significantly lower energy requirements for data routing.

The diverse range of applications, devices, interfaces, and radio technologies in healthcare systems complicates the modeling of IoT systems. Hence, developing a suitable system model could facilitate the design of IoT systems with optimized technologies tailored to specific requirements.

6 Big data in smart healthcare systems

Several remote healthcare monitoring applications require the continuous sensing of various signals and vital signs, resulting in substantial real-time data that must be processed, analyzed, recorded, and transmitted. Big data is defined as data whose complexity, diversity, and scale necessitate the development of new techniques, algorithms, structures, and analytics for management, visualization, and the extraction of hidden information.

Big data in healthcare refers to the large and complex datasets generated from various sources within the healthcare system. These sources include IoT sensors, biomarker data, clinical data, scanners, RFID data, genomic data, and wearable devices (Rumsfeld et al. 2016). Moreover, there are complex high dimensional data in healthcare systems such as medical imaging (photos, X-rays, MRIs, CT, ultrasounds, and slides), wave analysis such as EEG and ECG, audio files with associated transcripts, free text notes with NLP outputs all can be stored in EHR/patient records system (Rumsfeld et al. 2016). Additionally, connected medical devices, such as infusion pumps, cardiac monitors, and ventilators, generate streaming data on patient parameters, device status, and operational metrics, thereby contributing to big data analytics in healthcare systems. This big data should be analyzed to extract valuable information that can assist in diagnosis, treatment planning, medical emergencies, alerting systems, and the development of efficient real-time monitoring systems. However, this big data would generally be complex to process using regular database management tools or traditional data processing applications, as it exceeds the capabilities of conventional storage, processing, and analytical tools.

To this end, big data analytics has emerged as a feasible solution to overcome traditional data processing methods and achieve smart healthcare systems. To be specific, big data analytics is used to process vast amounts of data and enhance decision-making. The big data approach is characterized by six key attributes: value, volume, variety, velocity, veracity, and variability (Wang et al. 2018). The primary feature is volume, which denotes the enormous quantity of data analyzed to achieve the desired result. Variety refers to the different types of data stored and analyzed, including text, images, videos, sounds, and geometric data, as well as combinations of these types. Veracity indicates the accuracy and reliability of the data, ensuring it is suitable for big data analysis. These characteristics collectively enhance the ability to process and derive meaningful insights from complex datasets, ultimately improving decision-making processes in various fields, particularly in healthcare.

Properly managed big data can revolutionize the healthcare system by enhancing medical therapies and personalized medicine. This is why industries, including healthcare, are investing heavily in infrastructure to harness the potential of big data for better services and financial benefits. Effective management and analysis of this data are essential to extract meaningful insights. However, efficiently managing big data requires advanced solutions to achieve its full potential goals toward smart healthcare systems. Big data analytics must be integrated with advanced algorithmic methods and tools to retrieve meaningful information.

Therefore, the concept of using big data analytics in the healthcare industry suggested the use of various data mining, ML, pattern recognition, and neural network approaches to extract the most useful information from complex data. These methods can be used for managing and analyzing big data and are constantly evolving in real-time data streaming, capturing, aggregating, and analyzing. Therefore, these approaches ensure fewer errors while also ensuring quality and reducing costs. In addition, integrating such methods with big data analytics in medical imaging can lead to more accurate disease diagnosis, easier patient monitoring, faster decision-making, and enhanced patient treatment. Most of these insights are missed by traditional methods. Additionally, these approaches can also be utilized for disease investigation and drug discovery. Healthcare professionals use these methods to analyze data for specific abnormalities, effectively filtering structured information from raw data. Furthermore, using such advanced methods aims to improve the integration and utilization of electronic medical records within the healthcare sector. By enhancing data capture, analysis, and visualization, big data technologies can significantly improve healthcare systems.

Several works have investigated the role of big data in the healthcare system. For example, the work in Xu et al. (2014) addressed data heterogeneity and real-time access in IoT healthcare systems. To this end, a semantic data model called the ubiquitous data accessing (UDA) method-based IoT has been proposed for improved data accessibility. The proposed approach demonstrated the effectiveness of big data analytics in emergency medical systems, thereby enhancing decision-making. The work in Sakr and Elgammal (2016) discusses the role of big data analytics systems as an effective technique for improving network communication speed, computational capabilities, and data storage capacities, which can provide various advantages and characteristics that contribute to improving the efficiency and effectiveness of healthcare services. The work in Abdellatif et al. (2018) proposed an efficient data-specific transceiver design that leverages the inherent attributes of big data analytics at the physical layer, thereby reducing the transmitted data size with minimal overhead. The objective of the work was to minimize the data volume that needs to be transmitted, ensuring efficient communication and data storage while maintaining the QoS requirements. The work in Chen et al. (2018) proposes a 5 G-Smart diabetes system and data sharing mechanism to facilitate data sharing in big data networking for efficient patient monitoring and analysis. The work in Maktoubian and Ansari (2019) proposed an integrity monitoring framework and a data analytics module for ensuring visibility into medical devices and enabling the prediction of potential failures before they occur. The work in

Dumka and Sah (2019) proposed a novel approach with a smart ambulance and big data based on an IoT system. The work in Chergui and Verikoukis (2020) proposed a framework for implementing big-data-driven dynamic resource provisioning. This framework encompassed the development of low-complexity traffic prediction models for slices, resource allocation models, and enforcement of service-level agreements utilizing constrained deep learning techniques. The aim of the research was to efficiently manage resources in a manner that meets service-level agreement requirements, leveraging big data analytics to improve performance and reliability. The work in Ahad et al. (2020) explained the challenges posed by the vast and diverse data generated by healthcare applications. The work in Alotaibi et al. (2020) reviewed big data analytical systems and their role in healthcare and supply chain management. The work in Karatas et al. (2022) demonstrated the advantages and opportunities offered by big data analytics, highlighting its pivotal role in revolutionizing information gathering, sharing, and utilization within healthcare operations. The work in Nazir et al. (2019) investigated the role of big data for cardiology disease. The role of cloud computing in big data is explained in Sahoo et al. (2018).

6.1 Benefits of big data and key related points

Healthcare providers manage vast and diverse data, including patient histories, physical records, imaging information, laboratory results, and both digital and analog data. This data is massive and varied in type, format, and structure. Big data analytics can handle these massive amounts of diverse and varied data types and formats. Based on a patient's medical history, big data assists in diagnosing diseases for individual patients and can analyze vast amounts of data to extract important information for precise decision-making. Specifically, big data analytics helps identify relevant patterns in patient data, facilitating early detection and efficient management of chronic diseases.

Big data analytics has the potential to address the growing need for healthcare to manage the surge of clinical data, thereby enhancing the effectiveness and quality of healthcare delivery. It improves nursing capabilities. In addition, big data analytics is crucial for Industry 4.0 to enhance the efficacy and efficiency of healthcare systems (Karatas et al. 2022). Big data analytics is beneficial in enhancing and optimizing healthcare processes, resource management, and service delivery. Big data analytics provides advanced tools and techniques to analyze enormous volumes of health data, which dramatically accelerates medical research and development.

The integration and interoperability of various data sources, such as genomic data, clinical trials, EHRs, and wearable device records, in real-time health monitoring are key advantages for big data analytics. In particular, this integration enables a thorough understanding of patient health and treatment plans, ensuring that data from various systems and formats can be merged and examined simultaneously. Moreover, big data is essential for personalized medicine, as it facilitates the identification of genetic markers associated with diseases through the analysis of genomic data, thereby enabling the creation of individualized therapy regimens tailored to the genetic profiles of specific patients.

Since big data analytics can enhance disease prediction, it can lead to a reduction the healthcare costs by decreasing hospital readmission rates. Healthcare organizations often lack a clear understanding of the variables causing readmissions. By identifying these relationships, big data analytics enables healthcare providers to improve patient care and prevent readmissions.

The real-time insights provided by big data analytics also enable more proactive and preventive healthcare measures, which further contribute to cost reduction in healthcare. Big data analytics has the potential to facilitate smart and connected communities (SCC), a specialized form of smart cities.

Advanced computational tools and methods can be integrated with big data and effectively utilized to extract valuable information and address real-life issues, thereby achieving smart healthcare systems. For example, exploring the combination of machine learning (ML) and big data analysis can offer significant benefits to healthcare systems. Furthermore, investigating how to integrate big data and cloud computing to create a comprehensive healthcare services platform is crucial for effective healthcare systems. Overall, big data analytics can revolutionize healthcare systems by providing opportunities through its diverse applications. It will also drive digital disruption in the healthcare world and facilitate real-time decision-making.

6.1.1 Current challenges in big data and future research directions

As mentioned above, in real-time telemedicine systems, IoT devices are continuously generating massive amounts of data. High-speed data generation poses challenges in collecting, organizing, processing, and making decisions about patients. In addition, there are several challenges such as visualization, mining, analysis, capture, storage, search, and sharing. Traditional mechanisms might not perform efficiently in handling such large and diverse amounts of data. Big data thus necessitates innovative processing techniques to improve decision-making, uncover new insights, and optimize processes. Furthermore, several challenges exist in big healthcare data, including data aggregation, data format, incompleteness, timeliness, scaling, normalization, noise removal, and queuing. To fully benefit from big data in the healthcare system, the aforementioned issues must be addressed.

Additionally, sophisticated data analysis mechanisms are necessary to extract valuable patterns and insights from large datasets. Issues such as size, irregularity, and sparsity will also pose significant challenges to data management and processing in healthcare. In particular, the biggest challenge with big data is its management. Traditional software cannot handle such large volumes of data. Therefore, advanced applications and software with fast and cost-efficient computational power are needed. Data quality, integrity, and accuracy are other significant challenges in smart healthcare systems that need to be addressed through big data-based healthcare systems.

For future research, advanced techniques are necessary to aggregate, store, and manage large datasets. The attack on patient privacy is identified as a serious concern in the context of healthcare big data analytics. Hence, advanced technologies are also needed to address the privacy and security issues with big data.

Healthcare systems that use big data analytics can benefit from the use of cloud and edge computing. Computing technologies enable on-demand services to scale to large numbers of users, offering features such as virtualization, scalability, pay-per-use, and multi-tenancy. Cloud-assistive treatments can also be used to help medical professionals provide services irrespective of geographical location. It is necessary to conduct further research on the utilization of edge and cloud computing to support big data.

Implementation of AI algorithms and novel fusion techniques is essential for achieving efficient big data analytics in healthcare systems. For example, automated decision-making can be achieved through ML methods. Without proper software and hardware support, big data remains a significant challenge. Hence, developing enhanced techniques and smart web applications for efficient analysis of big data is crucial and is worth investigation in the future. In addition, an efficient recognition model is required to process big data in healthcare, which is useful for identifying features and performing classification for diseases.

Analyzing big data remains challenging even for the most advanced computers, primarily due to limitations in memory access rather than processing power. Memory capacity, bandwidth, and latency requirements often surpass computational needs. A possible solution could be the use of quantum computing approaches to address this challenge. Further investigation in this regard is needed.

Medical images often encounter technical issues, including various types of noise and artifacts. Mishandling these images can result in tampering, potentially misrepresenting anatomical structures, such as veins. To address these challenges, measures such as noise reduction, artifact removal, contrast adjustment, and post-mishandling quality enhancement are essential. These steps ensure the accuracy and reliability of medical images, which is critical for accurate diagnoses and effective treatment planning. Therefore, employing advanced image processing algorithms with big data analytics is worth investigating in the future.

Smart power systems, often referred to as smart grids, are fundamentally driven by the integration of advanced information and communication technologies (Ghorbanian et al. 2019). This integration facilitates real-time monitoring, optimization, and control of energy flow while simultaneously generating an unprecedented volume of data from diverse sources such as smart meters, sensors, and distributed energy resources. Similarly, the healthcare sector has experienced a parallel transformation, with digital health records, wearable devices, remote monitoring systems, and connected medical equipment contributing to a vast accumulation of complex and heterogeneous data. In both domains, the emergence of big data presents not only a challenge but also a significant opportunity (Ghorbanian et al. 2019). The convergence of big data analytics within smart power systems and healthcare infrastructures allows for predictive insights, improved decision-making, and enhanced operational efficiency. Therefore, understanding and managing this data deluge becomes critical, as both sectors increasingly rely on data-driven intelligence to support resilient infrastructure and proactive care models. In this context, the role of big data analytics is not only beneficial but also indispensable, bridging the gap between technological advancements and practical, scalable solutions in energy and healthcare management.

A critical question that can also emerge is how to develop a high-performance platform capable of efficiently analyzing big data, alongside designing suitable data mining algorithms to extract meaningful insights (Tsai et al. 2015). In this context, information fusion is anticipated to become a key direction for enhancing the outcomes of big data analytics. Consequently, ensuring data protection will also become an integral aspect of future research in this domain.

Despite the numerous advantages and wide-ranging applications of big data, several pressing challenges remain–particularly in the areas of analytics, management, and data privacy and security–that must be addressed to enhance the overall quality of service. In this regard, blockchain technology, with its inherent decentralization and robust security features, holds significant promise for advancing big data services and applications (Deepa et al. 2022). To this end, the work in Deepa et al. (2022) presented a comprehensive survey of the intersection between blockchain and big data, highlighting current approaches, emerging opportunities, and potential future research directions.

Ultimately, the next generation of wireless communication systems is expected to facilitate the analysis of big data, thereby enabling the realization of massive IoT. Further investigation into the role of big data analytics in future wireless communication networks, particularly in the context of smart healthcare systems, is required.

7 Computing techniques

A new paradigm shift has occurred in recent years, enabling the softwarization and virtualization of networks–that is, creating programmable, decentralized networks using various computing techniques (Chiang And Zhang 2016). This can be accomplished by segmenting the network into various communication tiers or levels (multi-tier architecture), starting from end-user devices and moving on to edge computing platforms, core computing platforms, and cloud (data center) platforms. The considerable propagation distance between the end user and the cloud center can be overcome with the help of these layers. This can, in fact, help alleviate the latency sensitivity problem in IoT applications. Figure 11 shows the multi-tier architecture for healthcare systems.

Computing techniques in the healthcare system have revolutionized the way medical data is managed, processed, stored, analyzed, and utilized, leading to significant improvements in patient care. A smart healthcare system leverages such computing techniques to provide fast, secure, efficient, and affordable healthcare services. The enormous volumes of medical data can be analyzed in a scalable and economical manner using computing techniques, which also enable healthcare practitioners to access and share patient data easily. Figure 12 illustrates the related computing techniques for smart healthcare systems.

7.1 Cloud computing

The concept of cloud computing is presented as an efficient concept for managing, storing, analyzing, and processing sensor data online. Cloud computing is a computing architecture in which all computing resources, including storage capacity, memory allocation, and computing power, are used collectively in the cloud-based environment. Through the Internet, cloud computing offers on-demand processing services and shared computer resources (Mosenia et al. 2017). Body sensor networks are utilized in numerous widespread healthcare applications, generating vast amounts of data that must be managed and stored for analysis. Specifically, a large volume of healthcare data is sent to the cloud platform for effective management, processing, storage, and analysis from sensors, actuators, embedded devices, wearable devices, and IoT devices. Cloud computing offers flexibility, self-healing capabilities, self-configuration, and ubiquitous access, presenting numerous advantages. Web applications in cloud computing platforms allow flexible access to resources, adjusting dynamically to scale up or down according to computing needs.

The adoption of cloud computing in healthcare referred to Healthcare as a Service (HaaS). HaaS is a model where healthcare services and infrastructure are provided through

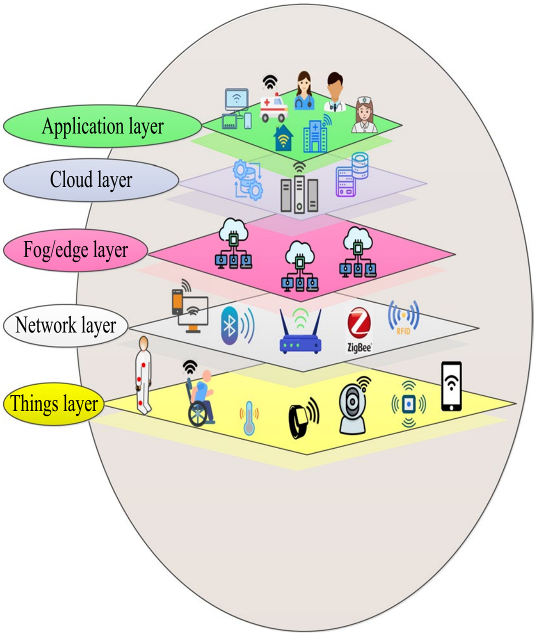


Fig. 11 Multi-layer architecture for healthcare systems

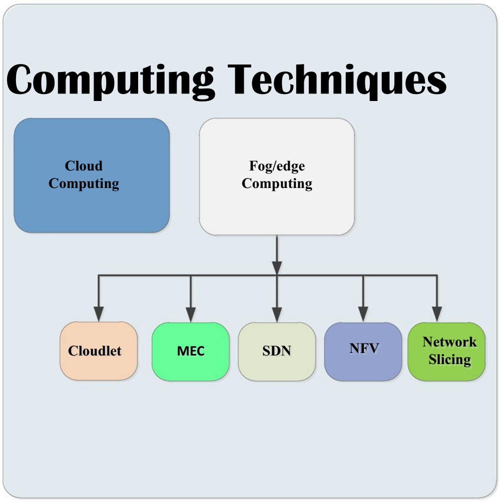


Fig. 12 An illustration of related computing techniques for the smart healthcare system

cloud-based platforms. It enables providers to access and deliver healthcare services, manage electronic EHRs, and perform various medical operations over the Internet. HaaS often includes telemedicine, RPM, and data analytics, enhancing patient care, improving access to medical resources, and optimizing healthcare management.

Various research works have investigated cloud computing techniques in healthcare systems. To this end, the work in Yang et al. (2017) discussed the role of cloud computing to tackling the challenges associated with big data for local to global digital earth science and applications. The work in Gupta et al. (2017) presented an IoT-based cloud-centric architecture designed for predicting users' physical activities in sustainable health centers. The proposed architecture depended on embedded sensors to collect health-related data. The architecture includes a cloud data center, and public and private clouds, and employs XML web services for secure and fast communication. The proposed architecture demonstrated consistent response times between the local database server and the cloud data center, regardless of the number of users. The work in Sun et al. (2018) investigated the importance of IoT in enhancing healthcare by enabling remote, continuous, and real-time monitoring of patient's health parameters, which are then processed and stored in cloud data centers. The work in Zandesh et al. (2019) presented an SLR on the legal aspects of the health cloud. The work in Ahmadi et al. (2019) investigated the characteristics of cloud-based architectures, as well as the security and interoperability challenges associated with IoT systems in healthcare. The paper in Dang et al. (2019) provided a review that discussed the implementation of cloud computing within the healthcare system. The work in Darwish et al. (2019) presented the idea of"CloudIoT-Health,"a concept that combines cloud computing and IoT to give the healthcare system networking, processing, scalability, and storage capabilities.

The work in Abba Ari et al. (2024) also investigated the integration of cloud computing and the IoT and introduced a concept known as the Cloud of Things (CoT) in the healthcare system. The proposed approach provides essential storage and processing capabilities to address the challenges of the massive data generated by IoT devices. The paper also explained various security and privacy challenges, such as jamming, collision, and flooding, that might affect the development of CoT. The work in Karaca et al. (2019) presented a mobile cloud system integrated with cloud computing to handle healthcare information, specifically for stroke patients. Using Virtual Dedicated Servers (VDS) and Android-based mobile phones, the study set up a system comprising a mobile application and a server application. The system employs an Artificial Neural Network (ANN) to classify two stroke subtypes, cardioembolic and cryptogenic, using big data. This approach ensures availability, security, and scalability, marking the first use of the Multilayer Perceptron Algorithm (MLP) with a stroke dataset in this context. The paper in Singh et al. (2020) focused on how IoT can offer medical care to orthopaedic patients during the COVID-19 pandemic. The paper demonstrated that integrating medical IoT devices with healthcare systems through networking technologies can facilitate various cloud-based and network services such as data sharing, report monitoring, patient tracking, information gathering, and analysis. Additionally, the study examined how these services can enhance the quality of care and patient satisfaction during the pandemic, particularly under lockdown conditions. The work in Sun et al. (2020) proposed a collaboration between the edge and cloud to achieve maximum utility in the healthcare sector. The work in Onasanya and Elshakankiri (2021) investigated the application of IoT-based technologies in healthcare, specifically for cancer care services. To this end, frameworks and architectures for integrating IoT and wireless sensor networks, leveraging business analytics and cloud-based services to enhance treatment options, and address associated security and operational challenges.

7.1.1 Benefits of cloud computing and key related points

Cloud computing serves as a foundational technology for innovative healthcare systems, as it enables the efficient processing and analysis of vast amounts of data generated by sensors. With the use of cloud-based technology, healthcare systems can address the challenges posed by the increasing volume of digital data and enhance healthcare management. Cloud computing achieves scalability, reliability, flexibility, and accessibility, hence allowing for centralized data management, processing, and advanced analysis in remote data centers. Cloud computing provides computing resources on demand, including processing power, storage, and networking, which are needed to extract useful insights about the patients and make effective decision-making (Sujith et al. 2022). Healthcare systems utilize cloud storage platforms to store information between the data originator layer (or sensor device) and the data analyzer layer (or end-user applications) for real-time disease analysis. Cloud computing offers an essential infrastructure for efficiently storing and managing patient medical data, particularly given the vast amount of data generated per patient, which can overwhelm traditional storage systems. Cloud-based solutions effectively try to address the challenges of handling large-scale healthcare data by utilizing technologies such as big data analytics to derive valuable insights and present information in a more suitable format for healthcare professionals. Such analytics will enable more informed decision-making and improve patient care. By leveraging cloud-based infrastructure, healthcare providers can efficiently handle and analyze large volumes of patient data collected from various devices, leading to an enhanced RPM system, timely interventions, and personalized treatment strategies and recommendations. Cloud-based platforms can play a crucial role in advancing computational deep learning applications. By leveraging cloud computing, it can address the challenges of managing vast data volumes, enhance efficiency, reduce costs, and provide the flexibility needed for training DL models.

Smart healthcare systems can manage the rapid growth of connected IoT devices by adopting scalable architectures, leveraging edge and cloud computing for efficient data processing, and implementing robust security protocols. In addition, using AI-based analytics and adaptive network management can ensure consistent performance, reliability, and real-time responsiveness, which is crucial for patient safety and care quality. The emerging cloud computing technologies are considered an appropriate technology to manage the smart grids with many IoT smart devices of healthcare systems in a reliable, secure, and scalable way (Ghorbanian et al. 2019). The cloud computing technology, with the help of IoT, provides access to various storage, computing resources, and applications from anywhere and at any time through connected IoT smart devices (Bera et al. 2014). Besides, due to the large amount of data produced in smart grids, traditional communication and information management approaches may not be sufficient, and cloud computing technologies can help deal with such massive data (Ghorbanian et al. 2019). Cloud services offer the illusion of infinite computing resources on demand, delivering various services such as Infrastructure as a Service (IaaS), Platform as a Service (PaaS), Database-as-a-Service (DBaaS), and Software as a Service (SaaS) (Jagadeeswari et al. 2018). Both public and private clouds can be efficiently utilized, where public clouds are accessible to the general public and private clouds are dedicated to a single organization, while hybrid solutions integrate both models. Cloud computing in the healthcare sector presents compelling opportunities to contain integration costs, optimize resources, simplify processes, and enhance service quality.

7.1.2 Current challenges in cloud computing and future research directions

Even with the apparent benefits of cloud computing, it introduces latency challenges due to data transmission over the internet for analysis and retrieval, which may not be suitable for emergency healthcare services and any latency-sensitive applications. Some further delays could be caused by transferring information from the application to the cloud and back, which might be intolerable, or even a short period of unavailability caused by cloud failure or a lost Internet connection may be life-threatening (Mosenia et al. 2017). To address latency issues, distributed computing such as fog/edge computing can be used as a feasible solution, which can enhance network edge data processing and response times. The system becomes more reliable and predictable when high-priority data analytics are implemented in the cloud or distributed smart gateways, and when crucial decisions are made within the local network promptly.

Efficient and rapid computing algorithms are required in smart healthcare systems to achieve lower latency and higher reliability. Healthcare systems utilize diverse IoT devices that require robust tools to process and analyze the vast amounts of data collected from various network platforms, which are transmitted to the cloud. Adopting hybrid cloud/edge computing facilitates fast and effective computations, and providing an intelligent computing infrastructure for IoT in healthcare systems could be a valuable solution that needs to be investigated. Furthermore, exploring advanced methods to reduce energy consumption in cloud computing could be worthwhile in the future.

An unpredictable network can lead to uncertainty and reaction latency in cloud computing, as raw data is sent from sensor nodes to the cloud. When streaming-based data processing (such as signal processing on ECG or EEG signals) is required, the issue becomes more difficult. Therefore, ensuring a reliable network connection that enables seamless service delivery is crucial. Further investigation in this regard is needed.

The primary challenge facing cloud computing is ensuring data security and privacy. Vulnerabilities enable hackers to access patients' private medical information without authorization, raising concerns regarding user authentication, data ownership, data protection regulations, and the potential exploitation of patient data.

Furthermore, several challenges still face cloud computing, including managing large-scale data storage and achieving efficient resource allocation. Future research directions should focus on enhancing security measures, developing advanced data management techniques, optimizing resource distribution, improving cloud service integration, and reducing latency to meet the growing demands of cloud computing applications. Additional issues with cloud computing involve handling the complexity of cloud infrastructure, maintaining interoperability between different cloud services, and addressing latency concerns in data processing. These issues need to be addressed to take the full advances of cloud computing.

7.2 Fog/edge computing

While cloud computing enables centralized data management, the trend is now towards decentralized systems using distributed fog/edge computing, which is positioned between hardware and the cloud (Asghari And Sohrabi 2024). Fog/edge can extend cloud computing capabilities to be closer to the things it supports and where data is generated, to the network edge. In particular, fog/edge computing can offer processing and storage services to devices (nodes) at the network's edge, replacing the need for all computation to be done in the cloud's core. Key features of fog/edge computing include low latency and location awareness, extensive geographical distribution, mobility support, handling numerous nodes, predominant wireless access, strong support in real-time and streaming applications, as well as heterogeneity. These characteristics make fog/edge computing ideal for various essential IoT services and applications. Wi-Fi, Bluetooth, Zigbee, and other communication protocols are among those that fog/edge computing can use, depending on the specific deployment requirements and connectivity options available at the edge.

A fog/edge computing node could be any network device capable of storage, computing, and network connectivity, such as routers, switches, cameras, and servers. These nodes can be installed anywhere with a network connection and gather data from IoT devices. High-priority data requiring immediate attention is processed on the nearest fog/edge computing nodes, while low-priority, non-delay-sensitive data is sent to aggregation nodes for further analysis. With the rise of powerful smart devices, fog computing is becoming an alternative to the cloud for wearable medical systems-based systems (Mosenia et al. 2017). Fog/edge computing facilitates networking, storage, and processing services between endpoints and conventional cloud computing data centers.

Edge and fog computing minimize latency by processing data closer to the source, providing faster response times that are critical for remote patient monitoring. While cloud computing provides greater scalability and centralized analytics, edge-based solutions are often more effective in time-sensitive clinical scenarios, making them a strong complement to traditional models (Wang et al. 2025).

In a large-scale sensor network, processing signals locally at the fog or edge layer can significantly reduce traffic between gateways and the cloud. Fog computing reduces network bandwidth burden and enhances data security by storing data locally. Fog/edge computing further minimizes latency by keeping data near the network edge or on localized servers. Fog/edge computing provides rapid feedback and enhances real-time data processing, making it more suitable for IoT applications in healthcare systems (Huang et al. 2023).

A fog-based middleware layer can be very effective for healthcare systems. The middleware layer offers several key benefits. For example, it works with a variety of platforms and operating systems, supporting a large number of apps. It facilitates distributed computing and enables interactions among heterogeneous networks, devices, and applications. Middleware also adheres to standard protocols, providing standard interfaces that ensure portability and interoperability. This makes middleware crucial for standardization. Additionally, middleware offers a stable, high-level interface for applications, enabling them to operate independently of hardware and operating systems. This feature is particularly advantageous for IoT, where numerous heterogeneous devices and networks are frequently updated or changed. The work in Elmisery et al. (2019) presented a framework for a cloud healthcare recommended service and introduced nodes serving as a bridge between cloud healthcare services and IoT devices. To this end, the fog nodes hosted a fog-based middleware that efficiently aggregates patient health data while ensuring privacy and confidentiality. The proposed middleware employs a two-stage concealment process, leveraging the hierarchical structure of IoT devices to alleviate them from intensive privacy-preserving tasks, thereby allowing patients to control their health data privacy by enabling them to share their data in a concealed form.

Numerous research projects have investigated the role of fog and edge computing in healthcare systems. For example, the work in Elmisery et al. (2016) proposed the use of personal gateways as fog nodes between IoT devices and cloud-based healthcare services to ensure accurate health insights while preserving user privacy. The fog nodes employ a privacy middleware that performs a two-stage concealment process, relieving IoT devices from intensive privacy-preserving tasks. The work in Kraemer et al. (2017) classified applications into different use case categories and outlined different tasks that fog computing can address. The work in Rahmani et al. (2018) suggested creating a geo-distributed intelligence intermediary layer between sensor nodes and the cloud using fog computing in healthcare IoT systems. The work in Oueida et al. (2018) offered a network architecture that optimizes critical performance indicators in healthcare systems, including patient waiting times, length of stay, and resource usage, by combining cloud and edge computing. By integrating cloud and edge computing, the proposed framework aimed to enhance reliability, efficiency, and security, leading to improved patient management and resource use. The work in Kumari et al. (2018) proposed a multi-layered healthcare architecture for efficient data handling, and highlighted the application of fog devices and gateways in the healthcare system. The work in Mahmoud et al. (2018) provided a review of the key fog-enabled cloud of things model and introduced an energy-efficient strategy for allocating application tasks to fog devices. The work in Sukhmani et al. (2018) discussed the application of edge computing for AR/VR and Tactile Internet applications. The work in Paul et al. (2018) suggested using fog computing, which allows for quick processing of the data gathered from monitoring devices, to monitor patients with chronic illnesses. The paper in Rahmani et al. (2018) highlighted the strategic role of gateways in IoT-based healthcare systems, particularly in facilitating communication between sensor networks and the Internet. The study suggested a smart eHealth gateway that would use fog computing to deliver cutting-edge services including embedded data mining, real-time data processing, and local storage. The work in Verma and Sood (2018) proposed a remote patient health monitoring system for smart homes using fog computing at the smart gateway. This model leverages advanced techniques, including embedded data mining, distributed storage, and notification services, at the network edge. The authors also employed an event-triggered data transmission methodology to process real-time patient data at the fog layer. The work in Farahani et al. (2018) provided a review of the topic of integrating ecosystem architecture with IoT in healthcare systems. The proper proposed a multi-layer architecture, which includes fog computing, device, and cloud layers. The paper in Shah and Chircu (2018) presented a review of the key application categories in healthcare technology, specifically focusing on wearables and connectivity, disease detection and treatment, patient care, and sensor networks.

The work in Alam et al. (2019) introduced a solution for optimizing the selection of virtual machines (VMs) in edge and cloud computing environments for patient monitoring systems. The authors proposed a portfolio optimization method for dynamic VM provisioning, aiming to improve efficiency in an Edge of Things (EoT) environment. The work in Ray et al. (2019) discussed the importance of edge computing in enhancing IoT-based user experiences. The paper reviewed the industrial benefits of combining IoT with edge computing, discussed taxonomical components, and presented two practical use cases addressing urban smart living. In addition, the paper proposed an edge-IoT architecture for eHealthcare and demonstrated its effectiveness in reducing reliance on cloud analytics and storage. The paper in Bhatia and Sood (2019) introduced a fog-cloud architecture to deliver efficient healthcare services in smart office environments. The paper also explained temporal healthcare predictive monitoring and alert generation in smart offices. The study in Naresh et al. (2020) proposed the use of a three-layer architecture for IoT-based healthcare applications, which makes the benefit of fog computing to address challenges such as fault tolerance, latency, energy efficiency, interoperability, and availability. The work in Saheb and Izadi (2019) introduced fog computing for emergency health systems, focusing on its suitability to achieve low-latency solutions. The work in Mutlag et al. (2019) categorized the work in fog computing into three classes: fog computing in healthcare applications, system development in fog computing in healthcare applications, and review and survey of fog computing in healthcare applications. The work in Pustokhina et al. (2020) presented an effective training scheme (ETS) based on a deep neural network (DNN), referred to as the ETS-DNN model, for edge computing in healthcare systems. The work in Sufian et al. (2020) explained the role of Deep Transfer Learning (DTL) on edge devices, including IoT, webcams, drones, intelligent medical equipment, and robots, which can be particularly useful in pandemic situations, such as the COVID-19 pandemic. The work in García-Valls et al. (2020) introduced a framework designed to enhance service times for activities requested by remote patients using fog servers. It leveraged current processors' ability to parallelize tasks on reserved cores and utilize data distribution platforms to ensure QoS and faster response times and communication. The paper in Tuli et al. (2020) presented the limitations of centralized cloud-based systems for latency-sensitive applications like health monitoring and proposed a framework called HealthFog using a fog computing approach–the proposed HealthFog integrated ensemble DL with edge computing for real-time heart disease analysis. The performance of HealthFog is evaluated using the FogBus framework, with a focus on metrics such as power consumption, network bandwidth, latency, jitter, accuracy, and execution time. The paper discussed the architecture of 5 G smart healthcare, including key technologies like Device-to-Device (D2D) communication, small cells, SDN, NFV, mmWaves, and edge computing (Poongodi et al. 2021).

The work in Ullah et al. (2021) explained the gap in fog-assisted secure healthcare data collection by categorizing and analyzing different data collection and transmission schemes. The work in Kaur et al. (2021) presented fog computing's role in reducing latency to achieve efficient health monitoring and enhance patient care. In addition, the paper investigated the challenges and security issues associated with fog computing, emphasizing its potential for enhanced analytics, improved treatment, and increased patient satisfaction. The study in Ullah et al. (2021) highlighted the benefits of adopting fog computing over cloud-only approaches in reducing transmission delays, particularly for emergency healthcare applications, and suggests future exploration of SDN to optimize data traffic and communication costs further. The work in Panja et al. (2023) proposed a real-time COVID-19 patient health monitoring system by making use of edge computing and fuzzy logic techniques. The work in Aggarwal and Kumar (2023) suggested the use of fog computing for Tactile Internet-based applications to achieve ultra-low latency. The following are some related techniques to edge computing.

7.2.1 Mobile edge computing (MEC)

MEC is presented as a pivotal technology in the 5 G network, capable of optimizing mobile resources by hosting compute-intensive applications and locally processing massive data before transferring it to the cloud. MEC is considered a significant platform for edge computing, situated on the periphery of a cellular network and capable of processing massive volumes of data quickly after it is received. It provides context-aware services using RAN information and brings cloud computing capabilities closer to mobile consumers or edge networks within the RAN (Pham et al. 2020). MEC can reduce backhaul network congestion and enhance resource optimization, user experience, and overall network performance. MEC, in particular, improves response times and conserves bandwidth by bringing processing and data storage closer to the point of demand. It enables applications and services to be hosted at the edge of the network, providing faster data processing and improved performance for IoT devices. Combining MEC and AI can enhance eHealth security and efficiency. Mobile Augmented Reality (MAR) has recently been introduced to enhance human perception by merging the real and virtual world (Siriwardhana et al. 2021). MEC is essential for MAR, as it can provide ultra-low latency and high data rates while offering enhanced computational power closer to users. MEC can also be integrated with energy harvesting techniques to achieve satisfactory computational performance and support green computing (Mao et al. 2016). All the essential tasks of MEC can be accomplished using a virtualized software-based platform that leverages network functions virtualization (NFV) and software-defined networking (SDN). These techniques would ensure an efficient, scalable, and adaptable network infrastructure. Using a control algorithm, MEC can be utilized to optimize the medical quality of service (QoS). Several key challenges that MEC may face can be addressed using the powerful tool of ML techniques. Edge caching, compute offloading, resource allocation optimization, privacy and security concerns, and big data analytics are examples of the challenges that MEC may face. Further discussion about these issues and the role of ML techniques in addressing them can be found in Pham et al. (2020).

7.2.2 Network function virtualization (NFV)

NFV involves transforming network functions like routing, load balancing, and firewalls into virtualizing networks through software applications. This enables network functions to operate on standard hardware, thereby removing the limitations imposed by the tight coupling of these functions to specific hardware units. NFV can improve network service deployment and management by enhancing reliability, scalability, and flexibility. NFV can be utilized to deploy and manage network functions, allowing for efficient resource utilization and rapid service deployment. With NFV, several mobile devices can access computing services from a single-edge computing server. This is accomplished by building various virtual computers with the capacity to perform multiple activities simultaneously. There are many advantages to NFV with blockchain integration. This integration helps in ensuring immutable, auditable, non-repudiable, consistent, and anonymous configuration (Dai et al. 2019). NFV enables flexible provisioning, deployment, and centralized management of virtual network functions. In addition, combining NFV with SDN enables the creation of a software-defined NFV architecture, enhancing agility in traffic management and optimizing network functions. Such an architecture supports diverse applications, such as service chaining, and is increasingly prevalent in NFV deployments.

7.2.3 Software defined networking (SDN)

SDN revolutionized network management by separating the control plane from the data plane, hence enabling centralized control and programmability of network resources. SDN enables more dynamic and efficient network operations, allowing for the dynamic configuration and management of data-forwarding rules in real-time. This virtualization of network functions optimizes energy usage, enhances resource allocation, and improves security and privacy measures. SDN provides centralized and intelligent control over network traffic routing through software applications. Key advantages include centralized control, network programmability, and abstraction. SDN technology can improve the flexibility and scalability of distributed IoT. However, the centralization of SDN can result in a single point of failure. In addition, incumbent SDN devices (such as gateways) are also incapable of conducting computationally intensive analysis on data traffic. This necessitates the integration of blockchain technology with SDN to overcome the disadvantages of SDN as suggested in Sharma et al. (2017). The MEC network can dynamically manage various services with the aid of SDN. SDN can give the edge computing network a logically centralized control platform. Utilizing a centralized control platform would simplify data traffic management within the network, optimize resource allocation, address the network's scalability issues, reduce overhead, enhance dependability, and–most importantly–lower latency.

7.2.4 Cloudlet

A cloudlet is a small-scale cloud data center or server cluster situated closer to end-users or devices at the network's edge. It offers computer resources and services comparable to those of bigger cloud data centers. Although data processing and application execution are closer to the point of need, cloudlets aim to minimize latency and enhance network performance, typically in mobile or edge computing environments. Cloudlet is ideal for applications needing environmental awareness, decision-making, and reliable connectivity, such as image processing, pattern recognition, and video analysis. They offer significant processing power and low communication latency, ensuring continuous service even without cloud access, and synchronize with the cloud when it becomes available (Asghari And Sohrabi 2024).

7.2.5 Network slicing

Network slicing is an emerging solution for service-oriented networking, which involves dividing a single physical network infrastructure into numerous virtual networks, each customized to fulfill specific demands and service attributes. Every network slice has its resources, configurations, and performance limitations, and functions as a separate logical network instance. As such, it can be configured to provide specific network features and capabilities that are tailored to a particular application or use case. In healthcare systems, network slicing plays a significant role in enabling the deployment of advanced healthcare services and applications that require diverse network demands. These demands are low latency, high security, high reliability, and bandwidth guarantees. Network slicing can be customized to meet the corresponding end-to-end service requirement (Chergui And Verikoukis 2020). Through specialized slices, network slicing may meet the diverse networking requirements of heterogeneous IoT applications. Network slicing can provide the flexibility and scalability required to support advanced healthcare services and applications. Furthermore, network slicing ensures that healthcare applications receive the necessary bandwidth and low latency required for critical tasks such as remote surgeries and telemedicine. Edge computing platforms can utilize network slicing to dynamically allocate resources and prioritize traffic according to the specific needs of IoT deployments, thereby facilitating low-latency and effective connectivity between cloud services and devices. There is growing interest in extending network slicing capabilities across multiple domains and heterogeneous network environments, including terrestrial and satellite networks, to provide seamless connectivity and services across different geographic regions and network technologies. Multi-domain network slicing enables the orchestration and coordination of resources and services across diverse network infrastructures, thereby facilitating end-to-end connectivity and seamless service delivery. Network slicing enables resource optimization by dynamically allocating network resources based on demand, ensuring efficient bandwidth utilization and minimizing congestion. Furthermore, the emergence of network slicing technologies brings the agility and flexibility of networks to support various functional and performance requirements.

7.2.6 Benefits of fog computing and key related points

Fog computing processes data locally from physical devices, hence reducing data transmission to the cloud and improving efficiency. Fog computing enables secure data transactions among millions of connected devices, ensuring high scalability and even data distribution across the network. The introduction of edge computing enhances data integrity and confidentiality in IoT healthcare by processing sensitive health information directly on devices, thereby reducing the likelihood of data leaks and unauthorized access. Big data analytics is a voluminous data mining approach that requires both cloud globalization and fog localization (Jagadeeswari et al. 2018). Fog computing enables real-time data processing by leveraging clients' devices for significant storage, communication, control, configuration, and management tasks. Fog/edge enables local decision-making and data analysis, which is performed closer to the data source, thereby conserving bandwidth, reducing delay, improving service delivery, and enhancing overall system efficiency. Fog/edge computing would greatly aid in providing real-time monitoring of patients' health problems and achieve a decentralized health monitoring system, as it brings data processing closer to the source, such as IoT devices and wearable sensors (Chiang And Zhang 2016).

Many nodes spread across a wide geographical area can be supported by fog or edge systems. By separating edge devices from the central network, edge computing also guarantees device interoperability. Fog computing seamlessly integrates various devices and networks. Furthermore, fog computing can be beneficial for wearable medical sensors, offering distinct advantages over cloud-based solutions, including low latency, mobility support, and enhanced privacy awareness for healthcare applications.

Fog/edge computing addresses the challenges posed by network latency and the need for fast response times. By processing data locally on IoT devices, fog/edge computing minimizes the time required for data to travel back and forth between the device and a distant cloud server. This enables real-time analysis of healthcare data, crucial in time-sensitive applications such as remote patient monitoring, emergency response systems, and continuous health monitoring.

Fog/edge computing is essential in healthcare IoT environments because it dynamically adjusts the location of computing based on the environment's needs, context, and applications. It ensures low and predictable latency for service delivery during emergencies, reduces the volume of data transferred to the cloud by processing and storing vital data locally, and conserves bandwidth and battery resources (Kashani et al. 2021).

As mentioned earlier in this paper, low latency is a crucial feature in healthcare systems, which needs minimal transmission time between sensors and the processing unit. This can be achieved by either reducing the distance between sensors or increasing the transmission speed. Fog/edge computing addresses both issues by providing local computation power within the network. This is essential as sensor nodes typically have limited energy and computational resources. By processing data at the edge or pre-processing it before sending it to a remote unit, transmission times are significantly reduced.

Fog/edge enables sensitive data to be processed closer to the source, thereby reducing the risks associated with cloud storage. Fog/edge computing provides a platform for securing sensor-collected data by bringing computational capabilities closer to the devices. This helps protect user-generated data from the things layer. The fog paradigm's distributed approach enhances security. Securing sensor devices serves as the first line of defense, preventing attackers from exploiting the system. In addition, encrypting monitoring data before transmitting it to the cloud, which can be implemented at the edge node, adds an extra layer of security (Qadri et al. 2020).

Furthermore, edge computing enables offline operation and resilience to network failures (Raeisi-Varzaneh et al. 2023). For example, in scenarios where internet connectivity is intermittent or unavailable, edge devices can continue to function and process data locally, ensuring uninterrupted healthcare services and critical operations.

7.2.7 Current challenges in fog computing and future research directions

Although fog computing offers various benefits in healthcare IoT systems, it is also evident that it presents some challenges related to resource management. Specifically, one possible challenge is how to manage fog/edge computing infrastructure and allocate available resources to IoT devices efficiently. In particular, future work in fog/edge computing should focus on developing efficient management strategies and resource allocation for IoT devices in healthcare systems. Critical factors with resource management should also be taken into account, such as service availability and energy efficiency. Fog/edge nodes face the challenge of managing a multitude of service requests with limited computing and storage capabilities, requiring optimal allocation across IoT devices. In addition, due to the sensitive nature of current applications, fog computing latency and response time remain challenges that must be overcome to achieve QoS in real-time healthcare applications.

A major challenge of efficient digital health services on the fog network is that they typically span multiple computational activities, hence performing big data processing over sensible information, which must be protected (García-Valls et al. 2020). To this end, using the capabilities of current processors can improve the servicing of remote patient nodes. However, to further improve productivity, enhance self-organization, and minimize runtime, additional optimization procedures are required. This could significantly help in achieving distributed and parallel learning in edge-based architectures. Therefore, further research is required in this regard.

Healthcare systems are integrating IoT and fog computing by adding or replacing medical devices and sensors to enhance patient care. However, these devices often have different communication protocols, making integration challenging. Further investigation is needed to address this issue. Furthermore, the variety of devices and their communication technologies creates connectivity challenges. Seamless integration is needed to handle mobility and delay constraints. Networks should be non-intrusive and allow for sharing resources or infrastructure (Kraemer et al. 2017).

Fog/edge computing introduces complex, dynamic relationships between system nodes, particularly sensors and fog computing nodes. This dynamic nature, along with concerns about privacy and safety, necessitates effective trust management between devices. While trust models exist, they must adapt to the complexity of fog computing in healthcare. Addressing these issues is crucial for optimizing fog computing and integrating health applications effectively across various use cases and deployment scenarios. Besides, addressing security and privacy concerns, including authentication, access control, and intrusion detection within fog/edge infrastructures integrated with IoT, remains a significant challenge that needs further investigation in the future.

Research should explore scheduling methods that can effectively handle the heterogeneity of edge environments, accommodating diverse end devices and varying resources. Mobility challenges, such as frequent link failures due to device movement, necessitate efficient task scheduling mechanisms that can dynamically adapt to changing conditions. Moreover, developing fault-tolerant scheduling strategies is essential to ensure reliability in dynamic edge environments prone to failures and faults, supporting robust 5 G and IoT applications (Raeisi-Varzaneh et al. 2023). Scaling edge computing capabilities to manage the increasing volume of data from diverse endpoints efficiently is also a critical area for future investigation.

The integration of fog/edge computing and DL near data sources can lead to what is known as edge intelligence. This edge intelligence approach supports dynamic, adaptive edge management and maintenance. Integrating DL into edge frameworks enhances intelligent edge capabilities, fostering mutually beneficial advancements in edge intelligence. Further investigation into the edge intelligence is required.

Future research could also include an investigation of the training of complex DL models using local computational resources across the fog/edge layer. This approach aims to optimize performance, reduce latency, and enhance efficiency in processing big data tasks across the edge or fog cloud. Understanding the dynamics of data movement, task distribution, and resource utilization from edge/fog nodes and the cloud is crucial. In addition, designing and implementing tailored solutions for resource-constrained edge and fog environments will be essential to optimize data processing platforms for diverse application scenarios and real-world use cases.

8 Artificial intelligence (AI) techniques

The most significant impact on healthcare systems in the future is anticipated to come from AI. AI generally enables machines to behave like humans, allowing them to accomplish specific tasks and make informed decisions (Alzubaidi et al. 2021). AI includes ML and DL, in particular, which enables computer agents to learn, reason, and make deft judgments using the facts they have gathered. Usually, ML acquires the necessary knowledge from datasets or through interactions with pre-existing contexts (Alzubaidi et al. 2023). Medical data in huge quantities can be analyzed using ML systems, from EHRs to genomic sequences, enabling precise disease diagnosis, personalized treatment plans, and predictive analytics to enhance the healthcare system. The effectiveness of ML algorithms largely depends on the integrity of the input data representation. A suitable data representation could significantly enhance healthcare performance. Consequently, feature engineering has been considered in ML, focusing on constructing features from raw data to improve the performance of ML.

DL computing paradigm is considered the gold standard in the ML community, and it has gradually become the most widely used computational approach in the field of ML, thus achieving outstanding results on several complex cognitive tasks (Alzubaidi et al. 2021). DL utilizes large, multi-layer ANNs and can learn from massive amounts of data. Feature extraction can be achieved automatically throughout the DL algorithms. In particular, DL models can learn from many types of input data, including numbers, text, or even combinations of input types. DL algorithms improve predictions and classifications by enhancing reliability and accuracy. In contrast to traditional ML and AI approaches, DL technologies have recently been progressing massively with successful applications to speech recognition, NLP, information retrieval, computer vision, and image analysis.

One of the standout features of DL is its ability to process data in real time, allowing for immediate analysis and responses in critical situations. This capability supports intelligent decision-making. Additionally, DL plays a crucial role in medical diagnosis by analyzing medical images, patient records, and other diagnostic data, resulting in faster and more accurate patient care. Google recently introduced a DL technique for detecting cancerous tumors, while Stanford University researchers are exploring DL for skin cancer detection (Khalil et al. 2021). Additionally, DL aids in HAR by identifying and classifying activities from video and sensor data, which is beneficial for health monitoring. Furthermore, DL techniques are used to develop secure methods for image encryption and decryption, enhancing data security and privacy.

Several works have investigated the role of AI tools in the healthcare system. For example, the work in Jiang et al. (2017) investigated the diverse applications of AI across various types of healthcare data. The work in Molanes et al. (2018) discussed the shift from data collection in the IoT to deriving value from massive datasets through AI and ML. To this end, the paper highlighted the role of AI and ML in managing and interpreting IoT data to create valuable insights. The paper in Mohan et al. (2019) presented a novel method for predicting cardiovascular disease by applying ML techniques to identify significant features, thereby improving prediction accuracy. The work in Albahri et al. (2020) provided an SLR discussing the role of biological data mining and ML techniques in detecting and diagnosing COVID-19. The research in Chowdhury et al. (2020) investigated the utility of AI in the rapid and accurate detection of COVID-19 from chest X-ray images. The work in Zhou et al. (2020) demonstrated the role of DL in enhancing HAR for the healthcare system. The research in Rehm et al. (2020) examined the application of ML to intensive care unit ventilation control and patient diagnostics. The work in Wang et al. (2020) investigated the integration of the DL algorithm for detecting lung nodules with low-dose CT and its application in the healthcare system. The paper in Ismail et al. (2020) proposed a CNN-based model for RPM systems using IoT technologies to improve personalized healthcare and disease prevention. The paper in Ali et al. (2020) introduced a smart healthcare system for heart disease prediction using ensemble DL and feature fusion techniques. The work in Zhang et al. (2020) explained the role of ML in emotion recognition. The work in Zhang et al. (2020) introduced DL-powered triboelectric smart socks designed for IoT-based gait analysis and virtual reality (VR) applications. The work in Zhang and Tao (2020) discussed how AI can enhance the IoT by creating the AI of Things (AIoT). Furthermore, the paper investigated AIoT architecture within cloud, fog, and edge computing frameworks. The paper explored advancements in AI for IoT in four areas: perception, learning, reasoning, and behavior. The paper also highlighted promising AIoT applications that could transform various sectors.

The authors in Salman et al. (2021) highlighted the effectiveness of using different ML methods for healthcare systems by examining various algorithms, medical input data, and output results. Besides, the relationship between ML objectives and E-triage processes was explained. Finally, a cross-over taxonomy was provided that can link ML algorithms with telemedicine structures. The work in Williams et al. (2021) investigated the application of DL in image analysis for low-resource healthcare settings. The authors in Bajwa et al. (2021) provided a review that outlined the recent breakthroughs in the application of AI in healthcare and discussed how to build effective and trusted AI-augmented healthcare systems. The paper in Bharadwaj et al. (2021) investigated the application of machine learning (ML) in diagnosing cardiovascular and neurological disorders, forecasting disease stages, and assisting individuals with physical and mental challenges. The work in Bharadwaj et al. (2021) provided a detailed overview of various ML algorithms, analyzing their applications in diagnosis, prognosis, disease control, assistive systems, monitoring, and logistics. Additionally, the paper offers a comprehensive discussion of five primary medical applications, with a focus on modifying machine learning models to address issues in medical chemistry, wearable sensors, cancer, brain disorders, and medical imaging (Shehab et al. 2022). The authors in Ahmad et al. (2022) investigated the role of ML in heart disease diagnosis. The work in Rajpurkar et al. (2022) investigated the promising directions for medical AI research, such as novel data sources and collaborative setups between AI and human practitioners. The paper in Alshamrani (2022) discussed how AI and ML can be used for remote healthcare monitoring to collect and evaluate a variety of information, which in turn informs clinical decision support systems and the provision of healthcare services.

Traditionally, medical datasets with labeled inputs are used; however, obtaining such datasets is often costly and time-consuming. To address this issue, alternative learning paradigms like unsupervised, semi-supervised, causal inference, and reinforcement learning have been employed (Rajpurkar et al. 2022). Unsupervised learning, for instance, uncovers novel patterns in unlabeled data, which has helped identify clinically relevant subgroups in conditions like sepsis and breast cancer. In addition, supervised learning has enabled models to diagnose cancers from whole-slide images without needing extensive pixel-wise annotations, significantly reducing data annotation costs. Techniques like synthetic data generation have also enhanced image quality, such as improving low-quality MRI scans, demonstrating the potential of leveraging imperfect data to train high-performing models (Rajpurkar et al. 2022). The work in Rajpurkar et al. (2022) highlighted key developments in medical AI, focusing on prospective studies and advances in medical image analysis. A predictive analysis of heart diseases based on ML has been presented in Ramesh et al. (2022). The work in Koh et al. (2022) reviewed the development and integration of AI and ML tools for cancer imaging. The paper emphasized the requirement for multidisciplinary collaboration to create effective tools. The work in Vora et al. (2023) discussed how AI and ML transform drug discovery and pharmaceutical development. The paper demonstrated that by analyzing extensive biological data, AI helps identify disease targets and predict interactions with potential drug candidates, increasing the chances of successful drug approvals. The paper in Ashton et al. (2023) explored the potential of ML in paediatrics, particularly for long-term conditions with significant disease heterogeneity. The paper explained the foundational algorithms and emphasized the importance of proper data preparation and validation. Using examples such as preterm infant nutrition and pediatric inflammatory bowel disease, the paper highlights the promising role of ML in improving disease prediction and treatment. The paper in Ardahan Sevgili and Şenol (2023) emphasized the importance of preventing treatment-related complications in paediatric patients and highlighted advancements in early diagnosis, radiology, pathology, imaging technologies, and cancer staging and management supported by AI and ML. The work in Keles and Bagci (2023) reviewed recent ML and DL solutions in neonatology, evaluating their roles and methodologies, including algorithmic developments. The work in Choy et al. (2023) conducted an SLR of DL in image analyses for the healthcare system's skin disease diagnosis and monitoring. To this end, the paper explored new research avenues, including non-image data, unconventional problem formulations, and human-AI collaboration. The work in Si-Ahmed et al. (2023) discussed how ML-based intrusion detection systems can enhance the security of the IoT in healthcare systems. The work in Baydoun et al. (2024) investigated the application of AI in prostate cancer. The paper in Wassan et al. (2024) investigated the optimal number of hidden layers and activation functions for deep convolutional neural networks. The paper also compared different frameworks to accelerate neural network performance. The primary objective of the research was to investigate innovative techniques for accelerating neural network training without compromising accuracy.

Furthermore, advanced ML methods can be used to enhance healthcare analytics and patient safety. For example, the work in Zerkouk et al. (2023) introduced Deep Generative Model with Isolation Forest (DGM-IF), which is a new unsupervised anomaly-detection framework, for WSNs and IoT-based systems. The work presented a feasible approach for spotting outliers in large and streaming sensor datasets. Adapting DGM-IF to physiological signals from wearables or implantable monitors could enable early detection of subtle clinical deviations, thereby boosting the system reliability of patients in smart healthcare systems. Therefore, this work could hold significance as it presented advanced AI methods for identifying anomalies, which can be valuable in spotting irregular patterns within health monitoring data. Incorporating such concepts may enrich the data analysis of how AI contributes to greater system reliability and ensures better patient safety. In addition, the work in Gomes (2025) explored how intelligent tutoring systems (ITS), empowered by AI, offer personalized and adaptive learning experiences beyond traditional educational methods. To this end, the work presented a comparative analysis highlighting AI's role in delivering real-time feedback and dynamic learning adjustments. The paper also addressed key ethical considerations such as data privacy and algorithmic bias. The insights discussed in the paper can inform healthcare solutions where adaptive and data-driven systems could be significantly used to enhance user-specific outcomes in future smart healthcare systems.

8.1 Benefits of AI and key related points

AI tools can be effectively used to simplify the extraction of required data from redundant data. AI technologies can aid in the optimization of healthcare resource distribution, enabling healthcare institutions to do so more quickly and efficiently. The AI tools are key to the successful implementation of IoT-powered wireless sensor networks in healthcare systems, as a large amount of data needs to be handled intelligently. There are several different benefits of adopting AI tools in healthcare systems. Examples of these benefits include achieving healthcare security, enabling the implementation of mHealth and eHealth, providing efficient patient care, reducing power consumption, offering an effective real-time healthcare monitoring system, supporting a triage system, facilitating personalized treatment plans, and enhancing the quality of network service (Salman et al. 2021). Smart healthcare systems powered by AI tools can optimize resource allocation by providing information and communication technologies, as well as insights into patient admission rates, resource utilization patterns, and staffing needs, leading to cost savings and improved efficiency. AI tools can discover relationships, provide interpretation, and identify patterns from any healthcare data (Shehab et al. 2022). AI tools can be used to enable Clinical Decision Support Systems (CDSS) and allow RPM and chronic disease management.

Typically, in the healthcare system, data comes in various formats such as text, speech, time series signals, and images. Each of these formats offers a distinct perspective on the health and medical situations of individuals. Utilizing AI tools to handle and analyze data efficiently is crucial for managing the complexity of healthcare data. In particular, AI algorithms can analyze vast amounts of patient data to aid in early and accurate diagnosis of diseases. AL enables the creation of individualized treatment programs tailored to each patient's specific needs. This can be achieved by analyzing patient data and historical treatment results.

The early detection and classification of brain tumors are crucial for timely and effective patient treatment. The human visual cortex has limitations in distinguishing between different shades of gray in MRI scans. This necessitates the development of computer-aided diagnosis (CAD) and brain tumor classification (BTC) methods. To this end, AI tools can be effectively used for this purpose. The automated processes provided by AI tools can support radiologists by visualizing and defining tumor types, thereby reducing the need for invasive procedures like biopsies and enhancing the overall efficiency of patient care.

AI tools can accelerate the drug discovery process by analyzing molecular structures, identifying potential drug candidates, and hence, speeding up the development of new treatments and vaccines. AI systems enable the continuous remote monitoring of patients' vital signs and health data, facilitating timely interventions and reducing the need for repeat hospital visits.

Medical imaging, including X-rays, MRIs, and CT scans, plays a crucial role in the healthcare system for accurate disease diagnosis. AI tools are essential for diagnosing diseases, such as genetic disorders and early-stage cancers, which are often challenging to detect (Tuli et al. 2020). AI models can be trained using supervised learning algorithms, which can effectively classify and identify these diseases. Besides, adopting computer-aided diagnostic tools that utilize AI tools such as machine learning ML and DL methods can significantly enhance the performance of medical imaging applications, effectively overcoming the limitations of traditional diagnostic procedures that are time-consuming and susceptible to human errors. In particular, such tools have demonstrated their great potential in bio-medical image analytics for the early detection of chronic diseases by handling a large amount of health data to facilitate the provision of healthcare services (Shickel et al. 2017). Recent developments in DL have advanced computer-vision tools in medical imaging diagnosis. DL is gaining popularity in histopathological image analysis due to the high complexity of these images. Given the complexity and variability in medical imagery, traditional mathematical modeling can be very challenging; however, DL algorithms can effectively classify images as normal or abnormal, making it easier to handle and analyze (Shen et al. 2017). With the use of GPUs, DL can significantly reduce the time complexity compared to traditional algorithms. Clinical trials and research often take years to complete, but the use of DL can significantly reduce the time and costs involved in handling such tasks. DL-based data analytics help researchers identify potential clinical trials through data sources such as social media and prior doctor visits. This approach minimizes both the time and resources required for conducting clinical trials. Maintaining EHRs is time-consuming and costly, but DL can simplify this process considerably.

Vital signs, imaging data, and other physiological characteristics are among the health-related data that IoT devices, such as wearable sensors and remote monitoring tools, routinely gather. DL algorithms trained on large datasets can effectively handle this data and identify patterns, anomalies, and trends that indicate changes in health state. To detect arrhythmias or other cardiac abnormalities, wearable ECG sensor data can be processed and analyzed in real-time using DL algorithms. As DL models can continuously learn from new data, they can improve their predictive accuracy, thus enhancing health monitoring within the healthcare system. This dynamic, data-driven method demonstrates how DL converts unprocessed medical data into valuable insights that improve the delivery of healthcare (Pradyumna et al. 2024).

8.2 Current challenges in AI and future research directions

Despite significant progress in AI within the healthcare system, it may face considerable challenges, especially in gaining user trust and creating effective training datasets. For example, protecting user privacy in the healthcare system is crucial. To be specific, traditional ML requires all training data to be gathered in a centralized place. This centralized data indeed raises privacy and security issues and can be computationally demanding for training. A possible solution to address this issue is to utilize federated learning, which focuses on decentralized data processing and can be employed to enhance privacy through robust data aggregation, thereby ensuring a secure smart healthcare system (Antunes et al. 2022). Furthermore, federated learning could offer intelligent solutions while maintaining privacy by eliminating the need for data sharing. Further investigation into the application of federated learning in the healthcare system and comparison with traditional ML is required. In this context, the study in Mohammed et al. (2023) investigated the challenges of energy-efficient offloading and task scheduling in federated learning for healthcare systems operating within blockchain-enabled networks. Therefore, further research in this area is necessary. Besides, future work should focus on advancing ML and AI algorithms to enhance prediction accuracy and communication technologies. This includes exploring DL architectures for complex data analysis and addressing issues of bias and fairness in AI systems.

The use of large language models (LLMs) in medical contexts raises critical concerns, particularly regarding patient privacy and data bias. These models often require access to sensitive health data, making them vulnerable to privacy breaches if not properly secured. Furthermore, if trained on unbalanced or biased datasets, LLMs may inadvertently perpetuate existing disparities in healthcare, potentially leading to inaccurate or unfair clinical outcomes. Future research should explore robust frameworks for ethical data handling and develop mechanisms to audit and mitigate bias, ensuring that LLMs contribute equitably and securely to healthcare advancements.

While backpropagation (a supervised learning algorithm used for training ANNs) shows high training accuracy, its performance on testing data can be unsatisfactory. This is because backpropagation relies on local gradient information and random initial points, often leading to local optima. Additionally, with insufficient training data, ANNs can suffer from overfitting. Future work should investigate methods to enhance generalization and mitigate overfitting. In addition, DL in histopathology suffers from a problem of overfitting, which can be alleviated by using classical data augmentation techniques. Further investigation in this regard is needed.

With the advancement of big data analysis, DL can achieve significant performance improvements in scenarios involving large amounts of unsupervised data, achieving great success with vast amounts of unlabeled training data. However, when training data is limited, more powerful models are necessary to enhance learning capabilities. Therefore, designing deep models to learn from fewer training data effectively is particularly important for applications in speech recognition systems, visual recognition, and medical imaging diagnosis (Alzubaidi et al. 2024).

Because medical devices and sensors are integrated into the IoT framework, they generate vast amounts of heterogeneous data, which raises the complexity of ML algorithms in real-time applications. Extracting valuable insights from such a massive amount of data in real-time is crucial for making predictions and informed decisions. Pre-processing data, managing unbalanced data, and diagnostic analysis are all part of this procedure (Alzubaidi et al. 2023). Additionally, it is crucial to address several other key concerns, including network scalability, resource availability, and the energy efficiency of Internet of Things devices. In most DL algorithm modeling, achieving a trade-off between accuracy and computational complexity should be considered.

AI of things combines AI and IoT, driving advancements in fields such as smart healthcare and intelligent manufacturing by enabling real-time data acquisition and analysis through IoT sensors and networks. Despite its benefits, AI of things systems require substantial computing resources and electricity, which contributes to carbon emissions and increased energy consumption. As the world faces climate and energy crises, achieving carbon neutrality in AI of things systems is crucial. Sustainable, low-carbon computing for AI of things digital infrastructure, comprising sensors, edge devices, software, networks, and data, analytics are essential for developing energy-efficient and data-efficient AI of things applications. This effort is fundamental for sustainable technological progress. Further research is necessary on the application of AI in the healthcare system.

Applying ML methods in medical image classification depends on feature extraction and selection methods that are sensitive to different colors, shapes, and sizes of images. Hence, finding feasible methods for feature extraction and selection is a worthwhile area for future investigation.

Unsupervised, semi-supervised, and reinforcement learning approaches, which can overcome computational limitations, deserve further investigation compared to supervised learning methods. These approaches hold significant potential for future research and development. Efforts should also be directed towards developing efficient reinforcement learning algorithms to further the capabilities of AI systems.

Removing metadata, such as patient information, is insufficient to ensure patient privacy, especially in complex healthcare environments where multiple parties, including insurance companies and hospitals, utilize healthcare databases for work-related activities like data processing and analysis. Traditional AI methods that rely on a central server for analytics may be insufficient for the requirements of smart healthcare. Hence, further investigation is necessary to develop advanced AI methods that address this issue.

Future work should address the challenges posed by large image sizes, which demand significant memory as both model complexity and pixel count increase. Whole-slide medical images, often containing billions of pixels, frequently exceed the capacity of standard AI tools such as neural networks. Potential solutions include resizing images, which may compromise fine details, or splitting them into smaller patches, which can limit the system's ability to make holistic connections. Alternatively, identifying and cropping regions of interest can be done manually before inputting them into an AI system, though this introduces a manual step into an otherwise automated process. Developing efficient methods to manage large medical images without losing crucial information will be vital for future advancements.

For future work, it is essential to focus on enhancing the explainability of AI systems to build user trust. Ongoing research is needed to develop methods that interpret AI decision processes, quantify their reliability, and present these explanations clearly to users.

When creating training data, DL algorithms are susceptible to attacks (Al-Garadi et al. 2020). To produce undetected attacks, attackers can design DL systems that can identify DL-based detection techniques. This research area is still in development and requires further investigation to identify more feasible solutions. DL and ML methods need to be used to secure IoT devices in healthcare systems, which must be developed to be more robust against adversarial attacks. Furthermore, applying more advanced security algorithms is crucial, which requires further investigation.

One potential area of study for IoT security is the integration of AI techniques with other technologies. For instance, deploying ML and DL at the edge for IoT security can boost the scalability of lightweight IoT devices, enabling near-real-time detection and minimizing latency. Furthermore, integrating AI tools with blockchain can significantly enhance the security of IoT in healthcare systems. For instance, ML and DL can help blockchain technology achieve improved evaluation, data and device filtering, and intelligent decision-making. This enhances the deployment of blockchain, improving security and trust functions for IoT devices.

Interoperability in IoT refers to systems that seamlessly work together despite differences in hardware and software. The variety of standards and technologies used, along with solutions from different vendors, creates complexity and interoperability issues. Overcoming these challenges requires a standardized layered framework due to heterogeneous device connectivity. Hence, developing a heterogeneous networking paradigm is essential to support various technologies and ensure seamless communication across IoT in healthcare systems. Overall, the convergence of IoT devices, cloud and fog computing architectures, and AI algorithms offers unprecedented opportunities to transform modern healthcare systems. However, achieving a seamless integration of these technologies within existing healthcare infrastructures remains a challenge. This integration might not only require technical synchronization across disparate platforms but also organizational alignment to ensure practical deployment and adaptability in clinical environments. Healthcare institutions often operate using a wide array of legacy systems and proprietary technologies that do not communicate effectively with modern digital tools. The absence of standardized communication protocols hampers the smooth exchange of information between IoT devices, cloud services, and AI modules. In addition, real-time healthcare applications require low latency and high bandwidth, which places significant strain on network infrastructures, particularly in remote or under-resourced areas. While fog computing helps bring processing closer to the data source, mitigating some of these concerns, the integration still requires robust coordination and reliable data pathways. Moreover, integrating AI models into clinical workflows introduces its own set of complexities. These models must be capable of processing and interpreting data from various sources, including structured medical records, unstructured clinical notes, imaging data, and real-time sensor feeds. Ensuring that AI outputs are explainable, clinically valid, and aligned with established medical protocols is critical. Additionally, the dynamic and unpredictable nature of real-world clinical environments makes it difficult to maintain consistent AI performance across different contexts, patient demographics, and healthcare settings.

From an organizational perspective, the road to integration is equally demanding. Many healthcare providers remain hesitant to adopt emerging technologies due to concerns over reliability, lack of familiarity, and fears of disrupting established workflows. Furthermore, implementing these systems often requires significant investment in digital infrastructure, staff training, and long-term maintenance–resources that may not be readily available in all institutions. Regulatory uncertainty further complicates adoption, as healthcare organizations must navigate evolving standards for data protection, the ethical use of AI, and digital health compliance. To address these challenges, there is a growing need for collaborative frameworks that bring together technologists, clinicians, administrators, and policymakers to work together. Efforts should focus on building modular, standards-based architectures that support gradual integration and allow for interoperability across different vendors and platforms. Moreover, strong emphasis should be placed on the co-development of technologies alongside end-users to ensure alignment with real-world clinical needs. Such coordinated and interdisciplinary efforts could benefit the full potential of IoT, AI, and cloud-enabled healthcare.

To illustrate the real-world application of these AI methodologies, we present a case study in diabetic foot ulcer detection using thermal imaging.

8.3 Use case: AI-enabled thermal imaging for diabetic foot ulcer detection

Diabetic foot ulcers (DFUs) are among the most severe and costly complications of diabetes, affecting up to 34% of diabetic patients and contributing to high amputation rates and increasing global healthcare burdens (McDermott et al. 2023). Early detection is crucial for preventing progression and improving clinical outcomes. However, despite significant technological advancements, DFU diagnosis continues to face major limitations.

Most current diagnostic approaches rely primarily on visual inspection and manual diagnosis, which often miss early-stage complications such as sub-epidermal inflammation or ischemic changes. Even when aided by AI-based algorithms, these methods are constrained by their dependence on visible symptoms, lighting conditions, and clinician expertise. In essence, many AI systems merely automate what the eye can already see, leaving early, unseen physiological changes undetected (Alzubaidi et al. 2020; Fadhel et al. 2024).

To address these gaps, the Aptium.ai team (https://www.aptium.ai/) has developed the MultiScan4D with AI, a novel handheld scanner that redefines DFU detection through non-invasive, AI-integrated thermal imaging. The device (shown in Fig. 13) represents a technological leap in non-contact diagnostic imaging. MultiScan4D captures both high-resolution 3D surface maps and radiometric thermal data in real-time, providing multimodal insights into tissue integrity and subclinical abnormalities.

The integrated AI pipeline utilizes state-of-the-art DL models, enabling the system to detect abnormal thermal deviations, segment ulcerated or inflamed regions, and calculate precise ulcer area ratios for follow-up assessments. Unlike traditional systems, MultiScan4D can identify physiologic anomalies before they become visually evident, making it invaluable for pre-ulcer detection and preventive screening.

Importantly, the scanner's architecture is designed for robust scalability across clinical, research, and community health settings. The combination of MultiScan4D with AI represents a paradigm shift in DFU care, moving from reactive, late-stage diagnosis to proactive, AI-enabled early detection, enhancing patient outcomes, reducing amputations, and lowering costs. It is not limited to DFU; it can be used for breast and skin cancer detection.

Fig. 13 Aptium MultiScan4D scanner with AI integration

9 Techniques for maintaining security and privacy for smart healthcare systems

The mobility of IoT devices in smart healthcare, which often connect to various networks at different places, such as home, office, and public areas, increases their vulnerability to attacks. Furthermore, the increasing number of IoT devices in healthcare networks further complicates the task for developers to provide dynamic security updates and robust solutions for multiprotocol information exchange. These challenges underscore the need for advanced techniques to secure IoT devices across different network environments.

Ensuring security in healthcare systems involves verifying user identities by using strong cryptographic methods (authentication) and providing appropriate permissions (authorization). Non-repudiation ensures that messages are genuinely sent by verifying digital signatures and timestamps (Khan et al. 2020). Additionally, maintaining data integrity and confidentiality ensures that patient information remains intact and secure throughout the system. These combined measures are essential for protecting sensitive patient data and maintaining robust security in healthcare environments. Data encryption, strict access controls, and device security are essential security measures that protect against hacker attacks, unauthorized access, and data breaches. Furthermore, IoT in smart healthcare systems requires data confidence, integrity, privacy, service availability, immunity against Eavesdropping, resilience, unique identification, and efficient access control. Cybersecurity threats and vulnerabilities pose a significant risk to the healthcare system, as interconnected medical IoT devices are susceptible to attacks. Passwords that are too easy to guess or are weak can give hackers access to important information and allow them to take control of equipment. Regular software and security algorithm updates are necessary to thwart ransomware and malware attacks, which can corrupt devices and erase data. Due to limited bandwidth or processing capacity, denial of service (DoS) attacks are frequent and have the potential to delay patient care by flooding IoT in healthcare systems with traffic (Pradyumna et al. 2024).

Various techniques and tools have been developed to date to achieve healthcare data security, with the primary goal of protecting private medical information. These techniques aim to safeguard patient privacy, maintain data integrity, and minimize the risk of unauthorized access or security breaches in healthcare systems. Fig 14 provides a list of different techniques used to maintain security and privacy in smart healthcare systems, which will be discussed briefly in this paper.

9.1 Cryptography

Cryptography utilizes mathematical principles to encode and decode information for secure communication, thereby preventing unauthorized access or alteration of data. Cryptography techniques provide essential security elements such as confidentiality, integrity, and authentication of digital data (Singh And Kumar 2020). Cryptographic techniques provide security against illegal access, verify integrity and confidentiality, and allow owner authentication through hashing and digital signatures (Al-Haj et al. 2015). Cryptography techniques encode information into an unreadable code to prevent unauthorized access. Both asymmetric (using a public-private key pair) and symmetric (using a single key for both processes) encryption schemes are employed; however, both face challenges in key management and

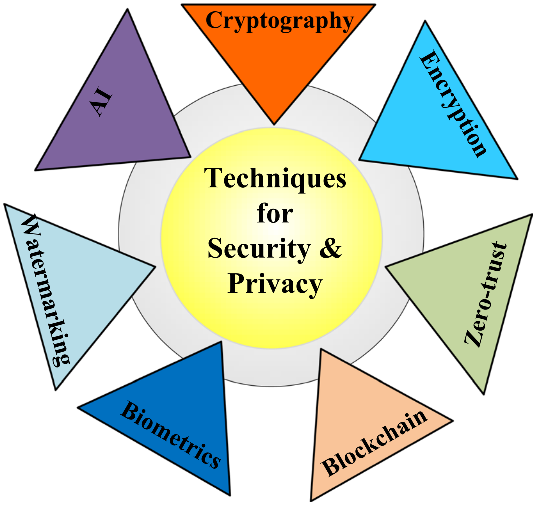


Fig. 14 A list of different techniques that can be used to maintain the security and privacy in smart healthcare systems

distribution, with symmetric encryption also being vulnerable to man-in-the-middle attacks (Gupta et al. 2020). Cryptographic methods can include random number generators, elliptical curves, logistic maps, and substitution and permutation schemes to enhance security (Gupta et al. 2020).

9.2 Encryption

Encryption is a specific technique of cryptography that involves converting plaintext (readable data) into ciphertext (unreadable data) using an algorithm and a key (Sasikumar And Nagarajan 2024). This process ensures that only authorized parties with the correct key can decrypt and access the original data. The basic goal of encryption is to keep data private while it's being transferred or stored. For data in IoT networks in the healthcare system to remain secure, private, and intact, encryption and data security are essential. Weak storage procedures and insufficient data encryption are additional risks to the healthcare system. Therefore, medical institutions can collaborate to create highly accurate prediction models by sharing encrypted datasets. To this end, the authors in Nadhan and Jacob (2024) investigated the use of a cryptography-based network for image encryption and decryption, as well as its potential application to the secure transmission of medical images by adopting a deep learning-based approach.

9.3 Zero-trust frameworks

Zero-trust frameworks have become a promising approach to protect IoT infrastructure in healthcare (Maji et al. 2025). Unlike traditional systems, zero-trust frameworks do not automatically trust any user or device. Instead, they require continuous verification to ensure each access request is secure and legitimate. This framework provides granular access control and authentication procedures to mitigate the risks associated with unauthorized access and lateral movement within IoT networks in healthcare systems. In addition, it continuously scans and evaluates network traffic to identify and address any security risks.

Firmware verification and secure bootstrapping are essential for preserving the integrity and security of IoT devices in the healthcare system (Nagarhalli et al. 2025). By supplying cryptographic keys and certificates before manufacturing, secure bootstrapping ensures that only trusted organizations can interface with the devices, potentially establishing a trustworthy starting state (Maqsood et al. 2025). Firmware verification prevents unauthorized access and maintains device authenticity.

9.4 Biometrics

Biometrics establishes an individual's identity based on physical, chemical, or behavioral characteristics, such as DNA, fingerprints, and iris patterns. Biometric systems operate in two primary modes: verification (a one-to-one comparison) and identification (a one-to-many comparison). These systems enhance the safety and privacy of patient data in healthcare applications. The work in Hamidi (2019) discussed the use of biometric data in IoT-based healthcare systems to significantly enhance the security and functionality of these systems by identifying individuals based on their unique physical characteristics, which are difficult to forge or lose. The application of biometric authentication schemes in medical devices offers significant advantages. Since many medical devices capture physiological measurements, real-time biometric key authentication can be performed by comparing data from wearable devices with signals from implanted sensors. This method provides distinct benefits over other security schemes, particularly in real-time IoT applications in healthcare systems (Sun et al. 2019).

9.5 Watermarking

To ensure accurate medical diagnoses, medical images must remain unaltered and intact. Health authorities and information security experts focus on using digital watermarking and steganography to enhance data security, image fidelity, authenticity, and content verification in eHealth environments (Qiao et al. 2025). These techniques embed data into various digital multimedia files, such as images, audio, video, and text, to protect and verify medical information during electronic storage, retrieval, and transmission. The embedded information serves various purposes such as copyright protection, authentication, and tracking the ownership or origin of the media. Watermarking techniques typically involve altering the data in subtle ways that do not degrade the quality or usability of the media but enable identification or verification when necessary. In healthcare, watermarking typically involves embedding information into medical images or documents to ensure authenticity, traceability, and integrity. This process helps prevent unauthorized alterations, confirms the source or ownership of medical data, and supports compliance with regulatory requirements. When embedding additional information into medical images through watermarking or steganography, it is crucial to avoid any distortion, as even slight changes can mislead forensic pathologists and affect clinical diagnosis. Techniques must strike a balance between payload and imperceptibility to maintain image quality. Several constraints must be met to prevent undesirable distortion in medical images. Healthcare professionals utilize watermarking to ensure the integrity and trustworthiness of patient records, medical images, and other sensitive information, thereby enhancing data security and patient privacy. Further discussion about the application of watermarking in the healthcare system can be found in Thakur et al. (2019); Anand and Singh (2021).

9.6 AI for secure healthcare system

ML and DL methods can play a crucial role in elevating IoT system security beyond secure communication between devices to creating intelligent security systems based on data analysis and predictive capabilities (Al-Garadi et al. 2020). A survey that may be utilized to create improved security techniques for IoT systems based on ML techniques and recent developments in DL can be found in Al-Garadi et al. (2020). The work in Ding et al. (2020) proposed a Deep Learning-Based image encryption and decryption network (DeepEDN) to facilitate the encryption and decryption of medical images. The work in Manimurugan et al. (2020) proposed a DL-based method and Deep Belief Network (DBN) to detect network intrusion in the IoT healthcare system. Federated learning can also be used for data security and privacy in healthcare systems (Hossen et al. 2022). The work in Mothukuri et al. (2021) proposed a federated learning-based anomaly detection method to proactively detect intrusions in IoT networks by utilizing decentralized on-device data.

The work in Martínez-Pérez et al. (2015) investigated the privacy and security issues in mHealth applications. To this end, the study was divided into three main parts: an examination of existing laws in the European Union and the United States regulating security and privacy in mHealth, a review of relevant academic literature, and a set of recommendations for app designers to ensure compliance with current security and privacy legislation. To mitigate security risks, the paper also examined privacy and security issues and proposed an intelligent collaborative security paradigm. The work in Yang et al. (2017) provided a survey on security and privacy challenges in IoT. To this end, the survey was divided into four parts: identifying key limitations of IoT devices and their solutions, classifying IoT attacks, discussing mechanisms and architectures for authentication and access control, and analyzing security issues across different layers. The work in Shah and Chircu (2018) emphasizes the need to ensure data security and privacy, as well as to achieve improvements in healthcare system efficacy and safety. The work in Gatouillat et al. (2018) reviewed recent efforts to enhance IoT in healthcare systems using formal methodologies from the cyber-physical systems, highlighting practical applications that benefit both individuals receiving medical care and medical professionals. To this end, the paper demonstrated how cyber-physical systems can enhance system robustness, security, and reliability, while also facilitating verification and validation processes. The work in Papageorgiou et al. (2018) provided an analysis of security and privacy issues in mHealth applications. The work in Chacko and Hayajneh (2018) investigated the privacy and security implications of IoT in healthcare, arguing that healthcare organizations must incorporate security measures into the design of IoT devices. This includes conducting risk assessments, implementing authentication protocols, and verifying firmware updates to ensure optimal security. Additionally, employing a defense-in-depth strategy with multiple security layers protects against specific threats. Unauthorized access to data, devices, and networks must be strictly controlled through effective access control measures. Continuous security testing and monitoring throughout the device lifecycle, along with fostering a security-conscious culture among employees, are defined as essential steps. The work in Yanambaka et al. (2019) proposed a device authentication scheme for IoT in healthcare systems, where IoT device data was suggested to be stored in server memory. To this end, physically unclonable functions (PUFs) with hybrid oscillators were used to enhance security for robust and efficient protection in healthcare systems. In Dang et al. (2019), an assessment of IoT privacy and security concerns was conducted, covering attack vectors, potential risks, and security measures from the perspective of healthcare. The work in Yaacoub et al. (2019) presented a framework for secure data transfer from patients to hospitals in mHealth IoT systems, divided into three tiers. Each tier included specific measures for securing data and protecting patient privacy. These methods ranged from physical layer security techniques for portable sensors to combining traditional encryption with additional procedures at the cloud data center where medical records are stored. The work in Sun et al. (2019) provided a review to highlight the potential of biometrics for securing IoT against cybersecurity threats in healthcare systems and discussed security schemes for implantable IoT devices.

The work in Yaacoub et al. (2020) proposed a secure key exchange method to maintain the security of the data. The authors in Habibzadeh and Soyata (2020) investigated the threats, vulnerabilities, and consequences of cyberattacks on smart healthcare systems and reviewed the proposed countermeasures. The work in Masud et al. (2020) investigated how IoT networks lack reliable mutual authentication and key establishment mechanisms. The paper (Kelly et al. 2020) highlighted the potential issues of IoT-based healthcare, such as privacy, security, interoperability, and data management. The work in Ghubaish et al. (2020) presented the recent advances in the security of IoT-based healthcare systems based on physical and network attacks. The work in Gull et al. (2020) presented a data encryption method to prevent IoT data in healthcare systems. The proposed approach involved using Huffman encoding as pre-processing and embedding the generated codebook. According to experimental findings, this method greatly increases the payload capacity for hidden data in IoT healthcare systems. The work in Azeem et al. (2021) proposed a framework to enhance the security of data transmission and aggregation in IoT-based healthcare systems. The Secure Message Aggregation (SMA) method at mobile nodes and the Secure Message Decryption (SMD) algorithm at fog nodes form the foundation of the suggested architecture. The results demonstrated how well the suggested architecture stopped replay and data fabrication attempts, guaranteeing strong data security in healthcare systems. The work in Alshehri and Muhammad (2020) investigated the critical issues of privacy and security associated with healthcare systems-based IoT devices (Alshehri And Muhammad 2020). The work (Muhammad et al. 2021) highlighted the importance of security and privacy in IoT-based healthcare systems. The work in Sadhu et al. (2022) provided a survey discussing the security and solutions for IoT.

The work in Alhaj et al. (2022) examined the use of IoT in healthcare systems, emphasizing the necessity to look into and resolve security concerns. The work in Mishra and Singh (2023) explained the challenges facing IoT in healthcare systems, such as equitable access, privacy, security vulnerabilities, and regulatory issues. The work in Mishra and Singh (2023) discussed how to develop a healthcare architecture prioritizing security based on the use of advanced technologies, such as big data, cloud computing, AI, and blockchain. The authors in Khan et al. (2023) proposed a secure IoT-based healthcare system using transfer learning techniques. The work in Akhbarifar et al. (2023) presented a RPM model that makes use of a block encryption method to secure health and medical data in a cloud-based IoT environment. The model predicted patients' critical health situations through data mining of biological data from smart medical IoT devices, ensuring the protection of sensitive patient information with secure block encryption techniques. The work in Abba Ari et al. (2024) focused on discussing the privacy and security issues in the cloud of things. The work in Adil et al. (2024) identified security challenges in healthcare-based IoT applications, emphasizing the need for a secure communication infrastructure due to the unstructured communication formats used by patient wearable devices and other networking entities.

9.6.1 Benefits of security and privacy techniques and key related points

Because healthcare data is sensitive, security and privacy are essential in IoT healthcare systems. Ensuring the security of medical images is essential since they contain valuable diagnostic information. From a privacy perspective, maintaining patient rights and confidence requires adhering to healthcare rules. Respecting human autonomy and maintaining the integrity of healthcare data raises ethical concerns, including getting informed consent and transparently using data and noting that trust involves managing credentials and controlling access to protect the confidential information of service providers and patients. Privacy can be achieved by preventing unauthorized access. Therefore, security and privacy are foundational issues in healthcare systems. By identifying and addressing potential vulnerabilities, healthcare organizations can proactively mitigate risks associated with cyber threats, ensuring the continuity of safe care and services. Ensuring the integrity of data prevents tampering and corruption of medical records, maintaining the accuracy of patient information, which is crucial for making informed medical decisions. By implementing strict access controls and monitoring mechanisms, we can protect healthcare information against threats, hence ensuring that only authorized personnel have access to sensitive information. Therefore, applying advanced security techniques could enable secure and private data sharing between different healthcare providers and systems, thereby improving coordination and collaboration in patient care.

9.6.2 Current challenges in security and privacy techniques and future research directions

In IoT healthcare systems, anonymizing patient data is essential for maintaining privacy. Finding a balance between data utility and privacy protection has been attempted in several ways, but it is still a challenge (Sun et al. 2019). Therefore, developing effective methods for anonymizing sources in privacy-preserving data-sharing schemes to ensure patient anonymity and access control is required. Furthermore, developing robust encryption techniques and enhancing authentication methods necessitates further research. Protecting against various cyber threats, such as malware, phishing, and ransomware, also requires further investigation. Due to resource constraints, developing advanced algorithms is crucial for countering security threats and maintaining privacy during data sharing. Cryptographic techniques can secure data access for authorized users; however, new threats necessitate innovative mitigation approaches. AI offers effective solutions for identifying and countering these threats. Further research about the exploitation of AI in the security of healthcare systems is required. Furthermore, the distributed nature of IoT in healthcare necessitates efficient key-sharing algorithms for authenticating node identities while conserving energy.

The limited computational power of miniaturized IoT devices makes implementing strong security schemes challenging. Therefore, several techniques for data security and encryption to address security issues in healthcare systems with minimal power consumption are still needed. Ensuring consistent security measures across diverse IoT devices in the healthcare system is challenging due to the need for seamless integration into the network. Thus, achieving interoperability, which allows such IoT devices to communicate and share data securely, is a significant hurdle. For the healthcare system to function efficiently and securely, it is essential to implement strategies that resolve compatibility issues, standardize communication protocols, and streamline data integration. Ensuring data security in real-time circumstances remains a challenging problem that requires attention. These difficulties present new research opportunities for the development of cutting-edge intrusion detection systems, secure communication protocols, innovative encryption techniques, and sophisticated data protection algorithms tailored to the specific challenges faced by IoT devices and healthcare networks.

Because IoT security and privacy are interdisciplinary, specialists in computer science and healthcare collaborate more often, which promotes interdisciplinary research and creative solutions. Researchers can develop novel strategies to safeguard private medical information and foster confidence in these systems as IoT technology advances, ultimately influencing the direction of healthcare security and privacy. For future work, an extensive analysis is needed, focusing on the stakeholders involved, the privacy assets at risk, the algorithms or models desired, the quality of these models, and their intended audience. Implementing the best healthcare modeling algorithms within privacy standards may be challenging due to tradeoffs and restrictions. To address these drawbacks, parallelizable methods such as ensemble learning can be employed to create reliable models. Efficient frameworks such as kHealth proposed in Sharma et al. (2018) can overcome privacy challenges by identifying and implementing appropriate privacy building blocks and scenarios. Balancing privacy and practicality in healthcare informatics will lead to the development of effective healthcare applications and their rapid adoption. Many security methods for IoT devices could potentially be applied to protect the healthcare system. However, wearable and implantable devices often lack the resources to implement such sophisticated methods due to their size and power constraints (Sun et al. 2019). To this end, ensuring the security of such IoT devices necessitates innovative solutions encompassing human, cyber, and physical elements. Many heterogeneous devices are connected to hospital networks without passwords or encryption, making them vulnerable to hacking. Hackers could disable individual devices, disrupt life-saving care, launch widespread attacks on specific device types, or steal data. Addressing these security gaps is critical to protect patient safety and privacy.

In connected healthcare systems, maintaining strong data security often introduces latency due to the need for encryption, authentication, and access controls. While these measures are essential to protect sensitive patient information, they may conflict with the need for real-time responsiveness in critical care. Therefore, system designers must carefully balance robust protection with performance efficiency, ensuring patient safety without compromising data confidentiality.

Furthermore, finding the appropriate hardware, software, and storage platform based on application needs is essential to ensure the reliability, robustness, and security of medical sensor networks and their data. This involves striking a balance between performance and security requirements to protect sensitive medical information effectively. Securing large-scale IoT in healthcare systems requires the development of real-time detection and protection solutions. Therefore, future research should focus on reducing computational complexity.

9.6.2.1 Blockchain Blockchain is considered one of the most important techniques for enhancing security and privacy in the healthcare systems. Blockchain is a decentralized technique that ensures the integrity and confidentiality of patient data by providing a tamper-proof record of transactions. Blockchain technique helps prevent data breaches and unauthorized access, ensures transparent and traceable medical records, and facilitates secure sharing of health information among authorized stakeholders. A list of transactions is contained in each block of the chain, and once data is recorded, it cannot be changed backward without also changing all blocks that come after it. Blockchain can be used to store immutable healthcare information. Thus, the immutable nature of blockchain, coupled with its decentralized structure, significantly reduces the risk of fraud and cyberattacks, making it a vital component for protecting sensitive healthcare data. Blockchain eliminates the need for a central authority, thereby enhancing network connectivity and accelerating the training process. Furthermore, robust security and privacy are provided in healthcare systems by fine-grained data access regulations in blockchain, such as smart contracts, which offer dependable authentication for federated health data processing (Nguyen et al. 2019). Blockchain technology can be a feasible solution for addressing the security issues associated with big data techniques, offering immutability, security, and traceability (Shi et al. 2020). Blockchain technology can operate as a regulator by transparently recording training-related transactions, guaranteeing cooperation and accountability. For large-scale healthcare collaboration, blockchain has emerged as a viable solution, particularly when integrated with federated learning. This combination creates decentralized healthcare systems with numerous medical entities acting as data contributors.

Numerous studies have examined the impact of blockchain technology on the privacy and security of healthcare systems. For example, the work in Shen et al. (2019) proposed a blockchain-based system for medical image retrieval with privacy protection. The work in Kumar and Tripathi (2021) proposes a blockchain network equipped with smart contracts and an integrated interplanetary file system (IPFS) cluster node for patient and device authentication. This cluster also serves as a distributed data storage layer, ensuring the secure transmission of authenticated data over the blockchain. The IPFS node provides security, authentication, and secure storage management, while the consortium network maintains data privacy using hash-based storage in IoT-enabled healthcare systems. The work in Xu et al. (2019) proposed healthchain, a blockchain-based privacy security method for health data. The proposed model utilized encryption to safeguard patient privacy by preventing unauthorized access to health data and ensuring the integrity of medical records by prohibiting their deletion or alteration during clinical procedures.

The work in Li et al. (2020) examined the security risks associated with blockchain technology and presented a comprehensive survey on the security of blockchain platforms. The authors have also examined genuine attacks on well-known blockchain systems and identified security risks, providing recommendations for strengthening blockchain security. The work in Khan et al. (2020) proposed a secure framework for encryption and authentication of Internet of Things (IoT)-based medical sensor data using enhanced Elliptic Curve Cryptography (ECC). The work in Rathee et al. (2020) offered a hybrid architecture that makes use of blockchain technology to process multimedia data in IoT healthcare. In particular, the work presented a security framework of healthcare multimedia data through a blockchain technique by generating the hash of each data so that any change or alteration in data or breach of medicines can be reflected in the entire blockchain network. The work in Farouk et al. (2020) discussed the impact of blockchain in the healthcare industry. The authors in Sengupta et al. (2020) outlined the recent blockchain-based solutions to address challenges in traditional cloud-centered applications, specifically focusing on key IoT and IIoT applications. The authors in Abu-Elezz et al. (2020) conducted a scoping review to explore and categorize the benefits of blockchain technology and its potential applications in healthcare systems. The authors in Shi et al. (2020) investigated the applications of blockchain in ensuring the security and privacy of EHR in healthcare systems. The authors in Tahir et al. (2020) employed a probabilistic model based on random numbers in conjunction with blockchain technology to provide IoT authentication and authorization in healthcare applications. The proposed approach successfully increased computational efficiency, lowered overhead, and strengthened mutual authenticity, according to the results presented. To address patient data privacy, the authors in Abou-Nassar et al. (2020) proposed a framework that incorporates enhanced encryption and IoT access control methods, improving security and interoperability to prevent tracking and profiling, while ensuring the confidentiality and integrity of patient data through IoT-based multi-cloud solutions.

The authors in Egala et al. (2021) employed a blockchain approach to develop a decentralized hybrid computing IoT system for healthcare applications, aiming to reduce latency and achieve low storage costs. The work in Hasan et al. (2022) examined privacy and security concerns, including entity authentication, integrity, and confidentiality, using access control methods and data authentication based on lattices that can be applied in quantum systems with limited resources. The authors noted that two of the biggest weaknesses in IoT devices are the absence of trustworthy authentication methods and network access controls. The authors also discussed a novel blockchain-based approach that can help in improving the confidentiality of IoT networks in healthcare systems. The work in Ktari et al. (2022) proposed an eHealth monitoring platform based on the IoT, utilizing smart sensors to collect patient health information. To guarantee the security and privacy of the data gathered, a blockchain approach is proposed. Data were processed and stored in a blockchain node after being encrypted and transmitted over an embedded Raspberry PI4 platform. The study highlighted the potential of blockchain in improving healthcare data management and monitoring while demonstrating the platform's efficacy as an affordable alternative for a secure EHR system. The work in Yaqoob et al. (2022) outlined the main advantages of blockchain technology for the healthcare sector, highlighting its useful applications and the prospects it offers. The work in Yaqoob et al. (2022) demonstrated how blockchain can provide secure, transparent, and efficient healthcare data management, substantially improving operational efficiency, data security, staff management, and cost-effectiveness. The work in Qahtan et al. (2023) provided a review and taxonomy that intersects 'security and privacy development' with 'blockchain-based healthcare industry 4.0 applications'. The work in Hussien et al. (2019) analyzed and mapped the recent technologies that use blockchain techniques in healthcare systems.

Benefits of blockchain and key related points

Blockchain technology has the potential to be crucial for enhancing healthcare system security and authentication. The blockchain aims to overcome the security issue of information communication, manage the quality of service of IoT services, and allow simultaneous resource allocation and authentication (Viriyasitavat et al. 2019). Blockchain, with its decentralized and trustworthy nature (Qian et al. 2018), holds great potential for securely sharing EHRs and managing data access across multiple medical entities (Nguyen et al. 2019). Blockchain enhances the security of IoT in healthcare systems by providing transparent data storage and management. Additional advantages of blockchain include the secure transfer of patient data between various platforms and the elimination of third-party access control, resulting in a quick, easy, and secure healthcare system. In healthcare systems, blockchain technology helps address security and privacy concerns by offering immutability, transparency, and decentralized consensus. In the context of IoT in healthcare, blockchain ensures data integrity, traceability, and secure transactions. By leveraging blockchain, patient data confidentiality and integrity are maintained through a tamper-resistant and transparent platform for the exchange and storage of medical data.

Current challenges in blockchain and future research directions

When implementing blockchain technologies, several practical considerations need to be taken into account. For example, implementing blockchain technology could face challenges such as high computing costs, large storage requirements, and significant bandwidth overhead. The participation of many organizations in the healthcare network can lead to issues with large volumes of health data, frequent requests, and concerns about stability. Hence, blockchain networks may suffer from slowness and congestion. In this regard, the scalability issue remains a main challenge that needs to be addressed. Therefore, further investigation to address this issue is required.

Blockchain networks may consume substantial amounts of energy. Hence, addressing the energy consumption issue of blockchain in the healthcare system is of particular interest. Different blockchain platforms often have different compatibility issues, which makes it a very challenging issue to transfer data or assets between them. Hence, achieving seamless interoperability between various blockchain networks is crucial for efficient utilization of blockchain technology.

In practical applications, encouraging miners to participate in the network is essential for maintaining a trustworthy and stable blockchain. Further investigation in this regard is needed. Further research is required in order to investigate the use of blockchain in protecting the integrity of medical information. Additionally, a major concern is finding the relevant technical talent to implement, manage, and resolve technical issues related to blockchain technology.

Blockchain can be efficiently combined with edge computing. This combination includes a comprehensive suite of services, encompassing pure data storage, device configuration, and governance, as well as sensor data storage and administration. This will be particularly effective for time-sensitive applications, such as healthcare monitoring systems. Further investigation into this combination could be worth considering in the future.

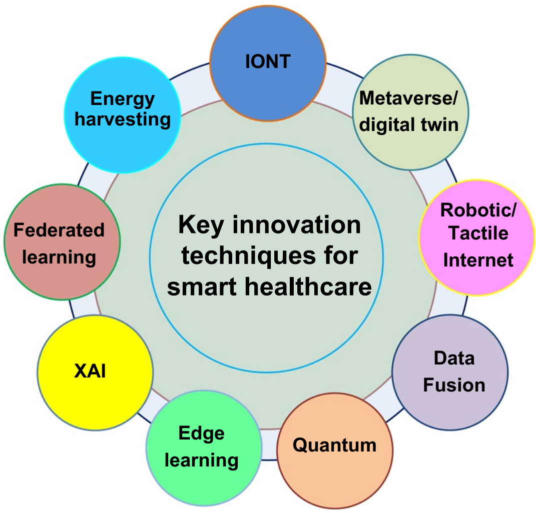
An intriguing area of research that warrants further study is the integration of intelligent blockchain services with e-health systems. For example, automatic clinical support platforms that utilize data mining or ML can be integrated into the cloud to analyze medical records and dynamically predict patients' health issues. This approach can assist healthcare providers, such as doctors, in offering instant healthcare services while ensuring data privacy and security. Solutions for efficient blockchain design in healthcare are essential to optimize data processing and communication, aiming to improve the latency.

In general, IoT devices often have limited computational resources, making it challenging to use advanced cryptographic mechanisms. These devices struggle with modern, secure public-key cryptography schemes due to constraints in memory and processing power (Hussien et al. 2019). Most blockchains are based on public-key cryptosystems, which face issues related to efficiency and security. Therefore, selecting appropriate cryptography is essential, as cryptosystems in blockchains must address post-quantum computing threats and seek energy-efficient, quantum-safe algorithms to ensure long-term data security. In addition, future work could also include the practical implementation of a cryptographic protocol model for encrypted transactions and evaluating the performance of an enhanced blockchain in decentralized healthcare applications. Addressing issues related to scalability, data privacy, and integration in blockchain-enabled healthcare IoT applications remains a significant challenge. Combining blockchain with SDN technology could offer a promising security solution through decentralized controllers that manage numerous IoT devices and secure sensitive health data. The ability to detect malicious activities can be significantly improved, hence, further investigation into this combination is required.

10 Key innovation techniques for smart healthcare systems

Key innovation techniques are essential for transforming traditional healthcare into smart, efficient, and effective systems. Such techniques also aim to achieve Hospital-to-Home (H2H) services, enabling patient-centric systems and improving the quality of life. Smart healthcare systems aim to achieve global accessibility, massive data analytics, low latency, high reliability, and high performance in real-time care systems. A list of the key innovation techniques for smart healthcare systems is provided in Fig. 15.

Fig. 15 A list of the key innovation techniques for the smart healthcare system



10.1 Development of nano things (IoNT)

The healthcare sector benefits significantly from the application of nanotechnology. The advancement of nanotechnology in healthcare allows the incorporation and development of nanoscale technologies to enhance diagnosis, treatment, and monitoring. These IoNTs can perform new types of systems at the nanoscale with unprecedented accuracy (Pramanik et al. 2020). Nanotechnologies enable the development of innovative nano-devices with advanced capabilities. Nanotechnology applications in biomedicine and medicine likely have the most significant impact on human health and well-being.

The concept of nanomedicine has also been introduced recently, defined as the combination of nanotechnology and nanoscience. Nanomedicine leverages the unique properties of nanomaterials to address health issues at the cellular level. Its applications span various areas, including targeted drug delivery, tissue regeneration, enhancement of magnetic resonance imaging (MRI), gene therapy, and biomarker detection. These applications are applied across a wide range of medical disciplines, including oncology, immunology, osteopathy, and urology (Chude-Okonkwo et al. 2017).

The wireless interconnection of nano-processors in massive multi-core architectures enables innovative parallel computing architectures. Moreover, the interconnection of nanomachine networks or nanonetworks with macroscale communication networks opens the door to transformative applications not only across scales (from nano to macro) but also across domains (biological and non-biological) and realms (classical and quantum).

The work in Chude-Okonkwo et al. (2017) highlighted the potential of molecular communication and nanonetworks in advancing targeted drug delivery within the context of smart healthcare systems-based nanomedicine, offering insights into how these paradigms could address challenges and facilitate the translation of targeted drug delivery from research to clinical practice, thereby contributing to the broader field of the Tactile Internet. The work in Pramanik et al. (2020) provided a comprehensive review of the potential of nanotechnology in the healthcare system, focusing on its applications such as nanomedicine, nanoimplants, nanobiosensors, and the IoNT. To be specific, the paper presented multilevel taxonomies and discussed the importance of nanotechnology. The paper particularly focused on the IoNT, outlining its general and communication architecture while critically examining the challenges associated with its application. The paper also investigated the integration of the Internet of Bio-Nano Things (IoBNT) to enhance compatibility with the human body. Recently, the work in Abadal et al. (2024) provided an overview of the progress in wireless nanocommunication technology, focusing on electromagnetic nanonetworking across microwave, terahertz, and optical bands. The paper compared the characteristics and potential of these technologies with the requirements and challenges of various nano-networking applications in the IoNT and on-chip networks, including quantum computing applications.

10.1.1 Benefits of IoNT and key related points

Nanomedicine, nano implants, nano biosensors, and the IoNT have the potential to revolutionize medicine and healthcare services. Nanotechnologies are providing the engineering community with a new set of tools to design and manufacture novel electronic components, a few cubic nanometers in size, that can perform specific functions, such as computing, data storage, sensing, and actuation. The integration of several nanocomponents into a single entity, just a few cubic micrometers in size, will enable the development of more advanced nanodevices. Through effective communication, these nano-devices will be able to accomplish complex tasks in a distributed manner. Smart drug delivery and nanoscale surgeries are prime examples of where the IoNT can be highly beneficial. With IoNT, various healthcare applications can be utilized, including nanorobots for precise drug delivery, nanosensors, precision medicine, minimally invasive surgeries, and swarms of nanorobots for accessing inaccessible body parts (Pramanik et al. 2020). IoNT is revolutionizing nanoscale networks with applications in sensing and precision medicine, paving the way for future advancements.

10.1.2 Current challenges in IoNT and future research directions

Creating effective communication protocols for nanoscale devices is challenging due to their unique physical constraints and properties. Traditional battery technology could be infeasible for IoNT devices, so there is a need for more efficient and reliable energy-harvesting solutions. Integrating the IoNT with existing IoT infrastructure is also challenging due to differences in scale, communication protocols, and processing requirements. Future research should focus on addressing these challenges to advance the field.

The primary challenge in the IoNT framework is creating effective information reservoirs and establishing efficient routing pathways within the network. Hence, there is an essential need to develop efficient information storage and routing mechanisms. Future research should consider electromagnetic nanonetworks, offering a thorough overview of the latest advancements in technology, theory, design, optimization, and applications. Key areas include developing techniques for wireless nanosensor networks and the IoNT, exploring wearable and implantable electromagnetic nanonetworks to bridge the nano-bio and macroscale worlds, and creating new communication and networking architectures for massive multi-core computing platforms enabled by Wireless Networks-on-Chip in both classical and quantum domains. The nanonetwork technology may operate at the terahertz band, which may incur high path loss due to molecular absorption (Promwongsa et al. 2020). Further investigation in this regard is needed.

Future research should also focus on several key aspects of electromagnetic nanonetworks, including channel modeling for specific environments such as in-body to out-of-body channels for implantable electronics and intrachip to interchip links. Future research should also explore multiple frequency bands, such as microwave, terahertz, and optical. Important topics include waveform and wavefront design, modulation and coding, interference modeling, node discovery, and medium access control solutions. Performance evaluation metrics, such as capacity, latency, and reliability, as well as novel key performance indicators specific to nanonetworks, are essential. Finally, energy efficiency, battery-less, and energy-harvesting solutions with IoNT should be investigated.

Future research should address the coexistence of electromagnetic nanonetworks with existing wireless technologies and their seamless integration into the current technology landscape. Important areas of focus include security and privacy aspects, standardization and regulatory efforts, and prototyping. Furthermore, exploring how electromagnetic nanonetworks can enhance existing applications and services, as well as enable new ones, will be crucial for future work.

10.2 Energy harvesting for energy efficient healthcare system

As the number of IoT devices continues to grow, there is an increasing need to achieve environmental sustainability. Environmental sustainability is a significant concern due to rising energy demands and the growing amount of electronic waste. Promising research areas include reducing energy consumption, utilizing renewable energy sources, minimizing device size to reduce non-degradable materials, and studying various effects on human health. Self-sustaining, or autonomous systems, are devices or networks that function independently without external power or human intervention for an extended period. These systems utilize renewable energy sources, such as solar, wind, or energy harvesting, to power themselves. Energy harvesting is an effective technique for obtaining energy.

Energy harvesting in healthcare systems involves capturing and storing energy from various sources, such as body heat, movement, or ambient light, to power medical devices. The energy harvesting approach could enhance the energy efficiency of healthcare systems, particularly for wearable and implantable devices, by reducing dependency on traditional batteries. Utilizing energy harvesting ensures continuous operation and reduces the need for frequent recharging or battery replacements. This is crucial for maintaining the functionality and reliability of devices that monitor vital signs, administer treatments, and support patient health.

Clean energy sources can fuel wireless communications networks thanks to the energy harvesting technique (Buzzi et al. 2016). Through the use of rectenna circuits, energy harvesting enables wireless devices to collect energy from radio-frequency signals. This energy can then be utilized to wirelessly recharge the devices' batteries wirelessly, maintaining their functionality.

An example of utilizing wireless power transfer for wearables is the development of symbiotics, meshed structures worn continuously with epidermal contact for high-fidelity biosignal acquisition. These devices are recharged using far-field power casting, which transfers tens of milliwatts over distances of up to 2 m, enabling frequent recharges for uninterrupted operation over weeks or months (Stuart et al. 2021). Other wireless power transfer methods include photonics and near-field power, supporting chronically wearable devices with high energy needs. In addition, ambient energy sources, such as perovskite solar cells harvesting indoor and outdoor light, can recharge batteries for daytime operation (Min et al. 2023). Power can also be extracted from sweat, motion, and thermal gradients. Meanwhile, existing power supplies, such as smartphone-emitted electromagnetic radiation, can power and communicate with distributed sensor nodes, thereby enhancing sensing fidelity and reducing maintenance needs.

Energy harvesting for smart healthcare systems has been extensively researched to address the issue of unstable energy supply in IoT devices. The healthcare system implementation can periodically monitor the physiological parameters of the in-hospital patients. Therefore, IoT-enabled devices actively participate in data gathering and analysis, while also enhancing the quality of care through routine monitoring and reducing the cost of care. The work in Mao et al. (2016) investigated a green MEC system that incorporates energy harvesting to address the limitations of the battery capacities of mobile devices. By offloading computationally intensive tasks to an MEC server, the proposed system aimed to improve execution latency and overall computation experience. To ensure sustainable performance, the paper explored using renewable energy sources to power mobile devices via energy harvesting technology. The work in Shaikh and Zeadally (2016) reviewed the energy harvesting in wireless sensor networks (WSNs) using distinctive environmental sources. The authors in Feng et al. (2019) investigated a resource allocation problem for nanonetworks to satisfy the high throughput and low latency required for haptic communications. To this end, the authors consider a scenario where multiple in-body nanodevices transmit haptic information to a human operator. The authors in Feng et al. (2019) also assumed that each nanodevice has the capability of harvesting energy to power itself. The authors aim to select the optimal sub-frequency bands, transmission power, energy harvesting, and packet drop rate for each device such that the average data rate at each time slot is maximized while satisfying the energy consumption and latency constraints. An efficient DL framework for intelligent energy management in IoT networks has been reviewed in Han et al. (2020).

10.2.1 Benefits of energy harvesting and key related points

Energy harvesting significantly extends the lifespan of medical devices by reducing the need for frequent battery replacements, enabling continuous health monitoring with real-time data, and improving patient care. Energy harvesting also lowers maintenance costs by minimizing manual interventions and ensures a reliable power source for critical medical devices, reducing the risk of power failure. In addition, energy harvesting can support the development of portable medical devices, enhancing patient mobility and convenience, while utilizing renewable energy sources, contributing to environmental conservation efforts. The technique of energy harvesting has the ability to lessen carbon emissions (Ulukus et al. 2015). Energy harvesting can be effectively used to power M2M devices, which refers to the ability of smart devices to communicate with each other and exchange data without human intervention, and hence may have the potential to substantially improve the efficiency and sustainability of IoT in healthcare systems. Energy harvesting can also be used to power RFID devices.

Widespread use of WSNs is found in healthcare monitoring. Because these networks require continuous connectivity and prolonged battery life, the overall power consumption in them is considered a significant issue. Energy harvesting can therefore be effectively used to support WSNs. Numerous safety and medical applications, as well as remote monitoring with increased communication ranges and longer battery life, can benefit from energy harvesting (Ulukus et al. 2015).

With advancements in communication techniques, energy is expected to be delivered alongside medical analysis tasks, necessitating new transmission protocols to adjust data size and transmission rates. This approach can significantly reduce the energy dissipation of edge servers and requires re-modeling the corresponding energy queue. The joint transmission of energy and information not only enhances spectrum efficiency but also reduces equipment maintenance costs.

10.2.2 Current challenges in energy harvesting and future research directions

Power and energy storage technologies must meet diverse requirements for IoT applications, a crucial issue that warrants attention. Energy constraints to date pose a significant challenge for IoT-based healthcare systems, especially for wearable and implantable devices. While wearable devices can be periodically recharged, implants require long-lasting batteries since they cannot be frequently replaced or recharged. Advances in battery technology and the development of safe materials for implants are crucial areas of research. Efficient power management systems and resource optimization are essential for enhancing the longevity of IoT sensors. AI tools such as ML and DL algorithms can further optimize computing processes to conserve energy. The use of distributed architectures, such as fog and edge systems, reduces power consumption by minimizing long-range transmissions. Battery-less operations and energy harvesting from the human body are promising solutions to prolong the lifespan of implantable sensors. Developing low-cost, efficient energy harvesting and fast-charging methods is essential for sustaining the IoT ecosystem in healthcare without harming users. The efficiency and reliability of harvesting energy from ambient sources need significant improvement. Powering nanoscale and implantable devices without frequent battery replacements remains a challenge that needs to be addressed. Ensuring seamless integration with existing IoT infrastructure, while maintaining low power operation and high performance, is worth investigating in the future. Current energy harvesting technologies must be refined to provide consistent and adequate power for continuous device operation. Future research should focus on developing advanced materials and technologies for more efficient energy harvesting, such as novel nanomaterials and improved photovoltaic cells. Additionally, designing new transmission protocols and data management strategies to optimize energy consumption in IoT networks is crucial.

Furthermore, energy efficiency and battery lifetime remain critical challenges in the development of smart healthcare systems. In this regard, the work in Sodhro et al. (2021) investigated a smart healthcare framework based on the IoMT, emphasizing the integration of IoT technologies with medical services. This IoMT-enabled system incorporates various medical sensors and devices to facilitate real-time patient monitoring and improve clinical decision-making. However, a key limitation lies in the energy demands of these sensor-based devices, particularly their limited battery life and inefficient energy utilization. This necessitates the requirements of energy-aware strategies, which underscores the emerging need for integrating smart grid technologies into healthcare systems to ensure sustained, efficient, and reliable operation of IoMT systems. In particular, advanced information and communication infrastructures are essential for successfully operating smart grids and providing efficient, reliable, and sustainable electricity to customers (Ghorbanian et al. 2019). A smart grid refers to a combination of power systems, communication, information, and control infrastructures, capable of monitoring, managing, and optimizing the operations of all parts of the power system (Fang et al. 2011). It aims at improving the economic performance, efficiency, flexibility, reliability, resiliency, safety, and security of a power system. The integration of smart grids for smart healthcare could represent shift toward advanced energy systems with healthcare infrastructure, aiming to enhance resilience, efficiency, and sustainability in the medical system. In the healthcare system, deploying smart grids could ensure a reliable power supply, which is crucial for continuous patient monitoring and the operation of critical medical equipment. These smart grids can integrate renewable energy sources, hence optimizing the energy usage and achieving energy efficiency solutions. Additionally, smart grids utilize a diverse range of IoT-enabled devices, including smart meters, sensors, and actuators, to enhance performance. Specifically, smart grids can use sensors, IoT devices, communication protocols, and data analytics to enable real-time monitoring, automation, and efficient energy management in healthcare systems. Smart grids can facilitate the implementation of IoT devices and advanced AI algorithms, which can enhance patient monitoring and data management in healthcare systems. In Singh and Paliwal (2022), the authors conducted an in-depth study of IoT-based smart grid systems, outlining a conceptual framework and highlighting key features. Additionally, the paper examined existing challenges in IoT implementation within smart grids, providing valuable insights for future research opportunities. Overall, solutions such as energy harvesting and smart grids in healthcare systems require further investigation in the future.

10.3 Metaverse and digital twin

The use of AI in future 6 G networks will revolutionize interactions with the real world by collecting and processing data for advanced analysis, enabling accurate daily decisions, and enhancing users' quality of experience (QoE) (Masaracchia et al. 2024). This interaction paradigm is referred to as metaverse (Aloqaily et al. 2022). The metaverse utilizes digital avatars, virtual representations such as digital twins, and interactive technologies like extended reality to support the analysis, optimization, and operation of various wireless applications (Khan et al. 2024). The metaverse enables wireless system operations through virtual modeling, allowing network designers, mobile developers, and telecommunications engineers to collaborate on monitoring, observing, analyzing, and simulating their solutions in a virtual environment. The metaverse is considered to be the next-generation mobile computing platform that will be widely used in the future.

The metaverse makes use of technologies like VR and augmented reality (AR), extended reality (XR), holographic communication, and mixed reality (MR) to make communication between the real and virtual worlds possible (Park And Kim 2022; Adil et al. 2025). VR, at the virtual end, provides synthetic views with additional information, while AR, at the reality end, enhances the physical view with supplementary data. MR merges virtual and physical worlds, allowing real-time interaction. XR integrates AR, VR, and MR, enhancing human-specific information perception. XR headsets, sensors, and embedded systems are key to accessing metaverse-enabled wireless systems. Investigating these technologies enables the development of novel therapeutic approaches, which are substantially less expensive, thereby revolutionizing the healthcare industry. The metaverse is a collective virtual shared space, created by the convergence of virtually enhanced physical reality and physically persistent VR. It includes all virtual worlds, AR, and the internet. Holographic communication refers to the use of holograms to facilitate interaction and communication between people in a three-dimensional (3D) space. This technology leverages AR, VR, and MR to create immersive experiences that can simulate the presence of individuals or objects in a real or virtual environment. Holographic communication is used for communication and interaction, enabling people to"be present"in a virtual space. With 6 G technology, more proactive oral guidance through holographic communication becomes feasible (Ahad et al. 2024). Surgeons can provide real-time guidance and adjust their holographic view of the surgical site. The metaverse for healthcare refers to a virtual environment where patients, healthcare professionals, and medical institutions interact using e.g., VR and AR. Noting that AR and VR are the primary technologies used in the metaverse, particularly in education. VR studies are more prevalent in emergency medicine, while AR shows greater potential for diagnostic and treatment applications. This digital space facilitates remote consultations, medical training, and therapy sessions, hence enhancing the accessibility and efficiency of healthcare services. It is a digital space where avatars create immersive experiences. The metaverse integrates advanced technologies, including AI, blockchain, robotics, computer vision, IoT, and cloud and edge computing.

The metaverse system comprises two primary spaces: the meta space and the physical space. The meta space is a logical environment implemented through edge, cloud, or a combination of both technologies. In contrast, the physical space encompasses all necessary physical entities, such as edge and cloud servers, and devices, which are essential for wireless systems (Khan et al. 2024).

The metaverse's immersive virtual environment can be significantly enhanced by integrating digital twins, which provide real-time data and simulations of physical entities, resulting in more accurate and interactive experiences. A digital twin is a lifecycle virtual model of a physical system, process, or product that is updated based on real-time data and supports decision-making through reasoning, ML, and simulation. A digital twin is used for monitoring, simulating, and optimizing real-world objects and systems. By integrating AI and big data analytics, digital twins can process real-time data from their physical counterparts to create accurate and comprehensive behavioral models. This capability facilitates real-time decision-making, enhancing the efficiency of product development processes. A digital twin utilizes a virtual representation, combined with security-related technologies (e.g., blockchain), communication technologies (e.g., 6 G), computing technologies (e.g., edge computing), and machine learning (ML), to enable smart applications. The digital twins can help clinicians predict various situations in a virtual environment before implementing actual changes, which will reduce the risks and save costs. It can also help clinicians make an accurate diagnosis, develop appropriate treatment plans, and predict treatment effects, thereby achieving personalized treatment in a smart healthcare system. Digital Twin offers a novel medical simulation method that provides precise and effective services by combining multidisciplinary, multiphysics, and multiscale models. Leveraging AI and advanced technologies, it can accurately diagnose diseases and enable personalized treatment.

Clinical decision-making in traditional systems is often based on retrospective data and periodic check-ups, which can delay the detection of anomalies or disease progression. Emerging technologies such as digital twins offer a dynamic and real-time virtual representation of a patient's health status. By continuously integrating sensor data and medical records, digital twins enable predictive modeling and personalized care, significantly improving response time and treatment outcomes. Furthermore, in traditional methods, decision support systems are generally rule-based or dependent on static data inputs, which can limit adaptability in rapidly evolving clinical scenarios. However, when integrating the emerging technologies, e.g., digital twins with AI and real-time data streams, this would provide dynamic simulation environments for testing interventions, optimizing workflows, and predicting outcomes more accurately. These capabilities remain unattainable through traditional systems. However, digital twins depend heavily on high-quality data and computational resources. Further exploration of this integrating is essential to fully understand their practical limitations and optimize their deployment in real-world healthcare setting, and hence, ensuring adaptation in clinical environments.

Typically, digital twin models simulate real-time patient states using integrated data, enabling highly personalized and predictive analytics. When combined with the ultra-secure and high-speed data transfer capabilities of quantum communication, these frameworks can significantly enhance the precision of treatment planning and ensure the safe exchange of sensitive health information. This is crucial for maintaining data integrity in real-time predictive analytics.

The authors in Yang et al. (2022) investigated the possible applications of the metaverse in medicine. To this end, the authors demonstrated that interactions between virtual and real clouds, using the metaverse, will enable experts and terminal doctors to perform medical education, science popularization, consultation, graded diagnosis and treatment, and clinical research. The work in Garavand and Aslani (2022) conducted a review to identify the application areas of metaverse technology in the health field. The work in Musamih et al. (2022) discussed the potential applications of the metaverse in healthcare, such as telemedicine and telehealth, medical education and training, medical marketing, healthcare supply chain, healthcare facilities, and fitness and wellness. The work in Xu et al. (2022) focused on overcoming implementation challenges in communication, networking, and computation to ensure the success of the metaverse as the next evolution of the Internet. The authors provided a tutorial on the metaverse, including its architecture and current developments. The survey addressed communication and networking hurdles, exploring solutions that leverage next-generation systems for seamless access and interaction with avatars. It also examined the high computation demands, discussing cloud-edge-end frameworks to enable metaverse functionality on resource-limited edge devices. In addition, the paper explored how blockchain technology can support the development of the metaverse, enabling the trade of virtual content and controlling physical edge resources in a decentralized, transparent, and immutable manner. The work in Wang et al. (2022) provided an overview of the metaverse's foundations, security, and privacy. The authors in Ali et al. (2023) proposed an architecture that includes three environments: the doctor's environment, the patient's environment, and the metaverse environment. In the metaverse environment, doctors, patients, and nurses interact through avatars, with blockchain technology ensuring the safety, security, and privacy of their data. All consultation activities, including images, speech, text, videos, and clinical data, are recorded and stored on the blockchain. The collected data is then used by XAI models for predicting and diagnosing diseases. The work in Wu and Ho (2023) explored the application of the metaverse in acute medicine by reviewing published studies and the clinical management of patients. More specifically, the work reviewed published articles on the metaverse roadmap and identified four additional domains: education, prehospital and disaster medicine, diagnosis and treatment applications, and administrative affairs. The work in Chengoden et al. (2023) provides an overview of the metaverse for healthcare, emphasizing the projects associated with it, its prospective uses, and the enabling technologies for adopting the metaverse in healthcare. The authors in Ferorelli et al. (2025); Jeong and Lee (2025) provided a review application the metaverse in healthcare systems.

10.3.1 Benefits of metaverse and key related points

Future wireless designs can be developed based on a metaverse that uses a virtual model of the physical world system, along with other schemes/technologies such as optimization theory, ML, and blockchain. A metaverse utilizes a virtual model to perform proactive intelligent analytics before a user requests analysis, enabling efficient management of wireless system resources (Khan et al. 2024). A metaverse can accomplish different activities with the help of avatars, which behave similarly to real human beings in the virtual world. Metaverse technology enables human beings to create a digital world while transforming the physical world for the better.

The metaverse offers substantial benefits in healthcare, including remote monitoring of critical patients, improved access to data, and enhanced understanding of clinical results, such as blood sugar and heart rate. Through the metaverse, the interaction between the virtual and real worlds could be improved. This interaction could be between doctors, patients, and their families (Yang et al. 2022). Such services offered by the metaverse may allow disease prevention, smart healthcare, physical examinations, diagnosis and treatment, rehabilitation, chronic disease management, in-home care, patient aid, outpatient care, and consultations. The metaverse could revolutionize medical education for students, and public health could be enhanced. The metaverse holds significant potential for medical imaging, fundamentally transforming the nature and quality of imaging. Metaverse technologies can help healthcare professionals in the effective planning and diagnosis of diseases (Petrigna And Musumeci 2022).

Additional advantages of the metaverse for healthcare and patient care include its expanding role in medical training through VR, digital therapeutic applications, AR for surgical procedures, enhancements in radiology, the use of medical wearables, and advancements in mental healthcare (Kye et al. 2021). Metaverse can be utilized in dentistry for educating students and professionals, as well as for oral health education. By offering 3D digital images, this technology enhances dental interventions and meets future healthcare needs.

The metaverse enhances communication by allowing a deeper understanding of people's emotions, creating a space that feels close to a real environment. This facilitates more effective and interactive connections, making it easier for people to understand each other and collaborate on tasks, even when they are distant. This feature can be very beneficial for RPM in the healthcare system.

Metaverse applications minimize the need for physical objects and spaces by enabling private messaging in specific locations, overlaying information on offline objects, providing virtual screens, displaying store inventory trends and sales volumes, and offering virtual displays for IoT devices and smart home applications (Park And Kim 2022). Faster and more secure healthcare facilities with a realistic experience can be made available in the digital space through the metaverse's combination of AI and blockchain (Ali et al. 2023).

10.3.2 Current challenges in the metaverse and future research directions

Many advantages and applications of the metaverse have been identified, but its sustainability is crucial. The metaverse can thrive and address issues when the user base remains stable; however, a decline in users can jeopardize its maintenance. Sustaining various social relationships is more vital than individual events or tasks, such as games and simulations. Continuous connectivity through low-spec mobile devices is essential for long-term engagement. Managing user logs with episodic memory can enhance user comfort and encourage prolonged use of the metaverse. Since storing all experiences has limitations in terms of capacity and utilization, research on effectively retrieving and reusing significant episodes is necessary. In addition, new platforms should consider methods for importing and exporting existing user experiences to ensure ongoing usability. Special focus should be given to the use of the metaverse in dental diagnostics, treatments, and education.

With the gradual dissolution of boundaries between the physical world and the Metaverse, the integration of metaverse technologies into forensic science and medicine is evolving from a conceptual possibility into a tangible objective. The future envisions a scenario where individuals operate within fully immersive virtual environments, disconnected from the physical realm, yet applying real-world principles and methodologies from medicine and forensic science. However, realizing this vision requires the establishment of robust ethical frameworks and technical standards, the validation and protection of virtual evidence, and rigorous safeguards for data privacy. Moreover, careful consideration must be given to the integration of AI systems within these environments, particularly in addressing potential biases and ensuring fairness and equity. As such, further interdisciplinary exploration is essential to navigate these complexities. To this end, advanced decision-making approaches may be worth investigating in the future for this purpose. Futhremore, future work should address enhancing the software sophistication challenges to improve the overall metaverse experience.

High-quality VR/AR experiences in the metaverse would require substantial computing power and advanced hardware, which may not be accessible to all users. Therefore, future work should consider the development of more efficient and affordable VR/AR hardware, as well as optimizing software to run on lower-spec devices.

Efficient management of wireless resources in a metaverse-enabled system requires intelligent interfaces for resource allocation and channel estimation. Centralized learning, performed in the meta space, can face privacy issues, while distributed learning, done at individual devices with aggregation in the meta space, offers better privacy but longer convergence times due to data and system heterogeneity. To address these challenges, novel distributed learning algorithms are needed for intelligent interfaces in metaverse-enabled wireless systems (Khan et al. 2024).

Implementing the metaverse would require an efficient edge computing design to manage the high number of interactive technologies with low latency. Although interactive experience technologies are well-researched, their application in the metaverse is unique and complex, necessitating careful design considerations. Key use cases include metaverse-assisted remote expert systems, real-time collaboration, and industrial maintenance. For instance, in a remote expert system for industrial maintenance, cameras and sensors capture images and annotations, which are sent to a remote expert for advice.

Different healthcare systems and devices would be required to communicate seamlessly within the metaverse. Therefore, future work should focus on creating universal standards and protocols for interoperability among various healthcare technologies and platforms.

The metaverse may face security threats due to the diverse components in patient healthcare systems, making it essential for providers to safeguard user-sensitive information. Modern systems rely on regular security patch updates; however, persistent cyber-physical attacks have the potential to undermine these. By incorporating security and privacy considerations from the design stage, endogenous security, such as quantum key distribution, offers a potential solution with self-protection, self-evolution, and autoimmune capabilities (Chengoden et al. 2023). This approach ensures that future metaverse systems can withstand both known and unknown security vulnerabilities and privacy threats (Wang et al. 2022).

Efficiently modeling and managing the mobility of avatars and mobile users in a metaverse-enabled wireless system is crucial for effective operation. This involves two key aspects: mobility modeling for avatars in the virtual space and the real-time mobility of users. Accurate modeling of device and user mobility is crucial for analyzing wireless systems before deployment, enabling tasks such as channel estimation and equipment deployment for new applications. In addition, the mobility of devices within the virtual space must be effectively managed through novel migration schemes. Live migration is suitable for real-time applications (Khan et al. 2024). One main challenge in developing digital twins with metaverse is acquiring and integrating diverse data, such as geometric, performance, and sensor data. EHRs are often fragmented and difficult to process automatically. The lack of real-time physical monitoring data during movement and difficulties in data fusion hinder the creation of a mature digital twins model for the human body.

10.4 Robotic and tactile internet

Robotic and automation technologies in healthcare refer to the use of robots and automated systems to perform tasks traditionally carried out by humans in medical settings. Such technologies aim to improve the efficiency, accuracy, and safety of the healthcare system (Chaudhari 2025). Surgical robots are advanced robotic systems that assist surgeons in performing minimally invasive surgeries with precision and control. These robots can enhance surgical accuracy, reduce incision size, minimize tissue damage, and shorten recovery times for patients. Telemedicine robots enable remote medical consultations and examinations by allowing healthcare providers to interact with patients remotely. These robots may include features such as video conferencing capabilities, medical device integration, and mobility to navigate healthcare facilities.

Robotics can be used to assist patients' rehabilitation in regaining motor function and strength following neurological injuries or conditions such as stroke or spinal cord injury. Laboratory automation technologies automate various tasks in clinical laboratories, such as sample processing, testing, and analysis. As such, laboratory automation technologies would reduce turnaround times for test results and minimize errors.

The term "tactile internet" refers to the essential technology that enables the steering and control of virtual or real-world items via the Internet (Fettweis 2014). The Tactile Internet represents a vision of an Internet that enables the transmission of a sense of touch. In that case, people would no longer need to be physically near the systems they operate, but could control them remotely. Specifically, the Tactile Internet is based on the concept of Internet sensorial, as defined by IEEE's communication standardization, which encompasses tools for simulating stimuli and senses, and developing digital world perception abilities (Kashani et al. 2021). The advent of wireless technologies has spurred the exploration of Tactile Internet applications, particularly in healthcare and robotics. The Tactile Internet can be utilized in healthcare for various purposes, including trauma rehabilitation, interactive medical education, remote surgeries, tremor suppression in Parkinson's disease, and training and rehabilitation based on virtual and augmented reality (Hamid et al. 2025). Physicians can assist with the surgery remotely by using haptic and tactile technologies.

The integration of robotics with the Tactile Internet in healthcare can revolutionize medical practices by enabling real-time, ultra-low latency communication and haptic feedback. This technology allows for precise and immediate control of robotic instruments, making remote surgery and telemedicine more feasible and effective. Surgeons can perform delicate procedures remotely, experiencing the same tactile sensations as if they were physically present. Additionally, automated systems in hospitals can interact seamlessly with medical staff and patients, thereby improving efficiency and reducing human error.

The last decade has witnessed a significant progressive adoption of robot-assisted processes in medicine. The work in Burgner-Kahrs et al. (2015) presented advancements in continuum robot manipulators and systems designed for use in interventional medicine. The work in Ciuti et al. (2016) provided frontiers of in-vivo microrobotics and reviewed the state-of-the-art of current endoscopic devices with a focus on research-oriented robotic capsule endoscopes enabled by microsystem technologies. The work in Kassahun et al. (2016) focused on ML techniques directly applied to surgery, surgical robotics, surgical training, and assessment. The application of dual-arm motion mapping for telerobotics in RPM was discussed in Zhou et al. (2019). The work in Esteva et al. (2019) discussed the application of reinforcement learning in robotic-assisted surgery. The work in Manickam et al. (2022) provided a discussion on how AI can be used to support robotic surgeries, improve detection accuracy, enhance decision-making, and facilitate risk assessment. The authors in Simsek et al. (2016) investigated Tactile Internet applications in robotics healthcare systems. The work in Gupta et al. (2019) examined the use of Tactile Internet for telesurgery in smart healthcare systems. Telesurgery is a physician-assisted remote surgery that relies on real-time communication for its success (Gupta et al. 2019). The work in Oteafy and Hassanein (2018) investigated a framework leveraging edge computing and IoT contextualization to enhance multi-tiered cognition in the Tactile Internet, aiming to improve haptic feedback and real-time operation.

10.4.1 Benefits of robotic and tactile internet and key related points

Robotic and automation technologies in healthcare are crucial in improving patient care, optimizing clinical processes, and advancing medical treatment options. Such advanced technologies could offer opportunities to streamline processes, increase accessibility to patient care, and deliver more personalized and precise healthcare services. The Tactile Internet plays a significant role in revolutionizing the industry by fulfilling the demanding standards of ultra-low latency and ultra-high dependability, especially in fields such as robotics, healthcare, and human–machine interaction. Numerous new applications, especially real-time interactive haptic ones, will be made possible by the tactile internet, greatly improving daily living. Such advanced technology will be essential in healthcare for telediagnosis and telesurgery operations. Using the Tactile Internet, a physician may be able to perform a remote surgical operation on a distant patient by perceiving not only real-time visual and auditory but also touching sense of the distant environment. These new possibilities will revolutionize the set of applications and services provided to date by the Internet and take the next-generation systems to an unprecedented level of human-like communication (Promwongsa et al. 2020).

10.4.2 Current challenges in robotic and tactile internet and future research directions

Robotic and Tactile Internet holds significant potential in healthcare, presenting valuable opportunities for future research. However, robotic and Tactile Internet applications, such as telesurgery, require ultra-low latency, a high level of reliability, and security to function correctly and safely. A significant hurdle in achieving the ambitious goal of the robotic and

Tactile Internet is the demanding need for ultra-low latency. This poses a formidable challenge for upcoming communication networks. Hence, the main problem currently limiting the Tactile Internet stems from the requirement of extremely low latency. This poses a real challenge for future communication networks. Therefore, advanced techniques to address this issue are required. Additionally, the latency requirements of robotic and Tactile Internet applications may vary depending on the type of application and the dynamicity of the environment (Maier et al. 2016).

Telesurgery demands URLLC and high bandwidth efficiency to ensure precise movements of specialized medical equipment, such as making incisions or stitching. 6 G is expected to further enhance network reliability and significantly reduce latency, thereby improving the overall telesurgery experience (Ahad et al. 2024). This advancement underscores the importance of developing robust communication networks for future telesurgery applications.

Furthermore, future research directions include developing efficient buffer management mechanisms that work in conjunction with the underlying channel access algorithm to achieve an effective robotic and tactile Internet system. Future approaches could aim to achieve ultra-high reliability for tactile Internet traffic without compromising delay (Promwongsa et al. 2020).

10.5 Explainable AI (XAI)

Recent years have seen the rise of complex decision systems, such as Deep Neural Networks (DNNs) (Alzubaidi et al. 2024). The success of DL models, including DNNs, is due to efficient learning algorithms and their vast number of parameters. With hundreds of layers and millions of parameters, DNNs are often viewed as black-box models (Castelvecchi 2016; Albahri et al. 2024). In contrast, transparency aims to provide a clear understanding of how a model operates. As black-box AI models are increasingly used for critical predictions, the demand for transparency from AI stakeholders is growing (Kalasampath et al. 2025). The concern is that these AI models may produce decisions that cannot be justified or explained in detail (Alammar et al. 2023). Therefore, clear explanations are essential, especially in fields such as precision medicine, where experts require more information than just a binary prediction to support their diagnoses (Mienye et al. 2024). Generally, people are hesitant to adopt methods that are not easily interpretable and trustworthy, particularly with the rising demand for ethical AI (Arrieta et al. 2020; Alzubaidi et al. 2023). While there is often a trade-off between a model's performance and its transparency, improving the understanding of a system can help correct its flaws.

Developing XAI is crucial for bridging the gap between complex model predictions and clinical trust. By providing transparent and interpretable insights into how decisions are made, XAI ensures that healthcare professionals can validate, understand, and confidently act upon AI-driven recommendations, ultimately fostering safer and more ethical integration of AI in clinical practice (Hossain et al. 2025). To mitigate biases in AI algorithms, mechanisms such as diverse training datasets, bias detection audits, and fairness-aware modeling can be employed to address these issues. XAI frameworks that utilize transparent model architectures and interpretable outputs can help clinicians understand decisions, thereby fostering trust and enhancing clinical acceptance (Sadeghi et al. 2024).

Reliable AI is crucial to the healthcare industry, as it enables biomedical experts to make informed decisions and provides valuable guidance. XAI techniques help create transparent and comprehensible AI systems (Sadeghi et al. 2024). Examples of these techniques are class activation mapping (CAM) and layer-wise relevance propagation.

Numerous studies have shown how effective XAI is in the medical field, demonstrating its ability to diagnose conditions such as glaucoma and identify pregnant women with gestational diabetes mellitus (GDM) who require specialized prenatal care (Du et al. 2022). In AI-assisted decision-making, XAI systems' explanations have raised perceived fairness and confidence. Decisions made by AI systems that are biased may be unfair and untrustworthy. This highlights the importance of XAI in reducing the likelihood of bias in AI systems and ensuring impartial healthcare decision-making (Angerschmid et al. 2022). The work in Wani et al. (2024) has demonstrated that integrating XAI and the IoT in healthcare systems enables a fundamental shift towards more transparent and interconnected systems, leading to significant breakthroughs in medical decision-making. A robust framework was developed in Wani et al. (2024) to address key concerns raised by end-users, especially regarding affordable training methods and clear decision-making. To incorporate end-user perspectives, it's crucial to find the right balance between the number of models used and the costs involved. Simpler models facilitate easier explanations, aligning with XAI principles that emphasize accuracy, quality, and cost-effectiveness during both training and deployment. The work in Minh et al. (2022) provided a review to address the gap in understanding the growing field of XAI. It starts by providing a solid background on XAI and then analyzes various methods, categorized into three groups: (i) pre-modeling explainability, (ii) interpretable models, and (iii) post-modeling explainability. The paper also focused on discussing the current approaches for interpreting and analyzing DL techniques. In addition, it systematically examined several challenges in XAI, including the balance between performance and explainability, evaluation methods, security concerns, and policy implications. A smart mobile architecture was developed in order to facilitate the quick prototyping of personal health monitoring applications while guaranteeing the reliability of the data gathered and alarms issued (Oprescu et al. 2022). The work in Müller et al. (2022) explored the application of XAI techniques in vitro diagnostic (IVD) devices to support medical practitioners in making well-informed choices. The paper in Viswan et al. (2024) addressed the challenge of adopting AI models in medical applications, particularly for Alzheimer's disease (AD) detection, by focusing on XAI. The review aimed to make AI models' predictions more interpretable and trustworthy to medical practitioners. The work in Angerschmid et al. (2022) explored how XAI and fairness affect human trust and perceptions of justice in scenarios where AI is involved in decision-making.

10.5.1 Benefits of XAI and key related points

Introducing trustworthy AI into healthcare offers numerous advantages, including helping to diagnose diseases more accurately, enhancing patient care, alleviating the burden on healthcare professionals, and strengthening privacy in the healthcare sector. Trustworthy AI also supports ethical standards and regulations while cutting down on treatment costs and time (Albahri et al. 2023). In particular, by providing clear explanations for AI decisions, XAI builds trust among users and stakeholders, making it easier to adopt AI solutions. XAI allows users to understand the decision-making process, ensuring that AI systems can be held accountable for their outputs. XAI promotes collaboration between AI developers and domain experts by enhancing the transparency and interpretability of AI models. Trustworthy and interpretable AI methods could significantly help make informed and justifiable decisions, thereby providing effective decision support. The benefits of digital technologies in the healthcare system may be increased by combining reliable AI applications with data fusion. This may lead to improved disease detection, more efficient treatment identification, and higher-quality healthcare services (Alkhanbouli et al. 2025).

XAI can empower biomedical professionals to take responsibility for their decisions, which requires the use of clear methods. By identifying and displaying the crucial components of inputs and representations in a neural network that led to a specific outcome, XAI approaches like layer-wise relevance propagation might be helpful (Chou et al. 2022).

XAI can also play a crucial role in cardiovascular disease prediction. XAI techniques empower clinicians by elucidating the importance of each feature in predictions, enabling informed decision-making and building trust (Tjoa And Guan 2020). Collaboration between healthcare providers and AI-driven systems can be facilitated through XAI, which translates complex model outputs into actionable insights, thereby enhancing clinicians' diagnostic and treatment capabilities. Furthermore, XAI was applied in Deperlioglu et al. (2022) to diagnose glaucoma through image analysis, aiding in reliable healthcare decisions. The research aimed to develop a hybrid image processing approach combined with XAI-enhanced deep learning to ensure the reliability of diagnostic outcomes.

10.5.2 Current challenges in XAI and future research directions

XAI has gained popularity in recent years, but researchers have only partially explained the underlying theory of AI. Further research work is indeed needed to fully understand and interpret some technologies in IoT-based healthcare systems. This effort is essential to make smart healthcare systems more transparent and trustworthy. Highly accurate models might be complex, hence making them harder to explain. Therefore, feasible methods for XAI need to be developed. Future research should focus on exploring how XAI can be designed to foster effective collaboration between humans and AI in the healthcare system (Zahid et al. 2025).

Future research should focus on developing advanced methods that maintain high accuracy while improving the interpretability of models, possibly through hybrid approaches that combine interpretable models with complex ones. In addition, future research should focus on scalable XAI methods that can handle large and complex models without compromising the quality of the explanations.

The Artificial General Intelligence (AGI) concept has recently been introduced, which refers to the development and application of highly advanced AI systems that possess the ability to understand, learn, and perform a wide range of tasks at a level comparable to human intelligence (Raman et al. 2025; Ren et al. 2025). Unlike narrow AI, which is designed for specific tasks, AGI aims to exhibit general cognitive abilities, enabling it to handle diverse and complex activities across various domains within healthcare. Research efforts should aim to enhance AGI's cognitive abilities and reasoning. Developing more universal explanation methods that are adaptable to various contexts and user needs could improve the generalizability of XAI techniques.

10.6 Federated learning

Federated learning is a collaborative ML approach where multiple decentralized devices collaboratively train a model without sharing their raw data (Nguyen et al. 2021, 2022). Federated learning enables data training by synchronizing various devices with a central server, all without the need to share the actual datasets. More specifically, it enables multiple clients, such as hospitals, to collaboratively train ML models without sharing raw data, thus maintaining data privacy and managing the scalability of modern healthcare networks. This contrasts with traditional ML, which often relies on centralized data storage and processing. Each IoT device in the healthcare system trains a model locally using its data, and then periodically, the locally trained models' updates are sent to a central server. The central server consolidates these updates to refine a global model, which is subsequently redistributed to the devices. This process repeats, gradually refining the global model with each iteration.

Several works have investigated concepts, architectures, applications, and related issues with federated learning. For example, the study in Yang et al. (2019) offered a detailed examination of the data processing techniques, algorithms, and architectures employed in federated learning systems. The work in Rieke et al. (2020) discussed the technical issues and requirements of federated learning in digital healthcare systems. The work in Lim et al. (2020) provided a survey exploring the integration of federated learning in mobile edge networks. The work in Khan et al. (2021) provided a survey on the utilization of federated learning in IoT networks. A security framework based on federated learning in hostile environments is considered in Kumar et al. (2021). The work in Nguyen et al. (2021) provided a survey that covers the concepts, technologies, and associated learning approaches related to federated learning. The work in Prayitno Shyu et al. (2021) conducted a systematic review of research employing federated learning in healthcare applications. The work in Li et al. (2021) discussed the application of federated learning for Alzheimer's disease to enhance privacy in the healthcare system. The work in Xu et al. (2021) presented a study on federated learning architectures and models, with a special focus on their application in the healthcare system. The work in Boobalan et al. (2022) conducted a survey exploring the combination of federated learning and the Industrial Internet of Things (IIoT), with a primary focus on technical issues related to federated learning implementation. The work in Nguyen et al. (2022) presented the motivation and requirements for using federated learning in smart healthcare systems and reviewed emerging applications of federated learning in key medical areas such as resource-aware federated learning, secure and privacy-aware federated learning, incentive federated learning, and personalized federated learning. The authors in Kumar et al. (2022) proposed a federated learning framework that uses blockchain technology and homomorphic encryption to protect medical privacy. Experiments with CT scan images showed that the proposed framework effectively balances privacy and accuracy. The work in Hossen et al. (2022) used federated learning for skin disease classification. The work in Dhiman et al. (2022) discussed the exploitation of federated learning to ensure the confidentiality and integrity of the information stored in big data healthcare systems. The authors in Taha et al. (2023)carried out a survey on federated learning in the healthcare system from a data perspective. Recently, the work in Rani et al. (2023) presented a survey on the security of federated learning in smart healthcare frameworks.

10.6.1 Benefits of federated learning and key related points

Federated learning is proposed to enhance privacy, reduce latency, and build intelligent and distributed resources for IoT systems. The federated learning approach is particularly beneficial for smart healthcare applications (Nguyen et al. 2022). Federated learning addresses the challenges of centralized data collection and processing in healthcare by enabling distributed collaborative ML. Healthcare systems can leverage federated learning to offer flexible and private EHR management (Nguyen et al. 2021). Federated learning enables secure healthcare collaboration by leveraging its distributed and privacy-preserving nature, thereby enhancing medical service delivery.

In terms of data management and privacy, conventional healthcare systems often rely on centralized data collection and storage. While this simplifies access and management, it raises significant concerns related to patient privacy, data security, and compliance with regulations. Emerging technologies such as federated learning addresses these challenges by enabling decentralized model training across multiple institutions without sharing raw patient data. This not only enhances privacy preservation but also facilitates collaboration among geographically distributed healthcare providers. Noting that in traditional machine learning models, the data are trained from a single hospital or institution, which often suffer from limited generalizability due to data heterogeneity across regions and populations. On the other hand, federated learning enhances scalability by allowing institutions to collaboratively train robust and diverse models without pooling data into a central repository. This approach helps include diverse populations without compromising control over their data.

Federated learning enhances data security by keeping patient data localized on devices, sharing only model updates rather than raw data. This minimizes exposure risks compared to centralized models. While it offers strong privacy benefits, federated learning may face slight accuracy trade-offs and slower response times due to decentralized coordination and communication overhead in clinical settings.

10.6.2 Current challenges in federated learning and future research directions

There are increasing challenges in handling multi-source and heterogeneous IoT data, especially in healthcare systems. Data fusion is a significant trend in federated learning, driven by the increasing variety and volume of data (Ji et al. 2024). Effective methodologies and algorithms are needed to address issues such as user privacy, universal model design, and stability of data fusion results. By eliminating redundant information and combining various data sources, more valuable insights can be gained. Therefore, advanced methodologies and applications for data fusion in federated learning should be investigated in the future.

While federated learning could offer considerable improvements in healthcare systems, it might come with limitations (Rauniyar et al. 2023). Federated learning requires a robust coordination infrastructure and standardization across participating entities. Therefore, continued investigation is needed to address these requirements and guide future implementations. Handling device heterogeneity, which encompasses differences in processing speed, storage capacity, and network connectivity, presents additional challenges in federated learning that must be addressed. To handle these variations effectively without sacrificing model performance, advanced algorithms and methods must be developed. Furthermore, maintaining efficient federated learning operations depends on streamlining data transmission overhead and improving communication methods, which could be worth investigating in the future.

Federated learning requires to use of robust encryption and authentication mechanisms to protect healthcare data during transmission. In particular, balancing data protection with real-time performance presents a significant challenge in implementing advanced technologies, such as federated learning and blockchain, within healthcare environments. Federated learning enhances patient privacy by keeping sensitive data decentralized, allowing models to be trained locally without exposing raw information. However, coordinating training across multiple devices or institutions introduces latency and demands substantial computational resources, potentially slowing down system responsiveness. Similarly, blockchain provides a secure and transparent framework for storing medical records and audit trails; however, the validation processes and consensus mechanisms involved can introduce delays, particularly when quick access to data is crucial. While these technologies offer strong privacy protections–federated learning by keeping raw data decentralized, and blockchain through immutable and transparent data records–they can also introduce latency and computational overhead. These performance limitations may impact the responsiveness required in time-sensitive clinical scenarios. As such, these trade-offs underscore the need for careful architectural decisions and the development of hybrid frameworks that can maintain high standards of privacy and security without compromising the speed and efficiency required in time-sensitive clinical settings. Furthermore, future work should focus on optimizing these frameworks to maintain robust security without compromising real-time performance, particularly in high-stakes healthcare settings where rapid data processing and reliability are essential.

Managing diverse hardware and software characteristics from different platforms and devices in the healthcare systems could be a possible future work that needs to be considered. Specifically, these various platforms and devices often employ different learning frameworks and store data in diverse formats, resulting in significant communication and integration challenges. Addressing this heterogeneity in data is crucial for designing federated learning frameworks that can efficiently train models and improve performance in IoT-based healthcare applications.

10.6.3 Solutions to federated learning issues: AI-To-data (ATD) learning

AI-To-Data (ATD) is a next-generation decentralized learning framework designed to tackle key limitations in current collaborative AI models (Alzubaidi et al. 2025) (see Fig. 16). Unlike traditional approaches that rely on central servers or aggregators, ATD facilitates fully connected, peer-to-peer weight sharing between different institutions without requiring the sharing of raw data. This framework supports multi-modal learning (e.g., audio, image, video, and text) and features bias-aware model aggregation to ensure fairness across nodes with diverse data. ATD is designed to be scalable, adaptable, and privacy-first, making it ideally suited for real-world applications in sensitive areas such as healthcare, finance, and security.

ATD effectively addresses the core limitations of Federated Learning. In Federated Learning, a central server coordinates the learning process, which introduces a single point of failure, increases vulnerability to security threats, and may lead to biased model updates if certain clients dominate the learning process. Furthermore, Federated Learning's reli-

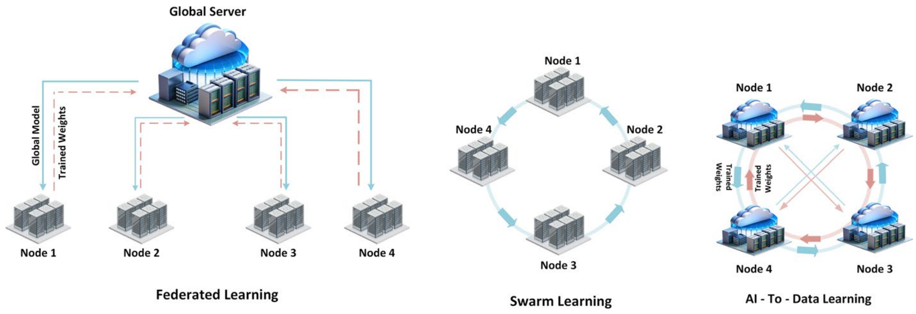


Fig. 16 Federated Learning Vs. Swarm Learning Vs. AI-To-Data (ATD) Learning (Alzubaidi et al. 2025)

ance on static client participation diminishes its adaptability in dynamic environments. In contrast, ATD entirely removes the central aggregator by adopting fully connected peer-to-peer weight sharing, which allows for the dynamic addition and removal of nodes. By decentralizing control, ATD also spreads the risk and boosts the robustness and scalability of the learning process.

Compared to Swarm Learning, which employs blockchain to manage updates and achieve consensus among nodes (Warnat-Herresthal et al. 2021), ATD offers a lighter and more flexible alternative. Swarm Learning experiences update latency, high communication overhead, and often lacks granular fairness controls in the learning process. ATD addresses these issues by utilizing adaptive ensemble techniques for bias mitigation, along with hybrid encryption and authentication mechanisms, to ensure secure and efficient model exchange. Consequently, ATD achieves faster convergence, enhanced data security, and equitable model contributions, establishing it as a practical and innovative solution for decentralized AI in real-world, multi-institutional environments.

Since the healthcare domain generates data from multiple modalities and is highly sensitive to issues of bias and privacy, ATD is exceptionally well-suited for its challenges.

10.7 Data fusion

The practice of combining data from multiple sources to provide more accurate, consistent, and practical information than can be found in any single source is known as data fusion. Data fusion or information fusion involves integrating multiple data sources to enhance the effectiveness of AI systems (Alanazi And Muhammad 2022; Chaabene et al. 2025). This process may include techniques such as data preprocessing, feature selection, and feature engineering. Data fusion enabled the integration of information from diverse sources, such as sensors, medical records, and clinical notes, to create a unified and comprehensive view of a patient's health (Chen et al. 2024). By combining heterogeneous data types, data fusion can enhance diagnostic accuracy, support timely interventions, and provide deeper insights into patient conditions across multiple dimensions. Data fusion is crucial for developing robust, explainable, and trustworthy medical AI systems because it facilitates the integration of diverse information sources. This integration enhances the accuracy and reliability of AI models, improving the transparency and explainability of their decision-making processes (Ahmed et al. 2024). Furthermore, combining multiple data sources can reduce potential biases, resulting in fairer, more reliable, and trustworthy AI models. Hence, data fusion is essential for developing more robust AI systems in healthcare (Angerschmid et al. 2022). Data fusion creates a more complete picture of a patient's health by integrating data from multiple sources within the healthcare system, such as imaging scans and lab findings. With this integration, medical professionals can diagnose patients more accurately and develop treatment regimens tailored to their needs. The accuracy of predictions is directly influenced by the quality, quantity, and relevance of the data gathered from IoT devices. Data fusion involves merging, reconciling, and analyzing data from multiple sources, such as sensors, databases, or other information repositories, using specialized approaches to gain deeper insights into the data. The work in Ahmed et al. (2024) evaluated data fusion techniques, emphasizing their role in enhancing healthcare systems. To achieve an inference for IoT-based healthcare systems, a novel data fusion approach utilizing type-2 fuzzy logic (T2FL) and Dempster-Shafer theory (DST) has been developed in Ullah et al. (2021). To this end, type-2 fuzzy logic efficiently assesses patient data membership values, while Dempster-Shafer theory (DST) adeptly integrates and processes evidence derived from these values within the decision-making system. Comprehensive computer simulations using datasets for heart disease and diabetes demonstrated that the proposed scheme significantly surpassed ontology-based and type-1 fuzzy logic approaches in terms of explainable decision accuracy. The authors in Jan et al. (2024) proposed a lightweight data fusion approach that is specifically designed for wearable devices in the IoMT to refine medical data transmitted by multiple wearable devices.

10.7.1 Benefits of data fusion and key related points

Data fusion has several specific applications in healthcare systems, including optimizing data quantity, reducing data size and dimensions, and extracting meaningful information (Ding et al. 2019). Data fusion helps eliminate data anomalies and prevents collusion of data detected from multiple sensors. However, the process generates abundant sensitive patient information, which poses a risk due to the potential untrustworthiness of both the collection terminals and processing centers. Data fusion has been shown to enhance the performance of ML models in numerous applications by combining heterogeneous data. More specifically, in the era of big data and AI, analyzing fused healthcare data has become increasingly complex due to the large volume and diversity of sources involved. Traditional techniques often fall short when it comes to processing and interpreting such high-dimensional, multi-sensor healthcare data at scale. To overcome these limitations, an innovative large model-driven strategy tailored for advanced healthcare data fusion in hyperscale, multi-sensor environments could be used (Lv et al. 2025). This approach might enable the seamless integration of data from a wide array of medical devices and sources, leveraging the power of large models to extract meaningful insights from both structured and unstructured health information.

By effectively reducing the volume of data, data fusion enables the system to minimize the energy needed for data transmission. A more comprehensive global knowledge can be attained by doing data fusion at the fog or edge layer. Cooperative data fusion at the edge, for example, can extract useful information from heterogeneous and varied data sources. Based on vital signs gathered from multiple sources, this method can provide a thorough picture of a patient's health status in medical applications.

Data fusion enhances the accuracy and reliability of AI models in healthcare by integrating more effective monitoring, and develop more effectiveng graph analysis and feature visualization. These techniques address the limitations of individual data sources, improving the overall precision of AI systems and leading to more accurate disease predictions (Albahri et al. 2023). The accuracy of AI models can be considerably enhanced using data fusion techniques (Alanazi And Muhammad 2022). Data fusion involves integrating data from various sources, such as sensors, wearable devices, electronic health records (EHRs), and other inputs, to create a more comprehensive and accurate representation. In the healthcare system, this technique can enhance decision-making by combining data from various medical devices to generate a comprehensive overview of a patient's health condition. This approach enables healthcare providers to make more accurate diagnoses, implement better monitoring, and develop improved treatment plans. Data fusion can significantly enhance AI-based disease diagnoses by integrating various data sources, including medical images, patient records, and demographic information. Moreover, integrating data fusion with advanced techniques, such as graph causal models and complex network inference, for verification and explainability, can lead to the development of more reliable and resilient AI systems.

Other forms of fusion, such as model fusion and decision fusion, can enhance the accuracy and reliability of diagnostic outcomes by leveraging complementary information from multiple sources (Alwzwazy et al. 2025; Abdulredah et al. 2025; Saihood et al. 2024; Alammar et al. 2024; Alzubaidi et al. 2024).

10.7.2 Current challenges in data fusion and future research directions

Data heterogeneity is considered a significant challenge in data fusion, as it originates from multiple sources with their own formats and standards. These sources, discussed earlier, include sensors, medical equipment, wearable devices, and Electronic Health Records (EHRs). This discrepancy may complicate the integration process. Therefore, future research should focus on developing robust frameworks and algorithms that can efficiently address heterogeneous data sources.

In general, the data collected contains socially constructed biases, inaccuracies, errors, and mistakes. However, the accuracy and quality of the data should be guaranteed. This is because inaccurate or low-quality healthcare data may influence patient treatment by leading to incorrect decision-making. Therefore, finding efficient methods for data pre-processing is essential to provide high-quality input for fusion algorithms. The accuracy and quality of the data should be established before training any model with the collected data. Additionally, advanced ML and AI algorithms can be utilized to enhance the accuracy and reliability of data fusion.

Additionally, given the sensitivity of patient data, data security and privacy are of paramount importance. To govern individual personal data, there must always be data protocols in place. These protocols should specify who is permitted to access the data and under what conditions. It's challenging to keep this data secure from breaches while still making it useful for fusion. Therefore, research should focus on finding advanced encryption and anonymization methods to ensure patient data throughout the fusion process. The distinction between centralized and distributed data fusion presents additional challenges related to privacy and explainability. However, distributed fusion introduces new issues. Distributed fusion, in particular, might pose additional security and privacy concerns.

Traditionally, ML applications in healthcare have relied on single-modality data, which limits their ability to replicate the holistic decision-making processes observed in clinical settings accurately. In contrast, clinicians draw upon a diverse array of information–ranging from demographic details and laboratory results to vital signs and multiple imaging modalities–to form a comprehensive understanding of a patient's condition (Krones et al. 2025; Sewify et al. 2025). This reliance on heterogeneous data underscores the need for computational models that can integrate and interpret such complexity. Recent advancements in ML have begun to bridge this gap, enabling more sophisticated integration of multimodal data and aligning more closely with how clinicians synthesize information in practice. In this context, Further investigation regarding the multimodal ML approaches in healthcare, focusing on the fusion of commonly used data types in clinical diagnosis–including imaging, textual records, time-series signals, and structured tabular data. This exploration not only highlights existing progress but also points toward promising directions for future research in creating models that more faithfully replicate the nuances of real-world medical reasoning.

Future research should also focus on assessing how automated ML techniques might affect healthcare services, as well as exploring the challenges and limitations of applying these techniques in health data integration. Ultimately, more efficient and safe data fusion techniques in the healthcare industry may be made possible by interdisciplinary approaches that integrate knowledge from cybersecurity, data science, and healthcare. Future data fusion approaches may thus consider endowing DL models with explainability by externalizing other domain information sources (Arrieta et al. 2020).

Furthermore, with the growing variety of technologies and communication protocols in healthcare, achieving smooth and reliable system interoperability remains a challenge. In particular, differences in data structures and system designs often obstruct the seamless flow of information between platforms. To overcome these challenges, several standardization initiatives have emerged. Among them are Health Level Seven (HL7), a framework for the exchange of clinical data; Fast Healthcare Interoperability Resources (FHIR), which simplifies the transfer of healthcare information through modern web technologies; and Digital Imaging and Communications in Medicine (DICOM), which standardizes the handling and sharing of medical imaging data (Gazzarata et al. 2024; Tölle et al. 2025). To achieve robust interoperability in smart healthcare systems, strategies should include adopting universal data exchange standards mentioned above and aligning system architectures through collaborative industry frameworks. Additionally, regulatory support and stakeholder consensus are crucial for ensuring seamless integration across diverse platforms and technologies. While these efforts mark important progress, a more in-depth investigation is needed to assess how effectively they are being adopted in real-world clinical settings and to guide the development of more adaptable and straightforward integration frameworks for the future.

10.8 Quantum computing for healthcare

Quantum communication and computing can play a crucial role in intelligent healthcare systems. More specifically, it has the potential to revolutionize the healthcare system by providing ultra-secure data transmission, enhancing patient privacy, and improving the reliability of telemedicine. Quantum communication and computing can improve the efficiency of computing and provide strong security for future 6 G networks (Zhang et al. 2019). Quantum communications offer robust security by utilizing a quantum key based on the quantum no-cloning theorem and the uncertainty principle. When eavesdroppers attempt to observe, measure, or copy data in quantum communications, the quantum state is disturbed, making the eavesdropping behavior easily detectable. This inherent property ensures the integrity and confidentiality of the transmitted information. Quantum key distribution ensures that sensitive medical information, such as EHRs and diagnostic data, is protected against eavesdropping and cyberattacks. This technology could facilitate secure remote consultations, real-time monitoring, and the safe sharing of critical patient data between healthcare providers.

Quantum computing, leveraging quantum superposition and entanglement, can significantly enhance computing capabilities through unitary transformations using qubits. Common digital computing uses binary digits to code data, whereas quantum computation uses quantum bits, or qubits. Therefore, qubits allow computer bits to operate in three states compared to two states in classical computation. This allows quantum computers to operate thousands of times faster than traditional computers. As such, this advancement enables the acceleration and improvement of AI algorithms that require extensive data processing and intensive training. Moreover, the integration of quantum theory with AI, known as quantum AI, has the potential to create more powerful and efficient AI algorithms, meeting the advanced requirements of 6 G technology (Zhang et al. 2019).

10.8.1 Benefits of quantum communication and key related points

Quantum communication offers substantial benefits for the healthcare system by providing robust security through quantum key distribution, which relies on the principles of quantum no-cloning and uncertainty. This ensures that any eavesdropping attempts can be easily detected. In addition, quantum communication enhances the accuracy and reliability of telemedicine, remote diagnostics, and patient monitoring, ultimately improving the overall quality and security of healthcare services. Quantum computing has the ability to address a wide range of computational challenges, enabling critical decisions to be made significantly faster than with classical computing methods(Chengoden et al. 2023). Quantum computing, when integrated with the metaverse, will provide solutions to the substantial challenges of security, extensive computational demands, and cyber threats. The immense computational power of quantum computing will significantly enhance healthcare applications within the metaverse. These applications rely heavily on computation and simulation, and quantum computing will improve both efficiency and user experience for doctors and patients. In addition, quantum randomness helps developers enforce regulations, preventing manipulation of healthcare applications by inhabitants and algorithms in the metaverse (Chengoden et al. 2023).

Quantum computing can be applied to big data analytics, offering a solution to the challenges posed by the massive size of datasets (Lloyd et al. 2016). Traditional computers struggle with memory access, leading to bottlenecks, but quantum computing's capabilities can significantly enhance data processing, making it a powerful tool for handling and analyzing big data (Dash et al. 2019). More specifically, quantum approaches can significantly reduce the information needed for big data analytics. For instance, quantum theory can maximize distinguishability in a multilayer network with a minimal number of layers. In addition, quantum methods need a smaller dataset to achieve highly sensitive data analysis compared to traditional ML techniques, thus reducing the computational power required.

Furthermore, quantum cryptography ensures the security of medical WBAN applications by leveraging the inherently unpredictable nature of quantum states for encryption (Cornet et al. 2022). The no-cloning theorem, which prevents copying or amplifying an unknown quantum system without altering its original state, makes stealing quantum information as challenging as stealing the quantum device itself (Cornet et al. 2022). Additionally, secure communication among a designated number of users is possible through quantum entanglement.

Quantum approaches have the potential to solve complex problems that are considered unsolvable using conventional computing. For instance, current encryption techniques such as Rivest–Shamir–Adleman (RSA), public key (PK), and Data Encryption Standard (DES), which are currently deemed secure, may become obsolete in the future as quantum computers will be able to break them quickly.

10.8.2 Current challenges in quantum communication and future research directions

The expensive and complicated nature of quantum communication hardware, such as quantum memory units and single-photon detectors, is regarded as a critical challenge that must be tackled. For wider deployment, these costs must be decreased, and the effectiveness and robustness of quantum components must be improved. Moreover, a major challenge is preserving the coherence of quantum states in real-world settings, where decoherence and other quantum disturbances are common.

Developing more effective quantum repeaters and error-correction techniques is crucial for increasing the dependability and range of quantum communications. Investigating hybrid systems that combine classical and quantum communication techniques may provide viable solutions for deployment and integration over time. Further investigation in this regard is needed. Future research could also focus on integrating quantum communication protocols with existing healthcare systems and exploring the use of quantum technologies to enhance the overall security and functionality of medical devices and networks. For example, integrating IoT, blockchain, and quantum ML could be a feasible solution for advancing multimodal data fusion in future smart healthcare-based AI systems (Dankan Gowda et al. 2025). Furthermore, the management and optimization of quantum networks using quantum computing may yield more scalable and effective solutions. Furthermore, research into quantum key distribution and quantum-safe cryptographic algorithms is crucial for ensuring the highest levels of security in the transmission of sensitive data.

10.9 Edge learning

Edge learning or edge computing with AI refers to the practice of performing data processing and ML functions on edge devices, which are located closer to the data source, rather than relying on centralized cloud-based systems. Edge learning complements cloud-based methods for big data analytics by enabling distributed edge nodes to cooperatively train models and conduct inferences using their locally cached data (Zhang et al. 2021). Edge computing allows real-time applications, including XR, holographic teleportation, and Tactile Internet. Integrating edge computing with AI techniques creates a new communication paradigm known as edge intelligence, poised to play a crucial role in the future of healthcare systems. By processing data locally, edge learning minimizes the latency associated with sending data to and from a central server. This is critical for real-time healthcare applications such as RPM and telesurgery.

A comprehensive survey of the recent research efforts on edge learning was provided in Zhang et al. (2021). The work in Zhu et al. (2020) proposed a new set of design principles for wireless communication in edge learning, referred to as learning-driven communication. The authors demonstrated that the introduced learning-driven communication techniques, including multiple access, resource allocation, and signal encoding, can break the communication latency bottleneck, leading to fast edge learning. The work in Mo and Xu (2021) examined a federated edge learning system where an edge server coordinates multiple edge devices to train a shared ML model using locally distributed data samples. The study focused on enhancing system energy efficiency by jointly optimizing communication resource allocation for aggregating global ML parameters and computation resource allocation for locally updating these parameters during distributed training.

10.9.1 Benefits of edge learning and key related points

Edge learning can reduce the need for continuous data transmission to the cloud, thereby conserving bandwidth. This is especially beneficial in environments with limited or expensive network resources. The primary motivation for advancing learning at the edge is to enable quick access to the vast real-time data generated by edge devices for fast AI model training, thereby providing these devices with human-like intelligence to react to real-time events (Zhu et al. 2020).

Edge learning has distinct advantages over cloud and on-device learning. First, it offers a balanced resource allocation, optimizing the trade-off between AI model complexity and training latency. Its proximity to data sources allows for real-time data processing, overcoming the cloud learning issue of propagation delay. Additionally, this proximity offers benefits in terms of location and context awareness. Compared to on-device learning, edge learning achieves higher accuracy by supporting complex models and aggregating distributed data from numerous devices. With edge learning, the data can be processed on local devices. This would help keep sensitive information within the device, thereby mitigating the threat of data breaches and ensuring better compliance with privacy and security regulations. Edge learning can provide more reliable performance in healthcare critical applications, where it can continue to perform a function even when the device is offline or has a poor internet connection. Edge learning can support the scalability of the healthcare system. For example, distributing the computational load across multiple edge devices can help scale the system more effectively than relying solely on central servers.

10.9.2 Current challenges in edge learning and future research directions

Edge devices typically have less processing power, memory, and storage compared to the cloud servers. Therefore, designing efficient AI algorithms that can address these challenges is essential. Edge devices often run on batteries, so energy-efficient models are crucial to avoid the rapid depletion of power resources. Keeping AI models updated on a multitude of edge devices can be complex and resource-intensive. Hence, ensuring consistent performance across all devices requires robust update mechanisms. Edge devices can vary significantly in terms of hardware capabilities and operating systems. Therefore, developing applications that function seamlessly across a wide range of devices becomes increasingly important- an area that may be worth further exploration in future research.

One of the primary goals of edge learning is to address the limitations of the limited data and processing power at each edge device. This can be achieved by utilizing mobile edge computing platforms and harnessing the vast amounts of data spread across numerous edge devices. In these systems, learning from distributed data and ensuring effective communication between the edge server and devices are critical challenges, presenting countless opportunities for future research.

11 Conclusion and future work

This comprehensive survey has reviewed and analyzed the latest advancements toward smart healthcare systems based on IoT. To this end, the key enabling and innovative technologies have been identified for the future of smart healthcare systems. After determining the most important, cutting-edge, and relevant technologies, this survey paper provides the fundamental definition of each technology, highlights its significant advantages, examines the most recent cutting-edge research, and discusses the related research challenges. By investigating the most influential research published over the last decade from top-tier publishers and top academic sources, this paper provides critical insights into the role of cutting-edge technologies including sensors and smart devices, RPM systems for telemedicine, and AAL, EHRs, ICT for networking and communication, big data analytics and management, computing methods, AI, and security and privacy measures. In addition, this survey paper identified emerging technologies, including IoNT, energy harvesting, the metaverse, digital twin, robotic and Tactile Internet, explainable AI (XAI), federated learning, data fusion, quantum communication, and edge learning, as reshaping healthcare systems. Furthermore, the survey identified persistent challenges and open research questions, offering a roadmap for future exploration in this rapidly evolving field. Overall, this paper is poised to serve as a valuable resource for academic researchers, engineers, healthcare professionals, and policymakers, providing them with the necessary knowledge to drive innovation and enhance the robustness of IoT-based healthcare systems. By addressing both current developments and future directions, this paper stands as a pivotal reference in the ongoing effort to revolutionize healthcare through technology.

In conclusion, several key directions merit further exploration in future research to enhance the development and implementation of smart healthcare systems. These could include the implementation of AI algorithms and novel fusion techniques, which are essential for achieving efficient big data analytics in healthcare systems. For example, automated decision-making can be achieved through ML methods. Future work may also explore the integration of advanced energy harvesting techniques to ensure sustainable and uninterrupted operation of smart healthcare devices, particularly in remote or resource-constrained environments. Future research could also focus on developing adaptive, context-aware security frameworks that preserve patient privacy while ensuring robust protection against evolving cyber threats in smart healthcare environments. Future research should focus on exploring how XAI can be designed to foster effective collaboration between humans and

AI in the healthcare system. Future research could also explore more adaptive and secure edge learning frameworks tailored for real-time healthcare scenarios, ensuring low-latency processing and enhanced privacy. Emphasis may also be placed on integrating federated learning to leverage distributed medical data without compromising patient confidentiality. To meet the specific requirements of multimodal AI, advancements such as federated learning, self-supervised learning, improved data sources, and system-level optimizations are essential. Federated learning, in particular, offers significant advantages by allowing models to train on diverse and distributed datasets while maintaining patient confidentiality, a critical factor when integrating various forms of healthcare data. Further investigation in this regard is needed. To address the dual challenges of data security and system responsiveness in smart healthcare systems, hybrid solutions integrating blockchain, federated learning, and complementary data protection technologies are essential. Blockchain ensures data integrity and transparency, while federated learning enables decentralized model training without exposing sensitive patient information. Combining these techniques with other encryption and authentication mechanisms creates a robust, multi-layered security framework that maintains real-time system performance. Furthermore, the progressive integration of advanced AI technologies, such as LLMs and Intelligent Autonomous Management (IAM) systems, can significantly enhance decision-making, data analysis, and operational efficiency in clinical environments. The inclusion of Generative AI (GAI) alongside LLMs offers a forward-looking perspective on the transformative potential of these technologies in revolutionizing smart healthcare, making systems more adaptive, secure, and intelligent. Data heterogeneity is also considered a major challenge with data fusion. Therefore, future studies could focus on developing more robust and context-aware data fusion techniques that integrate heterogeneous medical data in real time, enhancing diagnostic accuracy and decision-making. In addition, exploring scalable fusion models that maintain performance across diverse healthcare environments remains a promising direction. Future research could unlock the potential to realize the capabilities of general-purpose medical AI models across various clinical applications, ultimately improving both patient outcomes and healthcare delivery. This could be through further investigation and the development of artificial general intelligence. Future research could explore more intuitive XAI models that provide clear, actionable insights for both clinicians and patients. Emphasis should be placed on balancing model transparency with predictive performance to build trust and ensure ethical decision-making in real-world healthcare applications. Finally, while significant advancements have been made in the conceptual design and technical development of IoT-based smart healthcare systems, the transition from theoretical models to real-world clinical applications remains a crucial challenge. To bridge this gap, it is essential to promote experimental validation and pilot testing in real clinical environments. Conducting well-structured pilot studies allows researchers and healthcare professionals to assess the actual performance, reliability, and usability of these systems under dynamic and often unpredictable clinical conditions. Such studies help uncover practical limitations related to patient interaction, data integrity, latency, power consumption, and network stability. Furthermore, such studies iterative refinement of system components, ensuring that the proposed technologies align with the complex demands of healthcare workflows, regulatory compliance, and ethical standards. As a result, future research should prioritize collaborative efforts between technologists, clinicians, and policymakers to design scalable pilot projects that can demonstrate not only technical feasibility but also clinical efficacy, safety, and patient acceptance.

Furthermore, to validate the proposed solutions, well-structured pilot studies should be conducted in real healthcare environments. These studies should combine technical assessments–such as system reliability, latency, and energy efficiency–with clinical outcomes like patient safety, diagnostic accuracy, and user satisfaction. This approach will serve as a critical step toward translating innovative smart healthcare concepts into fully operational solutions that can be confidently deployed in hospitals and patient homes.

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Authors and Affiliations

Muntadher Alsabah[1] · Marwah Abdulrazzaq Naser[2] · A. S. Albahri[3,4] · O. S. Albahri[5,6] · A. H. Alamoodi[7,8] · Sadiq H. Abdulhussain[2] · Laith Alzubaidi[9,10]

✉ Laith Alzubaidi
l.alzubaidi@qut.edu.au

1 Medical Technical College, Al-Farahidi University, Baghdad, Iraq

2 College of Engineering, University of Baghdad, Baghdad, Iraq

3 Technical Engineering College, Imam Ja'afar Al-Sadiq University (IJSU), Baghdad, Iraq

4 University of Information Technology and Communications (UOITC), Baghdad, Iraq

5 Computer Techniques Engineering Department, Mazaya University College, Nasiriyah, Iraq

6 Institute of Innovation, Science and Sustainability (IISS), Federation University, Melbourne, Australia

7 Applied Science Research Center, Applied Science Private University, Amman, Jordan

8 GUST Engineering and Applied Innovation Research Center (GEAR), Gulf University for Science and Technology, Mishref, Kuwait

9 School of Mechanical, Medical, and Process Engineering, Queensland University of Technology, Brisbane, Australia

10 Centre for Data Science, Queensland University of Technology, Brisbane, Australia